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Central Bank Reserves for All: Welfare & Inequality in a THRANK Model

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Abstract

This paper examines the effects of a universal remunerated retail Central Bank Digital Currency (CBDC), dubbed “Reserves for All” (RFA), on distributional welfare as well as wealth and income inequality. Using a Three-Agent New Keynesian (THRANK) model built and calibrated to reflect Switzerland’s wealth and income distributions, I show that RFA provides an interest-bearing investment vehicle that fundamentally alters the wealth accumulation potential of households at the bottom of the wealth distribution. RFA reduces wealth and income inequality by 3.19 and 0.68 percentage points respectively, yielding a steady-state aggregate welfare gain of 1.16%. Additionally, stochastic simulations reveal that RFA provides a macroeconomic stabilisation effect, reducing the volatility of key aggregates such as consumption, inflation, and output. However, this comes at the cost of severe bank disintermediation, lowering bank profits and dividend income, which decreases investment and output in the long run. In the short term, the implementation of RFA imposes welfare costs driven by an initial savings-induced contraction in demand. Taking into account the full transition path, aggregate welfare gains reach 0.46%. Finally, tiered remuneration targeting excessive balances mitigates long-run disintermediation while preserving the redistributive benefits.

JEL codes: E42, E58, D31

Keywords: CBDC, Wealth Inequality, Welfare, Household Heterogeneity

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1 Introduction

Central bank reserves are a risk-free form of money currently restricted exclusively to commercial banks, which earn interest on these accounts. The declining use of physical cash has sparked the exploration of “Reserves for All” (RFA), i.e. extending access to central bank accounts to the wider public, often presented as a remunerated retail central bank digital currency (CBDC). While the existing literature debating CBDC has extensively modelled its aggregate macroeconomic implications, a critical gap remains regarding its distributional consequences. Would allowing the public to be exposed to the policy interest rate help the poorest build wealth through safer, higher-interest savings compared to traditional commercial savings accounts? This paper addresses this exact question, evaluating the impact of RFA on distributional welfare and wealth inequality.

To understand why RFA might impact inequality, we must examine heterogeneous returns and portfolio composition. While return heterogeneity is not the sole determinant of inequality, empirical evidence reveals that it does contribute to its increase (CAO & LUO 2017; XAVIER 2021), with returns to wealth found to be increasing along the wealth distribution. Differences in returns are tied to portfolio composition, as households across the wealth distribution hold different types of assets. Additionally, the differential fluctuations in asset prices have a major influence in shaping wealth inequality over time (AURAY, EYQUEM et al. 2025), with capital gains mechanically increasing savings rates of the rich as they are often retained (FAGERENG et al. 2019). This occurs as wealthier households possess the financial freedom to invest in high-yield, illiquid, or risky assets such as equities and real estate, as well as having access to exclusive wealth management services such as financial advisors and family offices. This portfolio heterogeneity across the wealth distribution also dictates exposure to aggregate risk. Because equity is heavily concentrated at the top, temporary shocks to equity returns have large, persistent effects on top wealth shares, acting as a primary driver of wealth inequality dynamics (CIOFFI 2021).

In contrast, constrained and low-wealth households face structural disadvantages. Deterred out of financial markets by participation, transaction, and information costs, they often keep their limited wealth in liquid forms such as basic bank deposits. While these provide direct utility, they offer persistently low yields as deposit rates are subject to monopolistic markdowns imposed by commercial banks. The increasing returns to wealth create an asymmetric wealth-generating mechanism that systematically limits the accumulation capacity of the bottom of the distribution. Because wealth inequality is driven by this disparity in asset access and returns, the introduction of a universal, risk-free, interest-bearing asset can serve to raise the yield floor. One such asset is conveniently found in central bank reserves, remunerated at the policy rate. If access is granted to all households, this may reduce the return spread between the top and bottom of the distribution, and perhaps serve to reduce inequality and improve welfare. This mechanism forms the core motivation behind my research question.

The debate around RFA is mostly framed around retail central bank digital currency, another way to designate household access to deposit accounts at the central bank. This literature emerged as the declining usage of cash and increasing reliance

on private payment systems sparked central banks’ interest to explore CBDC. The resulting debate spans the implications for macroeconomic aggregates, central bank balance sheets, monetary policy transmission, bank competitiveness (and the risk of disintermediation), financial stability, privacy, payment efficiency, and financial inclusion (BORDO & LEVIN 2017; INFANTE et al. 2024). Research in this area ranges from literature reviews and surveys, such as AUER et al. (2022), BINDSEIL & SENNER (2024) and INFANTE et al. (2024) to quantitative analysis using theoretical models, explored in section 2.

Because retail CBDC remains largely¹ theoretical, it is as of yet lacking a single defined set of design characteristics. As a result, researchers make widely differing assumptions, leading to inconsistent CBDC designs across macroeconomic models. BINDSEIL & SENNER (2024) criticises this literature, pointing out that these models are often out of line with the actual design features communicated by central banks working on CBDC. These are based on a “conservative” reaction to the evolution of payment systems, often featuring zero remuneration, quantity limits, restricted access, and linkage between CBDC wallets and commercial bank accounts. These design choices reflect central banks’ desires to prevent significant expansion of their balance sheets and bank disintermediation, a scenario in which banks lose deposit funding with possible consequences on bank lending. Hence, real-world CBDCs will likely be more akin to “digital cash”, rather than an investment vehicle. The design in this paper however, takes the opposite view, where I investigate the investment vehicle aspect. For this reason, I adopt the nomenclature of “Reserves for All” (RFA), taken from NIEPELT (2020), and following the idea in RICKS et al. (2020) that the simplest way to implement a CBDC would simply be to extend access of central bank accounts to the wider public.

I define RFA as a new universal asset, consisting of central bank reserves with unlimited access granted to households, and remunerated at the policy rate. In order for banks to keep operating in the case where they lose their deposit funding, central banks in this model offer unlimited funding though at a higher interest rate. While NIEPELT (2020) describes how RFA can, under certain equivalency conditions, be neutral to macroeconomic outcomes, this is not the case in this paper as the higher-rate funding breaks equivalency. Furthermore, RFA mechanically expands the central bank balance sheet because required bank reserves are not large enough to cover RFA demand.

This thesis contributes to the literature by placing the distributive effects of RFA at the centre of the analysis. To quantitatively assess the distributional impacts of RFA, this paper develops a Three-Agent New Keynesian (THRANK) model designed to broadly reflect the Swiss wealth and income distributions, a context characterised by high wealth inequality and vastly different asset compositions across wealth brackets (MARTÍNEZ 2021).

The framework splits infinitely-lived households into three groups based on their activities, financial access, and portfolio composition. Hand-to-Mouth (H) agents represent the bottom of the distribution, primarily earning labour income and holding

¹There have been a few real-world implementations such as the Sand Dollar in the Bahamas or the eNaira in Nigeria.

wealth solely in low-interest commercial bank deposits. Middle-Class (M) agents earn higher wages and participate in the housing market, holding housing wealth leveraged by mortgage debt subject to a collateral constraint. Saver (S) agents represent the wealthy capitalist class. They supply no labour, own the economy’s physical capital and firms, and are the sole recipients of dividends.

The model features an imperfectly competitive banking sector where banks possess market power, setting rate spreads for deposits, mortgages, and bonds. While steady-state rate markdowns are set depending on exogenous parameters rather than set endogenously (as seen in ASSENMACHER et al. (2024) or CHIU et al. (2023)), this setup does approximate the core implications of bank disintermediation. The policy experiment introduces RFA as an imperfect substitute for regular commercial deposits in the utility function. A preference shock transitions RFA from providing negligible liquidity utility to matching that of regular deposits, while remunerating it directly at the policy rate.

This analysis evaluates the introduction of RFA through steady-state comparisons, transition dynamics, and impulse response functions (IRFs) to stochastic shocks, with extensions exploring parameter recalibrations and tiered remuneration. The main outcomes of interest are welfare, measured via forward-looking (comparing steady-states) and backward-looking (focusing on the transition) consumption-equivalent variation (CEV), and inequality, quantified by Gini coefficients for wealth and income. Additionally, we report results on macroeconomic aggregates, volatility, and the distributive impacts on different agents.

The results demonstrate that granting all households direct access to the policy interest rate fundamentally alters wealth accumulation trajectories, leading to a reduction in both income and wealth inequality. Specifically, the policy particularly benefits the financially constrained H-agents, who vastly increase their wealth through higher compounding returns thanks to the higher yield. Calibrated to be as patient as the S-agents, this suggests that their baseline lack of savings stems from low yields rather than impatience. M-agents benefit similarly, while also taking advantage of lower house prices triggered by the S-agents’ portfolio downsizing. Indeed, the cost of this redistribution is sustained by the wealthy: RFA provokes severe bank disintermediation, which shrinks bank profits thus decreasing dividend payments. This forces S-agents to reduce consumption and downsize their portfolios. Ultimately, the income Gini lowers by 0.68 pp and the wealth Gini by 3.19 pp.

From a welfare perspective, the policy generates an aggregate forward-looking CEV (FL-CEV) gain of 1.16% when comparing steady states. These gains are not uniformly distributed: consistent with the underlying redistribution, H and M agents experience welfare gains, while S-agents suffer a large welfare loss. Furthermore, these steady-state gains do not take into account the perfect foresight simulations of the transition path, found to be costly with all agents facing welfare losses during the onset of the policy. The mechanism behind this is an economic contraction caused by a negative demand shock as agents increase savings to take advantage of the RFA interest rate, similar to a contractionary monetary policy shock. However, when taking into account the full transition, the aggregate backward-looking CEV (BL-CEV) reaches 0.46% indicating that the policy is still beneficial. Investigating a way to reduce the disintermediation effect, I propose an extended model featuring

tiered remuneration, where the RFA interest rate is penalised if RFA holdings exceed a certain calibrated threshold. In this extension, a trade-off appears in reducing disintermediation without overly constraining the wealth accumulation potential of RFA. A penalty calibrated to only affect the wealthy’s extreme RFA holdings leads to slightly increased welfare gains, while a penalty binding all agents reduces welfare gains.

Analysis of the stochastic IRFs shows that aggregate responses to a contractionary monetary policy shock change little with RFA. However, the transmission mechanism is modified, as H-agents are directly exposed to the interest rate hike. We see their consumption fall by less, supplemented by the higher rate which they previously did not benefit from. Evaluating volatility through the simulated standard deviations suggests a small macroeconomic stabilisation effect, with RFA reducing the volatility of interest rates, inflation, aggregate consumption, and output.

The remainder of this paper is organised as follows. Section 2 reviews the existing literature on the distributional impacts of monetary policy and retail CBDC. Section 3 details the THRANK model’s equations and calibration, outlining the behaviours of households, firms, and the financial sector. Section 4 presents the quantitative results, comparing steady states, analysing transition dynamics, and evaluating stochastic shocks. Section 5 discusses the limitations of this study. Section 6 concludes.

2 Literature Review

The use of heterogeneous-agent frameworks in studying the distributional impacts of monetary policies, including RFA, is highly important. BILBIIE (2008) splits agents into two groups by segmenting financial markets, finding that business cycle dynamics fundamentally change (going as far as inverting the IS curve’s slope) when a share of the population does not receive dividends and cannot participate in intertemporal substitution. KAPLAN et al. (2018) expand on this with HANK² models, showing that monetary policy shocks transmit to consumption primarily through general equilibrium effects in the labour market (the indirect channel) rather than through intertemporal substitution (the direct channel). BROER et al. (2020) separate workers and capitalists to isolate the labour and profit channels, highlighting the crucial role of wage stickiness in driving the effects of monetary policy. Collectively, this literature establishes that segmented markets matter in understanding macroeconomic shocks and policies and their distributional consequences. This paper opts for a three-agent model as a compromise between model complexity and tractability.

As the introduction of RFA alters available interest rates, some of its transmission channels closely mirror those of conventional monetary policy. AUCLERT (2019) identifies redistribution itself as one of these channels, amplifying transmission through heterogeneous marginal propensities to consume. Specifically, he outlines three mechanisms that affect aggregate spending: unequal distribution of gains from labour and profit income (the earnings heterogeneity channel), unexpected inflation revaluing

²While BILBIIE (2008) describes a Two-Agent New Keynesian (TANK) model, KAPLAN et al. (2018) expand to an infinite continuum of agents known as Heterogeneous-Agent New Keynesian (HANK) models.

nominal wealth between creditors and debtors (the Fisher channel), and differing asset exposures to real interest rates (the interest rate exposure channel). KOMATSU (2023) builds on this by decomposing the earnings heterogeneity channel into an income composition effect and an unemployment friction effect, concluding that expansionary monetary policy generally decreases consumption inequality. ESKELINEN (2021) captures these mechanisms in a more tractable three-agent framework and introduces a housing wealth channel amplified by collateral constraints, following the seminal work of IACOVIELLO (2005). Focusing specifically on this housing channel, HEDLUND et al. (2017) find that its redistributive consequences play a significant role in monetary policy transmission.

Empirical evidence confirms these redistributive mechanisms, though it highlights that the net effect on inequality depends on which channels dominate in a given environment. COIBION et al. (2017) uncover evidence for both labour-driven and financial-market-driven mechanisms, including earnings heterogeneity, portfolio composition, and financial segmentation. Looking at specific distributional outcomes, HERRADI & LEROY (2021) show that expansionary policy reduces the income share of the top one percent by lowering real asset returns, a finding consistent with the income composition channel. In contrast, ANDERSEN et al. (2023) demonstrate that interest rate cuts increase income inequality, with “non-labour” channels such as the portfolio channel as the main drivers. Because the literature indicates that monetary policy does have an effect on inequality, the interaction of these heterogeneous channels will shape the impact of RFA on inequality.

Apart from the interest rate channel, RFA fundamentally alters monetary architecture with the potential to cause bank disintermediation. The literature focusing on the question of disintermediation has found mixed results. Using a general equilibrium model featuring a monopoly bank, ANDOLFATTO (2021) concludes that CBDC need not have a detrimental effect on bank funding, even possibly increasing it. The mechanism consists of a rise in competitive pressure in the deposit market, leading to higher commercial bank deposit rates which lower profits but increase deposit quantities. CHIU et al. (2023) find similar results but in a model with Cournot-style imperfectly competitive banks where bank rates are forced to match CBDC rates. However, the mechanism reverses (i.e. CBDC causing disintermediation) if the CBDC rate exceeds a certain threshold. In a portfolio choice model where banks have market power, CHANG et al. (2023) again observe the same mechanism without disintermediation. In their case, bank disintermediation only occurs if accessing CBDC is less costly than regular deposits. In a DSGE model featuring bank runs, BIDDER et al. (2025) distinguish between “slow disintermediation” (a long-term decline in bank lending and funding) and “fast disintermediation” (sudden bank runs in the short-term). They argue that while a CBDC induces slow disintermediation, it can also improve financial stability by reducing the consequences of fast disintermediation, with the net effect on stability depending heavily on CBDC design. This sensitivity to design is further highlighted by NYFFENEGGER (2024), whose general equilibrium model shows that disintermediation is almost twice as severe when a CBDC serves primarily as a savings vehicle rather than strictly as a medium of exchange. This paper distinguishes itself from these by not focusing on modelling disintermediation, but rather on the effects to inequality and household welfare.

Despite the risk of disintermediation, the literature generally points to aggregate welfare gains from introducing a retail CBDC. KEISTER & SANCHES (2023), separating CBDC into cash-like or deposit-like, show that disintermediation implied by CBDCs would reduce bank loans, though often find welfare increasing despite it. In a DSGE model with monopolistic banks, PAUL et al. (2026) illustrate that disintermediation occurs since bank lending decreases due to the increase in competition from CBDC. This lowers bank profits, forcing them to raise deposit rates. Despite this, aggregate welfare increases (under the condition the CBDC rate remains below the policy rate) as households benefit from the higher rates and receive expanded liquidity services. When calibrated to the EU, welfare gains are smaller as the economy is more reliant on bank lending. BURLON et al. (2022) develop a DSGE model distinguishing between borrowing and saving agents to show that overall welfare improves despite bank disintermediation. They further demonstrate that welfare-optimising policy rules, such as adjustable interest rates or quantity limits, lead to additional welfare gains while mitigating disintermediation. In a two-country DSGE model, ASSENMACHER et al. (2024) find that CBDC unambiguously improves welfare without leading to disintermediation. Compared to this literature, the model present in this paper additionally decomposes welfare impacts across the wealth distribution.

While the broader literature primarily focuses on aggregate welfare metrics, a significant gap remains regarding distributional welfare and wealth inequality impacts of a remunerated retail CBDC. Splitting the population on anonymity-security preferences, AGUR et al. (2022) find that depositors emerge as winners and cash-holders as the losers, with aggregate welfare increasing. While endowments are equal in their analysis, they explain how, in practice, cash-holders are typically lower income households while higher income households more often rely on deposit payments, thus suggesting that a retail CBDC might be increasing inequality. Distinguishing households and bank owners, BACCHETTA & PERAZZI (2025) find small welfare gains mainly through the redistribution away from bank owners, without disintermediation affecting bank lending. They note that if bank owners are richer than the general population, CBDC could lessen inequality. However, by limiting heterogeneity to a binary division between bank-owners and standard working households, their framework abstracts from modelling the diverse portfolio compositions and segmented financial markets required to study inequality in greater detail. Separating agents into banked and unbanked households, MURAKAMI et al. (2024) find that CBDC reduces consumption inequality and increases welfare, but this effect is strictly confined to economies with low baseline financial inclusion. In economies with full financial inclusion (no unbanked), the model collapses into a standard representative-agent framework where welfare gains and redistributive effects vanish. While not being their central focus, YANG et al. (2024) analyse the impact of remunerated CBDC on wealth inequality using a HANK model, finding that it increases inequality. However, their framework focuses on the impact of CBDC on cash and financial intermediaries, which is reflected in their design choices. The model features three assets (cash, deposits, and CBDC), with adjustment costs of switching between them. Inequality increases through unequal income gains sustained by these adjustment costs. When the CBDC rate rises, wealthier households receive direct income gains on their large CBDC holdings, while poorer households see a much smaller gain as switching to CBDC is costly due to the adjustment costs.

This paper addresses a critical gap in the literature by placing the distributive consequences of a remunerated retail CBDC, dubbed RFA, at the centre of the analysis. In contrast to YANG et al. (2024), whose focus is on the impact on cash and financial intermediaries, this model is built paying specific attention to the composition of the wealth distribution and the transmission channels affecting it. Households are split into three with heterogeneous access to assets, in such a way to replicate the Swiss income and wealth distributions. This split additionally allows to study the difference in welfare changes across the lower, middle, and upper wealth classes.

3 Model

This section builds upon the baseline New-Keynesian model described by SIMS (2017) in several ways so as to feature both income and wealth inequality, and the transmission channels discussed in the previous section. These channels are paramount in the analysis of a change in monetary architecture, the introduction of RFA. The full list of variable and parameter definitions is provided in appendix A.1, and all equilibrium equations in appendix A.2. For detailed derivations of the model's equations, see appendix B.

The first major extension is the inclusion of three heterogeneous, infinitely-lived agents, in an attempt to replicate, in broad strokes, segments of the Swiss wealth-income distribution found in MARTÍNEZ (2021) and calibrated to target the reported Gini coefficients. These agents are separated not only by their position within the distribution but also by the composition of their income and wealth. These include labour wages, deposits, housing, mortgage loans, bank bonds, physical capital, and dividends. In order to have heterogeneous, sticky wages for more realistic labour market responses to the shock, workers supply differentiated labour services through unions, which set wages subject to quadratic adjustment costs. All agents have money in the utility function, which combines liquidity services provided by regular bank deposits and RFA deposits, both of which pay interest.

The second major extension is the presence of financial intermediaries, which endogenously set rate spreads for the different financial assets. This further contributes to creating inequality and is necessary to include the potential effects of disintermediation. These intermediaries are rate-setting commercial banks adapted from GERALI et al. (2010), which face quadratic adjustment costs. They are funded by cheap deposits, higher interest bank bonds, and unlimited, expensive lending from the central bank. They are constrained by a reserve requirement on deposits, and offer high-interest mortgage loans.

Using language from the CBDC literature, RFA could be considered as a universal, remunerated, retail CBDC. Money enters the utility function, and consists of a CES aggregator combining regular deposits and RFA. The experiment is designed in such a way to have both deposit types be identical in every way apart from the interest rate. As such, this framework does not consider RFA to offer any improvements in payment services, liquidity services, transaction costs, or other similar features. This serves to purely assess the benefits of RFA as an investment vehicle. However, they are imperfectly substitutable, such that agents must hold some amount of the two. Because cash is not modelled and the central bank's only asset are bank loans, RFA

mechanically expands the central bank balance sheet if its quantity in the economy exceeds existing bank reserves. Finally, the policy experiment consists of shocking the CES preference weight parameter (ϖ) for deposits going from (approximately) one to one-half. This represents equal weights for deposits and RFA such that, if they provided the same interest rate, agents would hold an equal share of each.

The rest of the model includes standard firms with sticky prices (CALVO 1983), with the addition of capital producers facing investment adjustment costs (GERTLER & KARADI 2011). Finally, the model features a central bank which sets the policy rate according to a Taylor Rule (TAYLOR 1993). It also provides unlimited funding to banks, albeit at a cost above the policy rate. I omit transfer and tax schemes in order to isolate the redistributive effects of RFA to only those stemming from market mechanisms. For the same reason I do not rebate central bank profits to households, thus removing the seigniorage welfare channel present in BACCHETTA & PERAZZI (2025). There is no government.

3.1 Households

The economy consists of three types of households, differentiated by their activities and access to financial markets and asset ownership: Hand-to-Mouth (H) agents, Middle-Class (M) agents, and Saver (S) agents with population shares:

$$\lambda_H + \lambda_M + \lambda_S = 1 \quad (1)$$

This heterogeneity allows to replicate in broad strokes the income and wealth composition of the different income-wealth classes of Switzerland based on the findings of MARTÍNEZ (2021). For income, MARTÍNEZ (2021) finds that it is largely composed of labour income for the majority of the population, gaining particular importance at the bottom of the distribution. However, she mentions that the composition changes considerably within the top 10%, where the share of labour declines strongly and capital income reigns dominant. The bottom of the distribution's other main source of income is transfers, though this is mainly from social security pensions. For wealth, MARTÍNEZ (2021) finds that the bottom of the distribution holds practically none of it. The little they have is not in real estate, but in financial assets, which include "securities, credit balances, cash on bank deposits, [and] gold." Explicitly excluded are "personal retirement savings accounts from the 3rd pillar [...] [and] wealth held within the occupational pension system (2nd pillar)." I make the assumption that the bottom only holds bank deposits. The middle-class' wealth is mainly tied to their real estate, although this is highly leveraged by mortgage debt. This is the only type of debt included in this model, as around 90% of all Swiss private debt is mortgage debt (MARTÍNEZ 2021). Another unique characteristic of the Swiss housing market is the low rate of homeownership, which she finds to be only 36% in 2010. Hence, the middle-class in this model is heavily indebted, whereas the upper class owns real estate outright, without debt, using it as a relatively safe investment vehicle for diversifying their portfolios (MARTÍNEZ 2021). The rest of the upper class' wealth is composed of financial assets such as securities.

Based on this, I split the agents as $\lambda_H = 0.50$, $\lambda_M = 0.40$, and $\lambda_S = 0.10$. I equate the positions of the agents in the wealth and income distributions as MARTÍNEZ (2021)

finds that the correlation of said position is heavily tail-dependent. This essentially means that the poorest (richest) in income are often the poorest (richest) in wealth. I make the simplifying assumption that this also holds for the middle class. The H-agents represent the bottom of the distributions, deriving income almost solely from labour, and owning no wealth apart from a small amount in bank deposits. While their ability to save through deposits means they are not strictly “hand-to-mouth”, I retain this terminology as these agents are the most financially constrained and thus resemble the hand-to-mouth agents of other heterogeneous-agent models (ESKELINEN 2021). The M-agents earn a higher wage, and can invest in housing as well as deposits. Following the representative agents of IACOVIELLO (2005), they can also borrow albeit subject to a collateral constraint. The S-agents receive capital income through real estate gains, interest on deposits and bonds, renting capital to firms, and dividends. The fact that these agents own all the capital, firms, and do not provide any labour is typical in capitalist-worker models. This is done to remove the unrealistic worker-profit transmission mechanism, where the transmission of demand shocks hinges on income effects on labour supply induced by countercyclical profits (BROER et al. 2020; CANTORE & FREUND 2021).

Each agent maximises their discounted expected utility function, which depends positively on consumption, real deposits, and housing, and negatively on labour, subject to their respective budget constraints:

$$V_{i,t} = \mathbb{E}_0 \sum_{t=0}^{+\infty} \beta_i^t U \left(C_{i,t}, \frac{\widetilde{M}_{i,t}}{P_t}, H_{i,t}, L_{i,t} \right), \quad i \in \{H, M, S\} \quad (2)$$

where β_i is the discount factor for agent i , $C_{i,t}$ is real consumption, $\widetilde{M}_{i,t}/P_t$ is the real money bundle (with P_t denoting the aggregate price level), $H_{i,t}$ is the real housing stock, and $L_{i,t}$ denotes hours worked.

The constant relative risk aversion utility function is separable in each variable:

$$U(\cdot) = \frac{C_{i,t}^{1-\sigma}}{1-\sigma} + \Psi \frac{(\widetilde{M}_{i,t}/P_t)^{1-\psi}}{1-\psi} + \Theta \frac{H_{i,t}^{1-\eta}}{1-\eta} - \chi_i \frac{L_{i,t}^{1+\varphi}}{1+\varphi} \quad (3)$$

where σ is the coefficient of relative risk aversion (inverse elasticity of intertemporal substitution), Ψ and ψ are the weight and inverse elasticity of the real money bundle, Θ and η are the weight and inverse elasticity of housing utility, and χ_i and φ govern the agent-specific weight of labour disutility and the inverse Frisch elasticity of labour supply, respectively. The housing term is omitted for H-agents, while the labour term is omitted for S-agents.

Money enters the utility function as a CES bundle composed of both regular bank deposits and RFA, featuring imperfect substitutability as in BACCHETTA & PERAZZI (2025):

$$\widetilde{M}_{i,t} = \left[\varpi^{\frac{1}{\epsilon_c}} (M_{i,t})^{\frac{\epsilon_c-1}{\epsilon_c}} + (1-\varpi)^{\frac{1}{\epsilon_c}} (M_{i,t}^c)^{\frac{\epsilon_c-1}{\epsilon_c}} \right]^{\frac{\epsilon_c}{\epsilon_c-1}} \quad (4)$$

where $M_{i,t}$ denotes nominal regular bank deposits, $M_{i,t}^c$ represents nominal RFA deposits, and $\epsilon_c > 0$ is the constant elasticity of substitution between the two liquid assets.

Because the two deposit types are identical in every aspect except their interest rates, imperfect substitutability is necessary to prevent corner solutions where agents fully switch to the higher-yielding asset. Economically, this imperfect substitution captures unmodelled differences between the two assets: for example, regular deposits may offer payment services, whereas RFA provides safety. Unlike some of the existing CBDC literature (ASSENMACHER et al. 2024; BARRDEAR & KUMHOF 2022), this model deliberately abstracts from improvements in payment efficiency, liquidity services, or reductions in transaction costs in order to isolate the role of RFA as a universal investment vehicle. The policy experiment is modelled as a structural shock to the preference weight, shifting from $\varpi \approx 1$ (where agents hold effectively zero RFA) to a symmetric steady state of $\varpi = 0.5$. This baseline value of 0.5 ensures that the two deposit types are treated as equivalent in the utility function, guaranteeing that any welfare changes stem entirely from financial returns rather than mechanical changes in preference (BACCHETTA & PERAZZI 2025). Indeed, assuming equal interest rates, the utility derived from the ex-ante bundle ($\varpi = 1, M_{i,t} = 5, M_{i,t}^c = 0$) is identical to that of the ex-post symmetric bundle ($\varpi = 0.5, M_{i,t} = 2.5, M_{i,t}^c = 2.5$). This cleanly isolates the gains attributable to the introduction of RFA in scenarios where rates differ.

For the agents that supply labour, all labour decisions are outsourced to unions which charge them a fee to cover the wage adjustment costs. Wage heterogeneity is implemented through the share (α_l) of each agent's labour type in the composition of aggregate labour.

3.1.1 Hand-to-Mouth

Hand-to-Mouth (H) agents represent the fraction of the population that earns low wages and is financially constrained. They cannot borrow and their sole available investment vehicle are deposits, which provide a low interest rate. Once RFA is introduced, these agents will be able to hold RFA deposits, functionally identical to regular deposits albeit being remunerated directly at the policy rate. These agents are calibrated to be patient in order for their difficulty in building wealth to not be driven by impatient behaviour.

Their choices must obey the following budget constraint, in nominal terms:

$$P_t C_{H,t} + M_{H,t} + M_{H,t}^c + P_t T_{H,t} \leq W_{H,t} L_{H,t} + (1 + i_{t-1}^d) M_{H,t-1} + (1 + i_{t-1}) M_{H,t-1}^c \quad (5)$$

They earn a nominal wage $W_{H,t}$ for their labour $L_{H,t}$, pay a labour union membership fee $T_{H,t}$, and receive gross interest on their previous period's deposits at rate i_{t-1}^d and on RFA at the policy rate i_{t-1} .

Solving their maximisation yields Euler equations for deposits and RFA. These are applicable to each agent group:

$$\frac{C_{H,t}^{-\sigma}}{P_t} = \Psi \frac{\widetilde{M}_{H,t}^{-\psi}}{P_t^{1-\psi}} \varpi^{\frac{1}{\epsilon_c}} \left(\frac{\widetilde{M}_{H,t}}{M_{H,t}} \right)^{\frac{1}{\epsilon_c}} + \beta_H \mathbb{E}_t \left[(1 + i_t^d) \frac{C_{H,t+1}^{-\sigma}}{P_{t+1}} \right] \quad (6)$$

$$\frac{C_{H,t}^{-\sigma}}{P_t} = \Psi \frac{\widetilde{M}_{H,t}^{-\psi}}{P_t^{1-\psi}} (1 - \varpi)^{\frac{1}{\epsilon_c}} \left(\frac{\widetilde{M}_{H,t}}{M_{H,t}^c} \right)^{\frac{1}{\epsilon_c}} + \beta_H \mathbb{E}_t \left[(1 + i_t) \frac{C_{H,t+1}^{-\sigma}}{P_{t+1}} \right] \quad (7)$$

3.1.2 Middle-Class

Middle-Class (M) agents represent the working middle class. Unlike the Hand-to-Mouth agents, they participate in the real estate market by owning housing, which provides utility and serves as collateral to access mortgage credit. They consume, hold deposits and RFA, supply differentiated labour to unions, and borrow from banks.

Their choices are constrained by the following nominal budget constraint. In addition to wages and deposits, M-agents purchase housing at the nominal price S_t and take on nominal mortgage debt $N_{M,t}$ at the mortgage interest rate i_t^n :

$$\begin{aligned} P_t C_{M,t} + S_t H_{M,t} + M_{M,t} + M_{M,t}^c + (1 + i_{t-1}^n) N_{M,t-1} + P_t T_{M,t} \leq \\ W_{M,t} L_{M,t} + S_t H_{M,t-1} + N_{M,t} + (1 + i_{t-1}^d) M_{M,t-1} + (1 + i_{t-1}) M_{M,t-1}^c \end{aligned} \quad (8)$$

M-agents are relatively impatient compared to other agents, as they represent the economy's borrowers. Their ability to borrow is limited by a collateral constraint tied to a fraction Γ (the loan-to-value ratio) of the expected discounted future value of their housing stock (IACOVIELLO 2005):

$$N_{M,t} \leq \Gamma \mathbb{E}_t \left[\frac{S_{t+1} H_{M,t}}{1 + i_t^n} \right] \quad (9)$$

Maximising their utility subject to the budget and borrowing constraints yields the optimal demand for housing and debt, represented by the following Euler equations:

$$S_t \frac{C_{M,t}^{-\sigma}}{P_t} = \Theta H_{M,t}^{-\eta} + \beta_M \mathbb{E}_t \left[S_{t+1} \frac{C_{M,t+1}^{-\sigma}}{P_{t+1}} \right] + \Gamma \Omega_{M,t} \mathbb{E}_t \left[\frac{S_{t+1}}{1 + i_t^n} \right] \quad (10)$$

$$\frac{C_{M,t}^{-\sigma}}{P_t} = \Omega_{M,t} + \beta_M \mathbb{E}_t \left[(1 + i_t^n) \frac{C_{M,t+1}^{-\sigma}}{P_{t+1}} \right] \quad (11)$$

where $\Omega_{M,t}$ denotes the Lagrange multiplier on the collateral constraint (9), expressed in terms of marginal utility.

Finally, because M-agents allocate their liquidity between deposits and RFA under the same CES aggregator as H-agents, their optimal deposit allocations yield identical deposit Euler equations to those defined in (6).

3.1.3 Savers

Saver (S) agents represent the wealthy capitalists of the economy. These agents are totally unconstrained, own the physical capital stock, invest in bonds, buy housing, and receive all dividend profits. They do not supply labour. Their discount factor indirectly pins down the steady-state interest rates in the economy.

Their nominal budget constraint is given by:

$$\begin{aligned} P_t C_{S,t} + S_t H_{S,t} + Q_t I_{S,t} + B_{S,t} + M_{S,t} + M_{S,t}^c \leq \\ R_t K_{S,t-1} + D_{S,t} + (1 + i_{t-1}^b) B_{S,t-1} + S_t H_{S,t-1} + (1 + i_{t-1}^d) M_{S,t-1} + (1 + i_{t-1}) M_{S,t-1}^c \end{aligned} \quad (12)$$

S-agents purchase new physical capital $I_{S,t}$ at the nominal price Q_t and buy bank bonds $B_{S,t}$ paying interest i_t^b . Their income stems from renting out their capital stock $K_{S,t-1}$ at the nominal rental rate R_t , alongside interest income and dividend profits $D_{S,t}$ from ownership of the firms.

Physical capital evolves according to a standard accumulation process with depreciation rate δ :

$$K_{S,t} = (1 - \delta)K_{S,t-1} + I_{S,t} \quad (13)$$

Because S-agents face no borrowing constraints, their asset pricing equations reflect standard, unconstrained intertemporal optimisation. The Euler equations for housing, capital, and bank bonds are, respectively:

$$S_t \frac{C_{S,t}^{-\sigma}}{P_t} = \Theta H_{S,t}^{-\eta} + \beta_S \mathbb{E}_t \left[S_{t+1} \frac{C_{S,t+1}^{-\sigma}}{P_{t+1}} \right] \quad (14)$$

$$Q_t \frac{C_{S,t}^{-\sigma}}{P_t} = \beta_S \mathbb{E}_t \left[\frac{C_{S,t+1}^{-\sigma}}{P_{t+1}} (R_{t+1} + (1 - \delta)Q_{t+1}) \right] \quad (15)$$

$$\frac{C_{S,t}^{-\sigma}}{P_t} = \beta_S \mathbb{E}_t \left[(1 + i_t^b) \frac{C_{S,t+1}^{-\sigma}}{P_{t+1}} \right] \quad (16)$$

As with the M-agents, the S-agent deposit Euler equations mirror those of the H-agents, adjusting only for the S-agent specific consumption and discount factor.

3.1.4 Labour Market

The labour market is built quasi-identically as in GERALI et al. (2010), where the patient and impatient agents forming aggregate labour are replaced by the H and M agents in this model. Each worker provides a unique labour type, sold by unions to perfectly competitive labour packers who assemble them in a CES aggregator governed by the elasticity of substitution ϵ_w :

$$L_{i,t} = \left[\int_0^1 L_{i,t}(j)^{\frac{\epsilon_w-1}{\epsilon_w}} dj \right]^{\frac{\epsilon_w}{\epsilon_w-1}} \quad (17)$$

and sell the homogeneous labour to intermediate goods firms. This is essentially the same Dixit-Stiglitz framework for monopolistic competition used in the goods market (DIXIT & STIGLITZ 1977); labour packers can be seen as the “final labour producers” while unions are the “intermediate labour producers”. For each labour type $j \in [0, 1]$ there are two unions, one for the H-agents and one for the M-agents (indexed by i). Each union (i, j) sets nominal wages $W_{i,t}$ for its members by maximising their utility subject to the demand function given by labour packers:

$$L_{i,t}(j) = \left(\frac{W_{i,t}(j)}{W_{i,t}} \right)^{-\epsilon_w} L_{i,t} \quad (18)$$

To introduce nominal rigidities in wages, wages are subject to quadratic adjustment costs scaled by the parameter κ^w . The unions charge each member household lump-sum fees to cover these costs. Letting $\pi_{i,t}^w(j) \equiv \frac{W_{i,t}(j)}{W_{i,t-1}(j)} - 1$ denote individual wage

inflation, the unions maximise the following function:

$$\max_{\{W_{i,t}(j)\}} \mathbb{E}_0 \sum_{t=0}^{+\infty} \beta_i^t \left[\Lambda_{i,t} \left(W_{i,t}(j) L_{i,t}(j) - \frac{\kappa^w}{2} (\pi_{i,t}^w(j))^2 W_{i,t} \right) - \chi_i \frac{L_{i,t}^{1+\varphi}(j)}{1+\varphi} \right] \quad (19)$$

where Λ_i is the Lagrange multiplier to the budget constraint, shown in appendix B.1.

Taking the FOC and imposing a symmetric equilibrium defines the Wage Phillips Curve for households $i \in [H, M]$:

$$\begin{aligned} \kappa^w \pi_{i,t}^w (1 + \pi_{i,t}^w) &= (1 - \epsilon_w) L_{i,t} + \epsilon_w \frac{\chi_i L_{i,t}^{1+\varphi}}{W_{i,t} \Lambda_{i,t}} \\ &+ \beta_i \mathbb{E}_t \left[\frac{\Lambda_{i,t+1}}{\Lambda_{i,t}} \kappa^w \pi_{i,t+1}^w (1 + \pi_{i,t+1}^w)^2 \right] \end{aligned} \quad (20)$$

3.2 Firms

Firms are split into three types, following standard practices in New-Keynesian models. Following the Dixit-Stiglitz framework, we have perfectly competitive final goods producers who aggregate intermediate goods and sell them to households. These buy from monopolistically competitive intermediate goods producers that set prices subject to a nominal rigidity *à la* CALVO (1983). Finally, we have perfectly competitive capital producers which build new physical capital and are subject to quadratic investment adjustment costs (GERTLER & KARADI 2011). All firms maximise their discounted infinite-horizon profits. Since they are fully owned by the S-agents, the stochastic discount factor:

$$\mathcal{M}_{t,t+k} = \beta_S^k \frac{\Lambda_{S,t+k}}{\Lambda_{S,t}} = \beta_S^k \frac{C_{S,t+k}^{-\sigma} / P_{t+k}}{C_{S,t}^{-\sigma} / P_t} \quad (21)$$

is defined by their marginal utility.

3.2.1 Final Goods Producers

Final goods producers operate in perfect competition to aggregate differentiated intermediate goods $Y_t(j)$ into a homogeneous final good Y_t . Their production technology is represented by a standard Constant Elasticity of Substitution (CES) aggregator:

$$Y_t = \left[\int_0^1 Y_t(j)^{\frac{\epsilon_y-1}{\epsilon_y}} dj \right]^{\frac{\epsilon_y}{\epsilon_y-1}} \quad (22)$$

where $\epsilon_y > 1$ is the elasticity of substitution between differentiated intermediate goods.

The final goods producers are identical and choose $Y_t(j)$ to maximise their profits, defined as the difference between the revenue from selling the final aggregate good and the cost of purchasing the intermediate inputs, which results in the following demand function for intermediate goods:

$$Y_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\epsilon_y} Y_t \quad (23)$$

Since there is perfect competition in the production of final goods, no profits are made.

3.2.2 Intermediate Goods Producers

There is a continuum of intermediate goods producers $j \in [0, 1]$ that operate in a monopolistically competitive market to produce $Y_t(j)$ according to a constant-returns-to-scale Cobb-Douglas production function combining labour and capital, where Z_t is aggregate technology and α is the capital share:

$$Y_t(j) = Z_t K_{t-1}(j)^\alpha L_t(j)^{1-\alpha} \quad (24)$$

Aggregate labour is a composite combining both union groups' offered work hours using a Cobb-Douglas technology, with α_l governing the share of H-agent labour:

$$L_t(j) = [\lambda_H L_{H,t}(j)]^{\alpha_l} [\lambda_M L_{M,t}(j)]^{1-\alpha_l} \quad (25)$$

Technology follows an AR(1) process:

$$\ln Z_t = \rho_Z \ln Z_{t-1} + \varepsilon_{z,t} \quad (26)$$

$$\text{where } 0 \leq \rho_Z \leq 1 \quad \& \quad \varepsilon_{z,t} \stackrel{\text{iid}}{\sim} N(0, \sigma_z^2)$$

These firms' behaviour can be broken down into three distinct parts. The first is a static problem in which they determine the optimal combination of labour types by maximising total labour worth ($W_t L_t$) minus the total wage costs. The second is a static problem in which they determine the optimal input ratio between aggregate labour and capital by minimising total costs subject to the production function. The third is an intertemporal problem in which firms set prices to maximise the expected discounted sum of future profits though, facing CALVO (1983) price-setting, only a proportion θ are able to reset the price in each period:

$$\max_{\{P'_t\}} \mathbb{E}_t \sum_{k=0}^{+\infty} \theta^k \mathcal{M}_{t,t+k} D_{t+k}^i(P'_t) \quad (27)$$

where future profits can be defined as a function of the new price:

$$D_{t+k}^i(P'_t) = P'_t Y_{t+k}(j) - W_{t+k} L_{t+k}(j) - R_{t+k} K_{t-1+k}(j) \quad (28)$$

The aggregate price level then evolves according to:

$$P_t = \left[(1 - \theta) P_t^{1-\epsilon_y} + \theta P_{t-1}^{1-\epsilon_y} \right]^{\frac{1}{1-\epsilon_y}} \quad (29)$$

As these three parts are relatively standard in New-Keynesian models, I relegate the derivations and the resulting equilibrium conditions to appendix B.3.2. For the intertemporal part, I follow the approach delineated in SIMS (2017) where the infinite-horizon problem is defined recursively.

3.2.3 Capital Producers

Capital producers operate within perfect competition to build new physical capital which they sell to the S-agents at the nominal price, Q_t . They face quadratic investment adjustment costs (κ^y) adapted from GERTLER & KARADI (2011). As with the other firms, they maximise their expected discounted sum of future profits:

$$\max_{\{I_t\}} \mathbb{E}_t \sum_{k=0}^{+\infty} \mathcal{M}_{t,t+k} D_{t+k}^k \quad (30)$$

where:

$$D_t^k = Q_t I_t - P_t I_t - P_t \frac{\kappa^y}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 I_t \quad (31)$$

The first order condition for investment yields:

$$\frac{Q_t}{P_t} = 1 + \kappa^y \left(\frac{I_t}{I_{t-1}} - 1 \right) \frac{I_t}{I_{t-1}} + \frac{\kappa^y}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 \quad (32)$$

$$- \mathbb{E}_t \left[\beta_S \frac{C_{S,t+1}^{-\sigma}}{C_{S,t}^{-\sigma}} \kappa^y \left(\frac{I_{t+1}}{I_t} - 1 \right) \left(\frac{I_{t+1}}{I_t} \right)^2 \right] \quad (33)$$

While perfect competition leads to zero profits in the steady state ($Q_t = P_t$), capital producers may generate temporary profits during transition dynamics when investment expands, as the adjustment costs drive the marginal costs above the average costs of production.

3.3 Financial Sector

The financial sector includes financial intermediaries and the central bank. Financial intermediaries are split between financial packers who offer composite products to households, assembled from the mortgage loans, deposits, and bonds managed by differentiated banks which models market power with a Dixit-Stiglitz framework. This approach for implementing financial intermediaries is adapted from GERALI et al. (2010), modified to fit the agents and assets of this model. Bank equity is omitted to simplify the model.

3.3.1 Financial Packers

To introduce interest rate stickiness without complicating the household's optimisation problem, perfectly competitive financial packers act as intermediaries that assemble and distribute financial composites. Their objective functions differ based on the direction of the financial flows. Loan packers purchase differentiated mortgage loans from banks at rate $i_t^n(j)$ and assemble them into a composite CES bundle to sell to households at the aggregate rate i_t^n . They maximise profit defined as the aggregate interest income received from households minus the interest paid to banks, $i_t^n N_t - \int_0^1 i_t^n(j) N_t(j) dj$.

Conversely, deposit and bond packers collect composite deposits and bonds from households, paying aggregate rates i_t^d and i_t^b , respectively, and distribute them to

individual banks to earn rates $i_t^d(j)$ and $i_t^b(j)$. Because these instruments represent household assets and bank liabilities, the packers' profit maximisation problem is inverted to reflect maximising the return passed back to households. For instance, deposit packers maximise the return from banks minus the aggregate interest paid to households, $\int_0^1 i_t^d(j) M_t(j) dj - i_t^d M_t$.

These yield the following demand functions for loans, deposits, and bonds:

$$N_t(j) = \left(\frac{i_t^n(j)}{i_t^n} \right)^{-\epsilon^n} N_t \quad M_t(j) = \left(\frac{i_t^d(j)}{i_t^d} \right)^{-\epsilon^d} M_t \quad B_t(j) = \left(\frac{i_t^b(j)}{i_t^b} \right)^{-\epsilon^b} B_t \quad (34)$$

where the elasticities of substitution ($\epsilon^n > 1$, $\epsilon^d < 0$, and $\epsilon^b < -1$) imply a downward-sloping demand curve for loans and upward-sloping supply curves for deposits and bonds.

3.3.2 Commercial Banks

As in GERALI et al. (2010), banks enjoy market power in intermediation, enabling spreads on interest rates relative to the policy rate. This market power is necessary to study the interest rate channel and the potential bank disintermediation effects of introducing RFA. Unlike GERALI et al. (2010), banks in this model do not hold equity capital. Abstracting from the bank equity channel simplifies the balance sheet and the already dense model.

A continuum of differentiated banks $j \in [0, 1]$ hold reserves at the central bank ($M_{B,t}^c$), issue mortgage loans (N_t), and fund themselves through remunerated deposits (M_t), issued bonds (B_t), and central bank lending (F_t). The balance sheet for bank j is defined as:

$$M_{B,t}^c(j) + N_t(j) = F_t(j) + M_t(j) + B_t(j) \quad (35)$$

Banks are subject to a reserve requirement on deposits, where ω is the reserve ratio:

$$M_{B,t}^c(j) = \omega M_t(j) \quad (36)$$

Banks choose their respective interest rates to maximise the expected discounted sum of future profits:

$$\max_{\{i_t^n(j), i_t^d(j), i_t^b(j)\}} \mathbb{E}_t \sum_{k=0}^{+\infty} \mathcal{M}_{t,t+k} D_{t+k}^b(j) \quad (37)$$

subject to the demand functions in (34), the balance sheet (35), and the reserve requirement (36), and where the profits are defined by the interest income from assets minus the interest outflow from liabilities. We introduce sluggish interest rate pass-through by making banks face quadratic adjustment costs when changing rates. These costs are subtracted from the per-period profit function:

$$D_t^b(j) = i_{t-1} M_{B,t-1}^c(j) + i_{t-1}^n N_{t-1}(j) - i_{t-1}^d(j) M_{t-1}(j) - i_{t-1}^b(j) B_{t-1}(j) - i_{t-1}^f F_{t-1}(j) \\ - \frac{\kappa^n}{2} \left(\frac{i_t^n(j)}{i_{t-1}^n(j)} - 1 \right)^2 i_t^n N_t - \frac{\kappa^d}{2} \left(\frac{i_t^d(j)}{i_{t-1}^d(j)} - 1 \right)^2 i_t^d M_t - \frac{\kappa^b}{2} \left(\frac{i_t^b(j)}{i_{t-1}^b(j)} - 1 \right)^2 i_t^b B_t \quad (38)$$

When rates are fully flexible, solving this maximisation problem yields the banks' target interest rates, which are set as markups (or markdowns) over their marginal costs of funding:

$$i_t^n = \frac{\epsilon^n}{\epsilon^n - 1} i_t^f \quad i_t^b = \frac{\epsilon^b}{\epsilon^b - 1} i_t^f \quad i_t^d = \frac{\epsilon^d}{\epsilon^d - 1} (\omega i_t + (1 - \omega) i_t^f) \quad (39)$$

These target equations outline the core intuition of the banking sector. Because the marginal source of funding is the central bank, the marginal cost for both mortgages and issued bonds is the central bank lending rate, i_t^f . Consequently, banks set the mortgage rate at a markup over i_t^f (since $\epsilon^n > 1$) and the bond rate at a markdown below i_t^f (since $\epsilon^b < -1$). For deposits, the marginal cost of funding is a weighted average of the policy rate i_t (earned on required reserves) and the central bank lending rate i_t^f . Crucially, while the introduction of RFA may alter the bank's overall composition and average cost of funding, it does not alter the *marginal* cost of funding. Therefore, RFA affects bank profitability and disintermediation directly through quantities, rather than shifting the rate-setting schedules. This is only the case in the steady state, as rates change according to monetary policy in the short run.

Factoring in the adjustment costs yields the following non-flexible rate-setting equations:

$$i_t^n = \frac{\epsilon^n}{\epsilon^n - 1} i_t^f - (1 + i_t^b) \frac{\kappa^n}{\epsilon^n - 1} \left(\frac{i_t^n}{i_{t-1}^n} - 1 \right) \frac{(i_t^n)^2}{i_{t-1}^n} \\ + (1 + i_t^b) \mathbb{E}_t \left[\mathcal{M}_{t,t+1} \frac{\kappa^n}{\epsilon^n - 1} \left(\frac{i_{t+1}^n}{i_t^n} - 1 \right) \frac{(i_{t+1}^n)^2}{i_t^n} \frac{N_{t+1}}{N_t} \right] \quad (40)$$

$$i_t^d = \frac{\epsilon^d}{\epsilon^d - 1} (\omega i_t + (1 - \omega) i_t^f) + (1 + i_t^b) \frac{\kappa^d}{\epsilon^d - 1} \left(\frac{i_t^d}{i_{t-1}^d} - 1 \right) \frac{(i_t^d)^2}{i_{t-1}^d} \\ - (1 + i_t^b) \mathbb{E}_t \left[\mathcal{M}_{t,t+1} \frac{\kappa^d}{\epsilon^d - 1} \left(\frac{i_{t+1}^d}{i_t^d} - 1 \right) \frac{(i_{t+1}^d)^2}{i_t^d} \frac{M_{t+1}}{M_t} \right] \quad (41)$$

$$i_t^b = \frac{\epsilon^b}{\epsilon^b - 1} i_t^f + (1 + i_t^b) \frac{\kappa^b}{\epsilon^b - 1} \left(\frac{i_t^b}{i_{t-1}^b} - 1 \right) \frac{(i_t^b)^2}{i_{t-1}^b} \\ - (1 + i_t^b) \mathbb{E}_t \left[\mathcal{M}_{t,t+1} \frac{\kappa^b}{\epsilon^b - 1} \left(\frac{i_{t+1}^b}{i_t^b} - 1 \right) \frac{(i_{t+1}^b)^2}{i_t^b} \frac{B_{t+1}}{B_t} \right] \quad (42)$$

3.3.3 Central Bank

The central bank's assets consist entirely of loans to commercial banks, while its liabilities comprise banks' required reserves $M_{B,t}^c$ and the retail central bank digital currency (RFA) held by households. The balance sheet is given by:

$$F_t = \lambda_H M_{H,t}^c + \lambda_M M_{M,t}^c + \lambda_S M_{S,t}^c + M_{B,t}^c \quad (43)$$

Because the central bank charges a higher rate i_{t-1}^f on its loans than it pays on its liabilities i_{t-1} , it generates nominal profits $P_t G_t$ (where G_t denotes real profits):

$$P_t G_t = i_{t-1}^f F_{t-1} - i_{t-1} (M_{B,t-1}^c + \lambda_H M_{H,t-1}^c + \lambda_M M_{M,t-1}^c + \lambda_S M_{S,t-1}^c) \quad (44)$$

These profits are not rebated back to the economy and are assumed to be wasted resources. As no agents benefit from this waste, the measured welfare gains are slightly biased downwards.

The central bank also conducts monetary policy by setting the short-term nominal interest rate i_t according to a standard Taylor rule with interest rate smoothing:

$$\ln\left(\frac{1+i_t}{1+\bar{i}}\right) = \rho_i \ln\left(\frac{1+i_{t-1}}{1+\bar{i}}\right) + (1-\rho_i)\phi_\pi \ln(1+\pi_t) + \xi_t \quad (45)$$

where $\pi_t \equiv \left(\frac{P_t}{P_{t-1}} - 1\right)$ is the inflation rate, $\bar{i}$ is the steady-state policy rate, ρ_i dictates the degree of policy inertia, and ϕ_π is the response to inflation. Finally, ξ_t is an exogenous monetary policy shock which evolves according to an AR(1) process governed by persistence ρ_ξ and standard deviation σ_ξ :

$$\xi_t = \rho_\xi \xi_{t-1} + \varepsilon_{\xi,t} \quad (46)$$

$$\text{where } 0 \leq \rho_\xi \leq 1 \quad \& \quad \varepsilon_{\xi,t} \stackrel{\text{iid}}{\sim} N(0, \sigma_\xi^2)$$

The central bank also sets the lending rate to commercial banks which is 25 (quarterly) basis points above the policy rate:

$$i_t^f = \phi_f + i_t, \quad \phi_f = 0.0025 \quad (47)$$

3.4 Equilibrium

The decentralised recursive equilibrium is defined as a sequence of 61 endogenous variables and 61 equations that satisfy the agents' optimality conditions, firm pricing rules, market clearing constraints, and policy rules, given the sequence of exogenous shocks and initial conditions found in appendix A.2. The goods market clearing equation is omitted as per Walras' Law. Since the model features a nominal trend, the system is rendered stationary by expressing all nominal variables in real terms. Real variables are defined as $x_t \equiv \frac{X_t}{P_t}$ except for the collateral constraint Lagrange multiplier, where $\mu_t \equiv \Omega_{M,t} P_t$. The Dynare model additionally features a number of "measurement variables" which are described and explained in appendix A.3.

3.4.1 Market Clearing & Aggregation

As the model features multiple agents, all individual variables have an "aggregate" version which follows the format: $X_t = \sum_i \lambda_i X_{i,t}$. The full list can be found in appendix B.5.

The housing market clears as $H_t \stackrel{!}{=} \bar{H} = 1$ with a fixed housing supply normalised to 1. The goods market clearing, or the economic resource constraint, can be found by combining the household budget constraints, the bank balance sheet and reserve requirements, the central bank balance sheet, and all profit definitions. It equates total production to consumption, investment, central bank (CB) profits, and resources lost to adjustment costs, denoted Adj_t :

$$P_t Y_t \stackrel{!}{=} P_t C_t + P_t I_t + P_t G_t + Adj_t \quad (48)$$

3.4.2 Steady State

The model features a non-inflationary deterministic steady state. Price and wage inflation are zero ($\pi = 0$, $\pi_H^w = 0$, $\pi_M^w = 0$), which implies that price dispersion disappears ($\Delta = 1$) and optimal relative prices equate to one ($\tilde{p} = 1$). Without investment adjustment costs in the steady state, the marginal value of capital equals its replacement cost, yielding a Tobin's Q of unity ($q = 1$). All exogenous shock processes are assumed to be at their unconditional means, setting aggregate technology $Z = 1$ and the monetary policy shock $\xi = 0$.

The full baseline steady state does not admit a closed-form analytical solution and is therefore computed numerically by solving the simultaneous system of non-linear equations using Dynare's default solver.

3.5 Calibration

The model is built to reflect the Swiss economy, and calibrated with the same idea. The main goal of the calibration was to target the levels of wealth and inequality found in Switzerland, which are ≈ 0.40 for the income Gini coefficient before taxes, and ≈ 0.80 for the wealth Gini coefficient (MARTÍNEZ 2021). Other parameters are either calibrated to standard values, or are otherwise based on values found in GERALI et al. (2010)³ and PAUL et al. (2026).⁴ While not based on Switzerland, both papers base their calibration on the Euro Area, as studies for Switzerland specifically are quite rare. I chose these two papers as they are close to my model. GERALI et al. (2010) was used for the entirety of the bank and union sectors, while PAUL et al. (2026) builds a model with a similar research question. I provide a sensitivity analysis for a few key parameters in section 4.4. The full table of calibrated parameters along with what they are targeting or their source is provided in Table 1.

The population shares are aimed at simulating the wealth and income distributions found in MARTÍNEZ (2021), where around 50% of the population is low-wealth workers, around 40% is made up of middle-class leveraged homeowners, and the top 10% are the wealthy capitalists. The relative risk aversion parameter, σ , is set to a standard value of 1.5 showing a certain level of risk aversion. The other inverse elasticity utility parameters are calibrated symmetrically, apart from the inverse Frisch elasticity which is set to 1. The labour disutility parameters, χ , are set to target steady-state labour equal to one-third, while the money utility parameter, Ψ , is set to target the deposit-to-GDP ratio in Switzerland (WORLD BANK 2022). The discount factors are set with the logic $\beta_S = \beta_H > \beta_M$. The fact that S and H agents have the same patience level serves two purposes: First, so that the low level of wealth of H agents is not due to impatience and unwillingness to save, but by their inability to do so caused by financial constraints. Second, it allows to show the potential of the RFA policy by having the H-agents be willing to save much more when RFA is introduced and thus build up relatively significant wealth. The M-agents are more impatient as they take the role of borrowers in the economy. I do study the case where $\beta_S > \beta_H = \beta_M$ in the sensitivity analysis in section 4.4. The S-agents' discount factor is the most

³Calibrated values based on GERALI et al. (2010) are taken from both their calibration, and their posterior estimation (mean).

⁴For calibration based on PAUL et al. (2026), I use their Euro-Area calibration found in their appendix.

important one as it pins down the steady-state policy rate of the economy. It is set at 0.994 in order to have an annualised policy rate equal to 1.5%. While high for Switzerland, it is necessary to solve the model and establish the benefits of RFA, as with a rate of zero, RFA would represent no difference from the baseline case. The rate of 1.5% was chosen according to the Swiss National Bank’s decision on the 23rd of March 2023 (SWISS NATIONAL BANK 2023).

The rest of the economy’s interest rates are based on markups/markdowns on the policy rate, depending on the elasticity parameters. In 2023, the savings account deposit rate in Switzerland was about 0.6% (SWISS NATIONAL BANK 2026a) motivating a calibration of ϵ^d such that i_t^d is around 1 annualised percentage point (pp) lower than the steady-state policy rate. ϵ^n is set to target a mortgage rate to be about 1.5 annualised pp higher, for a mortgage rate of around 3% (SWISS NATIONAL BANK 2026b). ϵ^b is set such that the bond rate is practically identical to the lending rate (thus higher than the policy rate). The lending rate is set to be 1 annualised pp above the policy rate (ϕ_f). This is higher than the real-world value of 0.25, but is necessary for the model to solve as otherwise spreads are too thin.

The LTV ratio, Γ , is set according to GERALI et al. (2010), and aligns closely with the down-payment requirement for mortgages in Switzerland. The reserve requirement, ω , is set to 0.025 according to Swiss banking law in 2023 (SWISS NATIONAL BANK 2024). As mentioned previously, the shock sets ϖ to 0.5, in order for deposits and RFA to have the same liquidity, as inspired by BACCHETTA & PERAZZI (2025). The baseline value for the elasticity of substitution between deposits and RFA, ϵ^c , is also taken from this study.

With the exception of the labour share of H-agents (α_l), all production-side parameters are set to standard literature values. α_l is calibrated to target an effective wage premium of about 1.75, matching the observed ratio of gross monthly wages between middle/upper management (11’174 CHF) and non-management positions (6’368 CHF) in Switzerland (FEDERAL STATISTICAL OFFICE 2024).

Table 1: Parameter Values

Prm.	Value	Description	Target or source
$\lambda_{H,M,S}$	{0.5, 0.4, 0.1}	Population Share	MARTÍNEZ (2021)
$\beta_{H,M,S}$	{0.994, 0.99, 0.994}	Discount Factor	1.5% Annualised policy rate
σ	1.5	Relative Risk Aversion	Standard
ψ	1.5	Inverse Elasticity of Money Demand	Standard
η	1.5	Inverse Elasticity for Housing	Standard
φ	1	Inverse of the Frisch elasticity	PAUL et al. (2026)
ϵ^c	6	Elasticity between Deposit varieties	BACCHETTA & PERAZZI (2025)
$\chi_{H,M}$	{17.6291, 12.758}	Labour Disutility	One-third S.S. labour

Table 1 – Continued

Prm.	Value	Description	Target or source
Ψ	0.0184	Money Utility	1.67 Deposit-to-GDP
Θ	0.2	Housing Utility	GERALI et al. (2010)
Γ	0.7	LTV Ratio	GERALI et al. (2010)
α	0.33	Capital Share of Production	PAUL et al. (2026)
θ	0.75	Prob. of keeping prices fixed	Standard
δ	0.025	Capital Depreciation Rate	Standard
κ^y	2	Investment Adj. Costs	PAUL et al. (2026)
ϵ^y	6	Elasticity between Good varieties	Standard
κ^w	100	Wage Adj. Cost	GERALI et al. (2010)
ϵ_w	5	Elasticity between Labour varieties	GERALI et al. (2010)
α_l	0.35	H Share in Labour Aggregator	$\frac{\lambda_M^w L_M}{\lambda_H^w L_H} \approx 1.75$
κ^n	10	Mortgage Rate Adj. Costs	GERALI et al. (2010)
κ^d	3	Deposit Rate Adj. Costs	GERALI et al. (2010)
κ^b	0	Bond Rate Adj. Costs	No lag
ϵ^n	6	Elasticity between Mortgage varieties	1.5 a.p.p. higher than policy rate
ϵ^d	-0.3	Elasticity between Deposit varieties	1 a.p.p. lower than policy rate
ϵ^b	-30	Elasticity between Bond varieties	Minimal markdown
ϕ_π	1.5	Taylor Rule Inflation Reaction	PAUL et al. (2026)
ϕ_f	0.0025	Lending Rate Penalty	Assumption
ω	0.025	Reserve Requirement	SWISS NATIONAL BANK (2024)
ρ_i	0.8	Interest Rate Smoothing	PAUL et al. (2026)

Note: a.p.p. refers to annualised percentage points.

4 Results

This section examines an unanticipated preference shock in which RFA transitions from providing negligible liquidity utility ($\varpi \approx 1$) to offering liquidity utility equivalent to regular deposits ($\varpi = 0.5$). The main mechanism is a sudden household

demand for RFA, coupled with an expansion of their aggregate deposit portfolios given higher available interest rates.

The analysis is organised as follows: First, we study the differences between the initial no-RFA steady state and the final RFA steady state. Second, we examine the transition dynamics between these steady states using a perfect foresight simulation that solves after approximately 600 periods. This highlights the initial impact of the shock, stabilising after about 70 periods, the short-term mechanisms, and the gradual shift to the new steady state. Third, we compare the impulse response functions (IRFs) to stochastic shocks in the baseline economy with those in the RFA economy, just as in MURAKAMI et al. (2024) and PAUL et al. (2026). Finally, we explore alternative calibrations and a specification that introduces tiered remuneration to RFA.

The main outcomes of interest are the impacts on wealth inequality, income inequality, and welfare. Inequality is measured using Gini coefficients, as detailed in appendix A.3. Welfare changes are evaluated through the multiplicative consumption-equivalent variation (CEV), expressed as the percentage change in consumption required to leave an agent indifferent to the introduction of RFA. I use both a “backward-looking” CEV (BL-CEV), following AURAY, DEVEREUX et al. (2026), which takes into account the entire transition path up to the new steady state, as well as a “forward-looking” CEV (FL-CEV), following PAUL et al. (2026), which isolates the infinite horizon of the new steady state while ignoring the transition path. The former accounts for the transitory costs of the policy, whereas the latter serves to compare two separate steady-state equilibria. Computational details are provided in appendix A.3, and a summary of CEV results can be found in the “Baseline” column of Table 3.

Overall, the policy experiment shows promise in improving welfare, reducing inequality, and stabilising the economy. However, these benefits come at the cost of financial disintermediation, significantly lowering commercial bank profits and dampening aggregate output.

4.1 Steady State Comparison

Here we discuss the steady-state values of the initial non-RFA economy and the final RFA economy, reported in Table 10. As we are comparing steady states, the relevant welfare measure is the FL-CEV, represented by the dotted lines in Figure 3.

Unlike some of the CBDC literature which finds no negative effects on bank deposit funding (ASSENMACHER et al. 2024; BARRDEAR & KUMHOF 2022), this model predicts severe bank disintermediation, characterised by a significant outflow of deposits. Due to the specification of the banking sector, this outflow does not directly restrict aggregate lending; it instead compresses banks’ profits. Agents heavily substitute towards RFA and expand their overall deposit portfolios, depriving banks of cheap deposit funding. Banks are forced to replace this funding with significantly more expensive central bank lending, cutting their profits by more than half.

This reduction in bank profitability directly penalises the S-agents, who are the sole recipients of bank dividends. Their total income falls by about 5%. Consequently, S-agents reduce consumption by a similar amount and downsize their portfolios across

all asset classes: capital holdings fall by about 0.5% (causing a parallel drop in investment), housing stock by about 2.5%, and bond holdings by 61%. The reduction in bonds is much larger than in housing or capital because housing provides direct utility, while capital is buffered by investment adjustment costs and decreasing marginal returns, meaning a lower capital stock naturally drives up its yield. Ultimately, S-agents' net wealth drops by roughly 5%. This translates to a severe welfare loss, with a FL-CEV of -4.37%. Intuitively, this means that residing in the RFA steady state is equivalent to a 4.37% permanent reduction of consumption in the no-RFA steady state.

In contrast, the drop in S-agent housing demand frees up supply for M-agents, reflected in lower house prices of roughly 3%. M-agents respond by expanding their housing stock by around 2%. This increase in housing stock is not enough to counteract the house price effect on the total value of their collateral, which tightens the borrowing constraint by about 1%. The forced reduction in debt, combined with an increase in total deposit holdings, raises their net wealth by 5.7%. Rather than dramatically increasing consumption, M-agents choose to consume only slightly more while reducing labour supply by 0.25%. In welfare terms, their FL-CEV is positive at 0.69%. This indicates a welfare gain as agents are indifferent between the RFA steady state and receiving a permanent 0.69% increase in consumption in the no-RFA steady state.

H-agents benefit the most. Having access to a higher interest rate, they can now accumulate wealth much more effectively, which they do by increasing their total deposit holdings. Their net wealth expands by 65%, indicating that their previous lack of wealth accumulation stemmed from restricted access to higher-yielding financial instruments rather than saving behaviour. With higher interest comes higher income, which rises by 0.75%. Consumption increases by the same amount, while labour hours decrease by 0.77%. The higher consumption and lower labour hours explain their substantial welfare gains, with a FL-CEV of 1.78%.

The reduced capital investment by S-agents lowers long-run output. This decline results in reduced aggregate labour and aggregate consumption, as the drop in S-agent consumption outweighs the increases for H and M agents. While some CBDC literature finds similar negative effects on capital and output (AGUR et al. 2022; PAUL et al. 2026), evidence remains mixed. Other models predict output increases (ASSENMACHER et al. 2024; BARRDEAR & KUMHOF 2022; CHIU et al. 2023), typically driven by the bank competition channel⁵ which stimulates lending. However, because steady-state rate spreads in this model are exogenous and fixed, banks do not competitively raise deposit rates in the new equilibrium. Without this rate adjustment to retain cheap funding, regular deposits do not see a rise in attractiveness, further emphasising the disintermediation mechanism, and driving the contractionary effect on long-run output.

The introduction of RFA successfully improves the wealth accumulation capacity of poorer households, leading to a reduction in both income and wealth inequality. Income inequality drops by 0.68 percentage points, while wealth inequality decreases by 3.19 percentage points. As shown in Figure 1, the effect on income takes relat-

⁵The bank competition channel refers to models with endogenous market power, where banks react to remunerated CBDC by increasing deposit rates.

ively few periods to materialise, as H and M agents gain immediate access to higher yields. Wealth inequality, on the other hand, decreases smoothly over a long horizon, reflecting the mechanics of compound interest. With a higher interest rate, the long-run wealth accumulation for the bottom of the distribution is augmented by exponential compounding. Additionally, part of the wealth inequality result stems from the reduction in bank profits which takes away from the S-agents. However, this is not a primary driver, as in the case where central bank profits are remitted to banks, preserving S-agents' dividends, there is still a similar sized decrease in wealth inequality. Ultimately, the aggregate FL-CEV is equal to 1.16% showing promising welfare gains in the long-run. Because the majority of the population benefits from RFA, and given the diminishing marginal utility of consumption for the wealthy, the welfare gains experienced by H and M agents significantly outweigh the welfare losses of the S-agents.

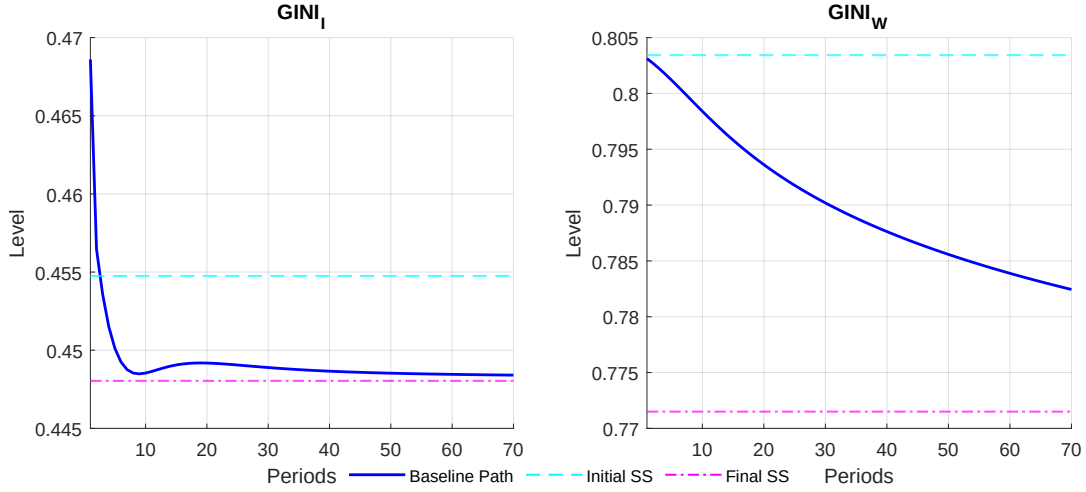


Figure 1: Transition paths and steady-states of income and wealth inequality. “Initial SS” refers to the no-RFA steady state while “Final SS” refers to the RFA steady state.

4.2 Impact & Transition

We now consider the immediate shock impact and the subsequent transition dynamics over the first 70 periods, illustrated in Figure 2 primarily through percentage deviations from the initial equilibrium. The paths of the main variables in levels are provided in the appendix (Figure 6). To study the transition, the relevant welfare measure is the BL-CEV which computes the CEV for each point of the transition compared to the initial no-RFA steady state, shown in Figure 3. The “terminal BL-CEV” is defined as the BL-CEV evaluated in the period immediately preceding the new RFA steady state, relative to the initial no-RFA equilibrium. This terminal measure captures the entire transition path.

The direct effect of the RFA introduction is on household deposit portfolios. Agents instantly substitute the majority of their deposit holdings to RFA and simultaneously increase their total deposit portfolio size, seen in their increased savings rate which spikes on impact. This comes at the cost of consumption, which drops on impact for each agent. The withdrawal of bank deposits forces commercial banks, which do not hold ample ex-ante reserves, to rely heavily on central bank lending. This substitution

reduces bank profits which permanently lowers S-agents' future expected income. Anticipating this decline, S-agents increase their savings rate much more aggressively than H-agents, to smooth future consumption.

The resulting drop in aggregate consumption triggers a standard negative demand shock. The firms that can adjust their prices do so, leading to deflation. As aggregate prices cannot fall by enough to absorb the shock, output falls. The central bank reacts by cutting the policy rate, which spreads to all interest rates in the economy. Inflation normalises quicker than the interest rates, causing the future real rate on bonds to drop. The increased savings of S-agents therefore occurs in the other investment vehicles as their housing and capital stocks rise, while their bond holdings fall. As a consequence, despite the negative demand shock, aggregate investment actually rises. The increased demand for housing increases its price, and drives out M-agents who are forced to sell. This lowers M-agent collateral, tightening their borrowing constraint ($\Delta^+ \mu$) and leading them to deleverage.

The initial investment boom leads to a gradual accumulation of capital in the periods following the shock. This larger capital stock eventually lowers marginal costs and causes output to overshoot. Thus, after the economic contraction of the demand shock, the build-up of capital leads to a temporary economic expansion which materialises through increases in labour, wages, investment, and aggregate consumption. These variables eventually peak and slowly decline toward the new steady state.

From a welfare perspective, the transition is costly for the economy. As illustrated on Figure 3, all agents suffer utility losses during the initial onset of the shock. H-agents experience the deepest initial drop in welfare, as they are still the most limited when it comes to consumption smoothing. However, their BL-CEV turns positive after approximately 150 periods, ultimately reaching 0.63%. M-agents suffer the least short-term pain, with the smallest decrease in consumption, likely thanks to the funds received from selling housing stock at elevated prices. This mechanism allows their BL-CEV to reach the highest terminal value of the three groups at 0.85%. S-agents are the only group with a consistently negative BL-CEV, steadily reaching -3.13%. As this value is smaller in magnitude than their FL-CEV (-4.37%), these agents do not suffer much from the transition itself, but rather from the permanent reduction in their wealth. Their extensive financial buffers and independence from labour market wages allow them to smooth the transition costs over the entire transition period. Overall, the aggregate terminal BL-CEV is positive reaching 0.46% and shows that the net effect, including the transition costs, is still positive. While the substantial gap between the aggregate terminal BL-CEV (0.46%) and aggregate FL-CEV (1.16%) is partially the mechanical effect of weighting heterogeneous discount factors over finite versus infinite horizons⁶, it does further emphasise the costs of the transition. For this reason, we later investigate tiered remuneration, a policy aimed at mitigating disintermediation. The underlying hypothesis is that by penalising excessive RFA holdings, we can restrict the quantity of RFA entering the economy, thereby dampening the initial shock and theoretically smoothing the transition path.

⁶This becomes clear in Table 3, where decreasing β_H to 0.99 significantly reduces the gap between both measures.

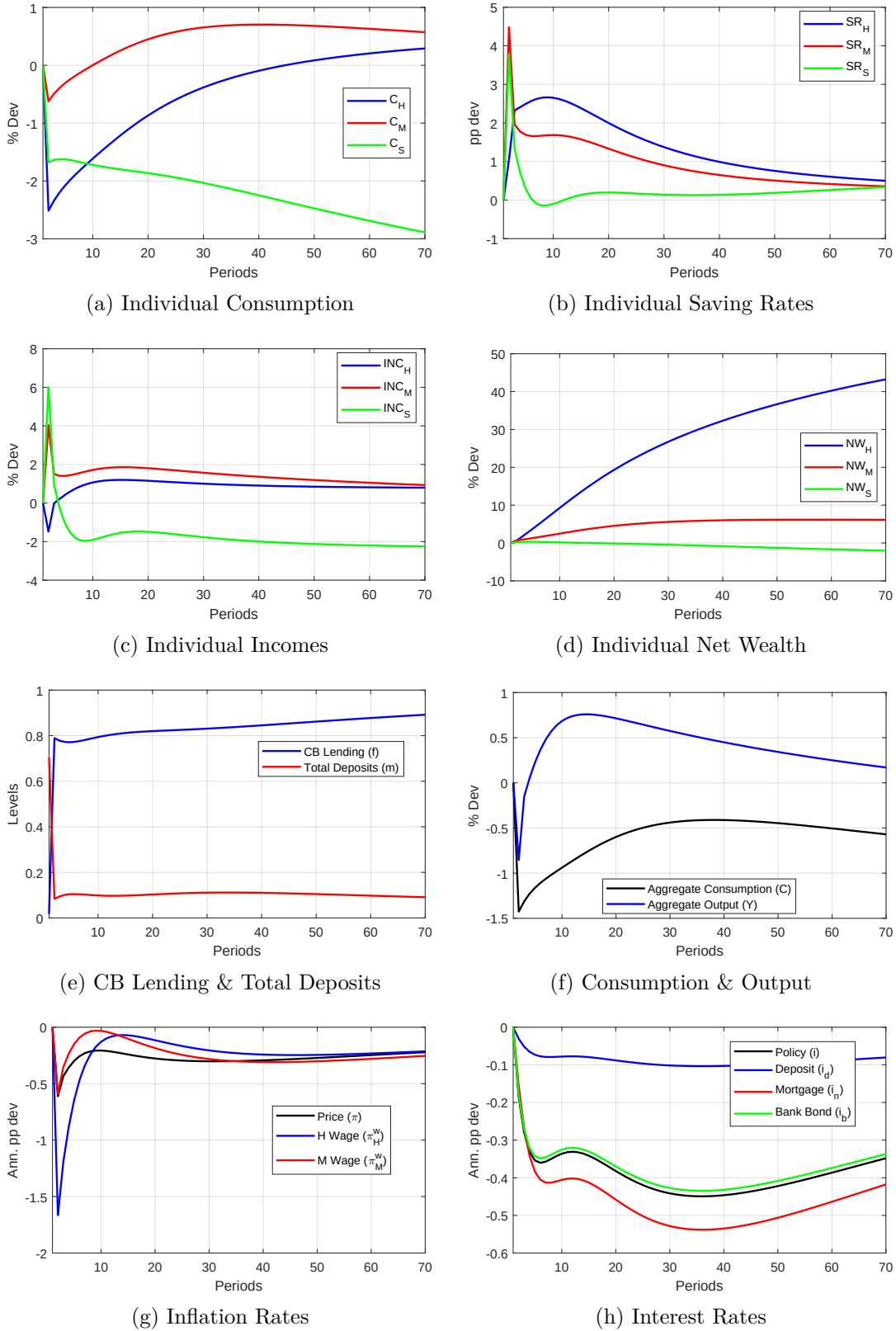


Figure 2: Transition (under perfect foresight) between the no-RFA steady state and the RFA steady state. Time periods are measured in quarters. Variables are either measured as percentage deviations from the steady state (% Dev), percentage point deviations (pp dev), annualised percentage point deviations (Ann. pp dev), or levels.

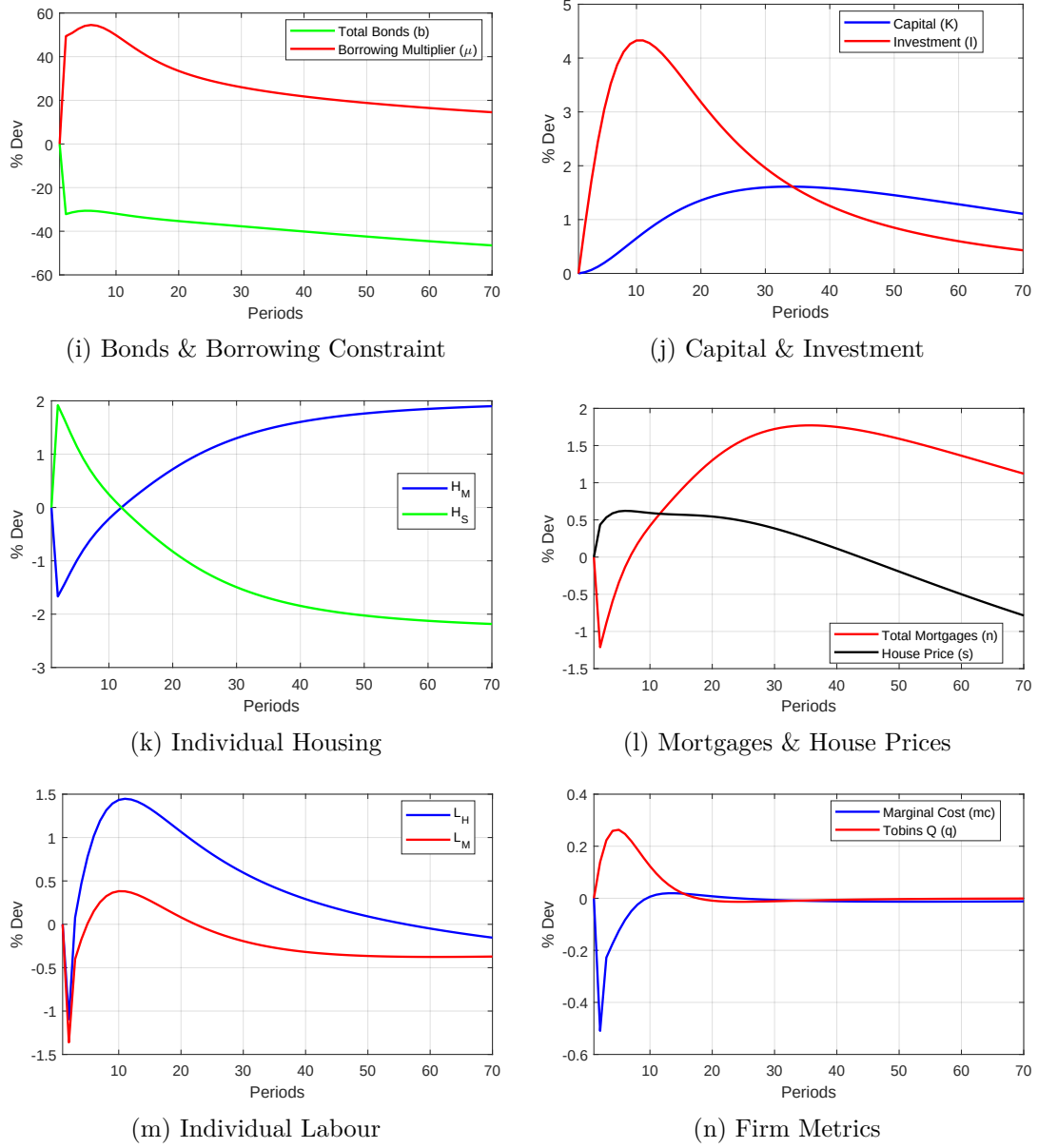


Figure 2: Transition (under perfect foresight) between the no-RFA steady state and the RFA steady state. Time periods are measured in quarters. Variables are either measured as percentage deviations from the steady state (% Dev), percentage point deviations (pp dev), annualised percentage point deviations (Ann. pp dev), or levels.

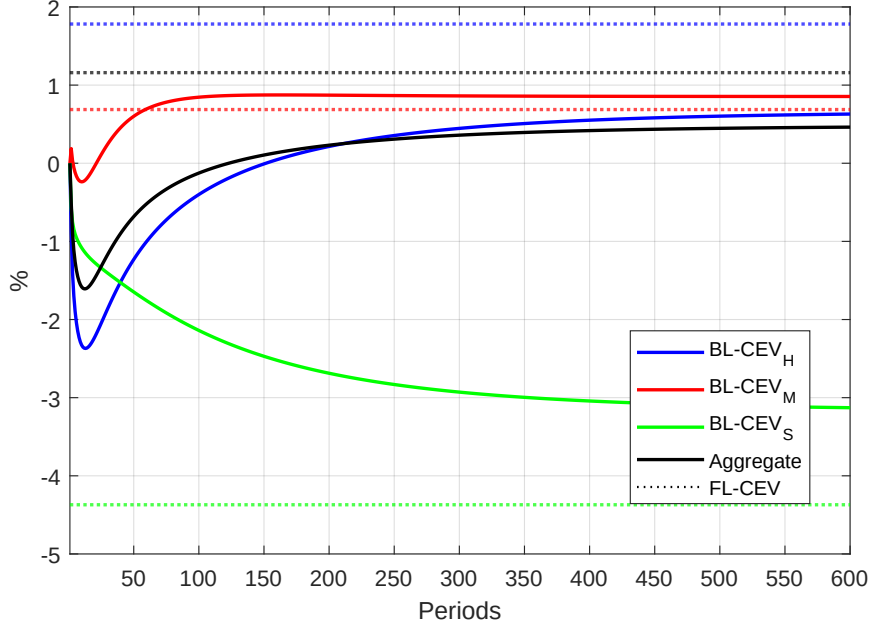


Figure 3: Welfare measurements across time. Full lines represent the BL-CEV (taking into account the transition), while the dotted lines represent the FL-CEV (only comparing steady states).

4.3 Stochastic Shocks

Having already examined how the economy reacts to the introduction of RFA in deterministic simulations, we now turn to analysing how the two economies differ in their response to transitory stochastic shocks around the respective steady states. Following MURAKAMI et al. (2024), the model is solved via a second-order perturbation around the deterministic steady states to generate stochastic IRFs and simulated volatility. Stochastic shocks consist of a 1% annualised monetary policy shock with a persistence of 0.70, and a 1% TFP shock with a persistence of 0.90. The analysis here focuses on the monetary policy IRFs for a selection of key variables. Additional monetary policy IRFs and the complete set of technology shock IRFs are provided in appendix C.2.

The contractionary monetary policy shock illustrated in Figure 4 exhibits standard behaviour in its impact on the economy, as the higher real policy rate depresses aggregate demand, lowers output, and triggers deflation. Unlike standard representative-agent models, the heterogeneous-agent framework allows us to dissect the distributional impacts and differences in transmission channels. In the no-RFA baseline, H-agents are virtually insulated from the direct interest rate change because bank deposit rates react sluggishly and only partially, due to the markdown. Instead, they are hit entirely via the labour market where falling labour demand forces them to work less and at lower wages. This depresses their income and wealth, driving down consumption. M-agents face the same labour market mechanism, but are additionally squeezed by the higher borrowing costs, resulting in the sharpest consumption drop out of all households. In contrast, S-agents are not directly affected by the depressed labour market and instead receive temporarily higher profits. Furthermore,

they directly perceive the rate change as bond yields react instantaneously and fully, prompting them to reduce consumption to increase savings.

In the RFA economy, transmission fundamentally changes. All agents instantly perceive the policy rate through RFA. For H-agents, the higher rate on their now-larger deposit portfolios provides an income boost that partially offsets their wage losses. This prevents their net wealth from dropping as deeply and smooths their consumption path. M-agents similarly benefit from higher yields, using the extra income to deleverage less aggressively than in the no-RFA case. These results align with those of both MURAKAMI et al. (2024) and PAUL et al. (2026), who find that aggregate impulse responses to a monetary policy shock change little under CBDC. However, the underlying mechanics diverge. Because the representative agents in PAUL et al. (2026) already receive the policy rate via bonds, they explain that the transmission channels remain unchanged, unlike in this model. My results thus align more closely with MURAKAMI et al. (2024), who demonstrate that previously unbanked households can now smooth consumption, thereby reducing consumption inequality in response to a monetary policy shock.

To assess volatility, I report the standard deviation for key variables in Table 2 when the model is subjected to both the technology and monetary policy shocks. The introduction of RFA generally reduces the standard deviation for most variables, suggesting a macroeconomic stabilisation effect. This effect is likely driven by more direct monetary policy transmission, as H and M households now perceive the policy rate directly and instantly, unlike the sluggish and partial transmission to the deposit and mortgage rates in the no-RFA case. This leads interest rates to be less volatile, then spreading the effect to inflation, wages, and consumption for H and M agents. In contrast, S-agents experience higher volatility in consumption, income, and net wealth. This is driven by bank dividends being more sensitive to the policy rate, caused by the high level of central bank lending in the RFA case. Nevertheless, aggregate consumption and output become more stable overall as the majority of the population sees lower volatility. These results somewhat align with MURAKAMI et al. (2024), who similarly find lower volatility for inflation and interest rates though report higher output volatility.⁷

4.4 Recalibration

In this section, we recalibrate certain key parameters and analyse the robustness of the results. These include calibrating β_H to 0.99, varying the substitutability of RFA to deposits, ϵ^c , and finding the welfare-maximising value of ϖ . The full results including the steady states and transition paths are provided in appendix C.3. Additionally, the appendix also includes results for a higher reserve requirement (ω) as well as log-utility in all variables (i.e. $\sigma = \eta = \psi = \varphi = 1$). For each recalibration, χ_H , χ_M , and Ψ are adjusted to hit the same steady-state targets for labour and deposits-to-GDP as in the baseline calibration.

As discussed previously, the goal of the original calibration of β_H equal to β_S was to

⁷They explain this by the fact that unbanked household consumption, which is counter-cyclical to TFP shocks in the absence of CBDC, becomes pro-cyclical when CBDC is introduced. This is due to the CBDC providing access to interest which the unbanked use to increase savings, thus lowering consumption when policy rates rise following a negative TFP shock.

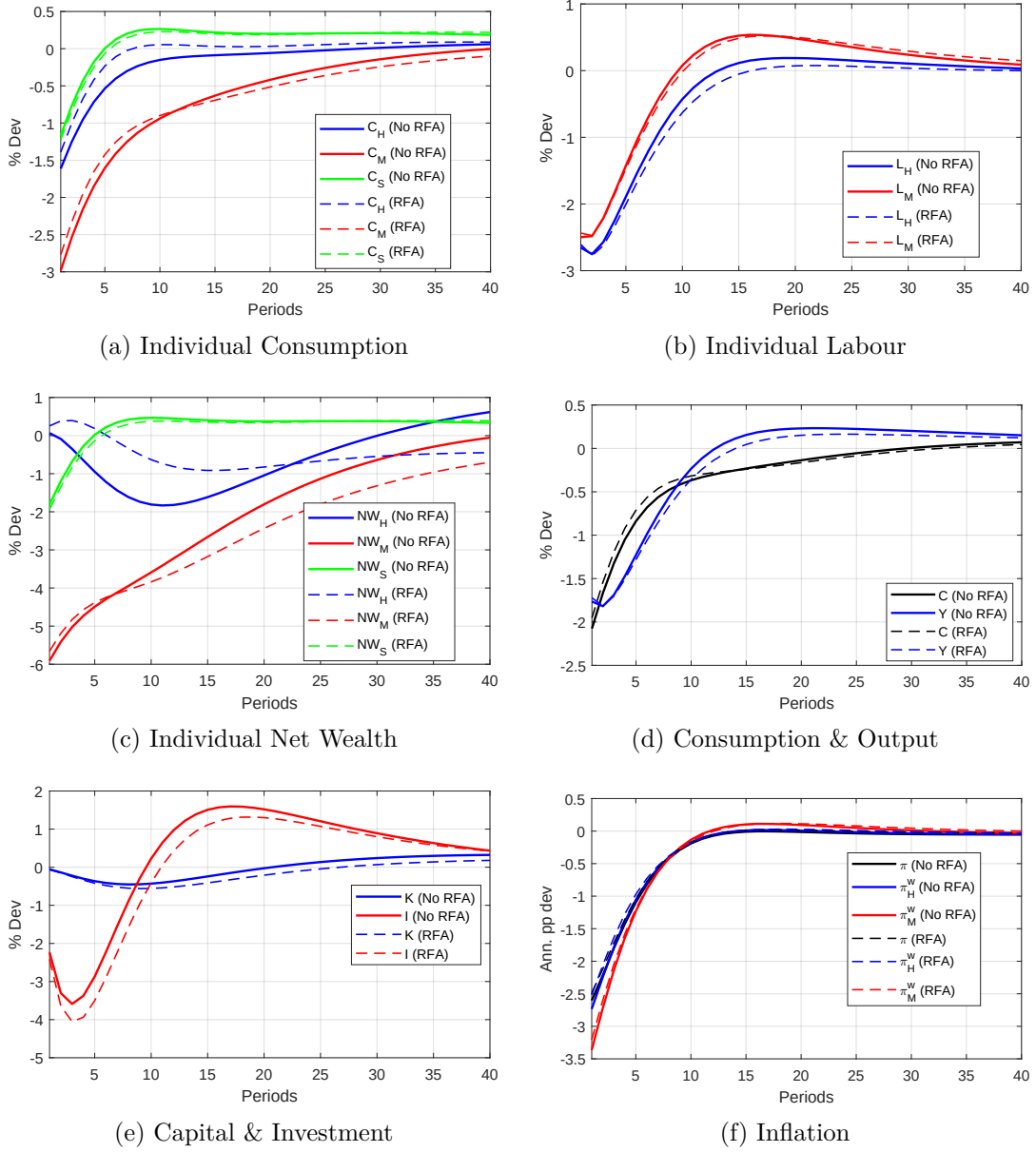


Figure 4: Impulse responses to a 1% annualised contractionary monetary policy shock. Time periods are measured in quarters. Variables are measured as percentage deviations from the steady state, except for inflation which is expressed through annualised percentage point deviations.

Table 2: Model Simulated Volatility

Variable	No-RFA	RFA
Standard Deviation (%)		
Individual Consumption ($C_{H,M,S}$)	{3.01, 6.30, 2.41}	{2.44, 6.03, 2.91}
Individual Housing ($H_{M,S}$)	{8.61, 9.90}	{7.07, 8.52}
Aggregate Consumption (C)	3.84	3.65
Aggregate Output (Y)	5.20	5.16
Capital (K)	3.53	3.41
Investment (I)	11.94	12.08
Labour (L)	6.76	6.61
Mortgage Debt (n)	9.75	9.20
House Price (s)	4.12	4.66
Bonds (b)	21.85	51.19
Dividends (d)	11.63	11.97
Wages (w)	2.75	2.68
Rental Rate (r)	7.48	7.15
Individual Income ($INC_{H,M,S}$)	{7.46, 31.67, 67.91}	{6.87, 32.35, 69.40}
Individual Net Wealth ($NW_{H,M,S}$)	{15.69, 18.06, 4.04}	{9.34, 19.09, 4.95}
Standard Deviation (Annualised pp)		
π	4.58	4.33
i	1.17	1.00
i^r	3.81	3.76
$i^{d,n,b}$	{0.24, 1.25, 1.14}	{0.19, 1.03, 0.97}
Standard Deviation (Absolute pp)		
$GINI_W$	1.29	1.47
$GINI_I$	15.55	15.97

Note: Standard deviations are based on model simulations. The model is solved and simulated via second-order perturbation about the deterministic steady state. Inflation and the interest rates are annualised.

analyse the effect of RFA on households that have patient preferences, but are not able to save due to limited financial access. This leads to them receiving particularly high welfare gains from RFA as they use the opportunity to increase their savings substantially. However, it also mechanically increases their CEV as they perceive more benefits from future gains which outweigh the transition costs. Table 3 demonstrates that lowering H-agent patience to the same value as the impatient M-agents leads to a much smaller FL-CEV, while the BL-CEV is only slightly affected. We see that the gap between the FL-CEV and BL-CEV narrows significantly, showing that it is partially mechanically driven by the gaps in weight differently-sized discount factors have over finite versus infinite horizons. Nevertheless, both the BL-CEV and FL-CEV are still positive, revealing that the long-term gains still outweigh the short-term costs.

Table 3: Welfare in baseline versus recalibration where $\beta_H = 0.99$

Agent	Baseline		Recalibration	
	FL-CEV	BL-CEV	FL-CEV	BL-CEV
Hand-to-Mouth	1.78%	0.63%	0.66%	0.52%
Middle-Class	0.69%	0.85%	0.89%	0.63%
Saver	-4.37%	-3.13%	-3.40%	-2.78%
Aggregate	1.16%	0.46%	0.47%	0.35%

The FL-CEV compares steady states. The BL-CEV takes into account the whole transition path.

We now move on to varying the elasticity of substitution of RFA to deposits. We first lower it to $\epsilon^c = 3$, where we observe a clear reduction in welfare gains compared to the baseline calibration (Table 4), which only benefits S-agents by protecting bank profits. As agents are now more constrained in their ability to substitute toward RFA, H and M agents enjoy smaller gains.

Table 4: Welfare in baseline vs recalibration where $\epsilon^c = \{3, 18\}$

Agent	Baseline		Recalibration			
	FL-CEV	BL-CEV	$\epsilon^c = 3$		$\epsilon^c = 18$	
			FL-CEV	BL-CEV	FL-CEV	BL-CEV
Hand-to-Mouth	1.78%	0.63%	1.52%	0.54%	1.67%	0.75%
Middle-Class	0.69%	0.85%	0.55%	0.74%	0.95%	0.92%
Saver	-4.37%	-3.13%	-3.66%	-2.63%	-4.47%	-3.32%
Aggregate	1.16%	0.46%	0.99%	0.40%	1.13%	0.55%

The FL-CEV compares steady states. The BL-CEV takes into account the whole transition path.

Interestingly, increasing substitutability to $\epsilon^c = 18$ yields ambiguous welfare effects: the aggregate BL-CEV rises, while the aggregate FL-CEV falls. Higher elasticity triggers more aggressive substitution into RFA, worsening bank disintermediation.

This harms the S-agents, whose FL-CEV drops to -4.47%. These agents then sell more housing, driving down prices further, and benefitting M-agents who purchase them thus raising their FL-CEV to 0.95%. These results diverge from BACCHETTA & PERAZZI (2025), who find that welfare gains strictly increase with the elasticity of substitution. Because both models feature a loss in deposit funding that is replaced by borrowing at a higher rate, it is not the modelling of disintermediation which explains the divergence. Rather, it stems from the consequences of the lost bank profits. In BACCHETTA & PERAZZI (2025), the reduction in profits acts as a transfer away from bankers to households, as seigniorage is redistributed. In this THRANK model, however, seigniorage is treated as a wasted resource and the lost bank profits affect S-agents, who are the sole source of aggregate investment. Consequently, higher substitutability leads to a higher output loss which, in this case, outweighs the welfare benefits of RFA. The importance in this result lies in rendering explicit the trade-off of how similar to design RFA compared to deposits.

Finally, we study the welfare-maximising value of ϖ . It must be noted that by varying ϖ , the equal liquidity assumption of RFA is dropped. Figure 5 plots both the FL-

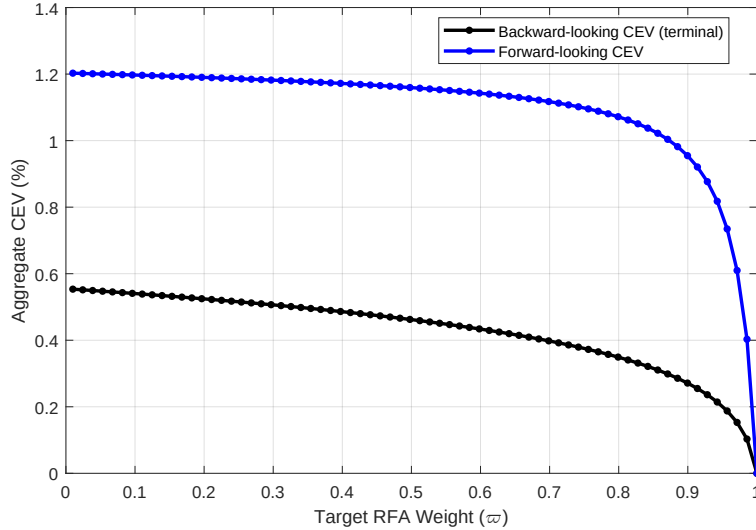


Figure 5: Aggregate welfare measures for varying CES weight of RFA (ϖ)

CEV and terminal BL-CEV against values of ϖ , where we find that the optimal level of ϖ is 0.01. This is the smallest value of ϖ tested, and the one that maximises the amount of RFA in the economy. The CEV values stemming from this optimal quantity are illustrated on Table 5, where we observe amplified gains for H and M agents but deeper losses for S-agents, effectively exacerbating the distributional mechanisms observed in the baseline calibration.

4.5 Tiered Remuneration

As the baseline model predicts significant disintermediation, introducing constraints on RFA holdings provides a relevant experiment to assess potential welfare improvements. This is also more realistic since, as BINDSEIL & SENNER (2024) note, central banks have communicated holding limits as a likely design feature for potential real-life CBDCs. I simulate these by introducing tiered remuneration as in ASSENMACHER

Table 5: Welfare in baseline vs recalibration where $\varpi = 0.01$

Agent	Baseline		Recalibration	
	FL-CEV	BL-CEV	FL-CEV	BL-CEV
H	1.78%	0.63%	1.79%	0.74%
M	0.69%	0.85%	0.88%	0.96%
S	-4.37%	-3.13%	-4.48%	-3.35%
Agg.	1.16%	0.46%	1.20%	0.55%

The FL-CEV compares steady states. The BL-CEV takes into account the whole transition path.

et al. (2024), where the interest rate paid on RFA suffers a negative penalty when RFA holdings exceed a certain threshold $\bar{m}^c$ according to the following rule:⁸

$$i_i^c = i - \frac{\tau}{2} \left((m_i^c - \bar{m}^c) + \sqrt{(m_i^c - \bar{m}^c)^2} \right) \quad (49)$$

where $\tau = 0.05$.

We first set the threshold $\bar{m}^c = 0.70$, calibrating it just high enough to avoid binding the H and M agents. This constraint specifically prevents S-agents from redirecting the majority of the economy’s deposits away from banks. Steady-state results and the full transition paths (in levels) are provided in appendix C.4.

In the steady state, tiered remuneration has a substantial effect on the retention of regular deposits, which are about four times higher than the baseline RFA equilibrium. Thus, banks have less need for central bank lending, which results in an increase of profits of 42.7% relative to baseline. As banks retain a larger share of their profits, S-agents, even though they are more constrained, actually suffer a smaller welfare loss with their FL-CEV equal to -4.19% compared to the previous -4.37% (Table 6). However, as S-agents retain more income, less housing is freed up for the M-agents to purchase. Because of this, M-agents receive a slightly smaller welfare gain relative to the baseline case. The retained income of S-agents also allows them to maintain a higher investment level, which translates into a smaller RFA-driven economic contraction. Overall, aggregate welfare is slightly higher suggesting that mitigating disintermediation through tiered remuneration could work to ease the adverse effects of the RFA policy.

Setting a stricter threshold of $\bar{m}^c = 0.5$ forces the limit to bind for all households, including H and M agents. Under this calibration, aggregate welfare gains drop significantly to a FL-CEV of 0.57%. This decline highlights a policy trade-off: while moderate tiered remuneration mitigates disintermediation in a welfare-improving way, excessively harsh constraints depress overall welfare. Notably, under this strict constraint, the gap between the BL-CEV and FL-CEV narrows compared to the baseline. This is driven primarily by the drop in the FL-CEV rather than a reduction in transition costs. Because the strict threshold now binds the patient H-agents, it directly

⁸This specification approximates $i_i^c = i - \tau \max(0, m_i^c - \bar{m}^c)$ for use in Dynare.

Table 6: Welfare in baseline model vs tiered remuneration extension

Agent	Baseline		Extension			
	FL-CEV	BL-CEV	$\bar{m}^c = 0.70$		$\bar{m}^c = 0.50$	
			FL-CEV	BL-CEV	FL-CEV	BL-CEV
H	1.782%	0.629%	1.786%	0.622%	0.924%	0.459%
M	0.687%	0.854%	0.649%	0.849%	0.432%	0.637%
S	-4.370%	-3.128%	-4.185%	-2.904%	-3.133%	-2.569%
Agg.	1.159%	0.462%	1.165%	0.469%	0.571%	0.324%

The FL-CEV compares steady states. The BL-CEV takes into account the whole transition path.

limits their long-term compounding wealth accumulation. As established previously, the FL-CEV mechanically overweights patient agents due to the infinite-horizon aggregation; therefore, capping their long-term gains disproportionately depresses the forward-looking CEV, artificially narrowing the gap between the two welfare measures.

While successful in mitigating long-run disintermediation, tiered remuneration fails to ease the short-term transition costs. The transmission dynamics of the RFA shock during the first 70 periods remain virtually identical to the baseline model under both calibrations of this extension. Because of this, alternative policies should be explored to specifically target transition costs such as a gradual introduction of RFA. This is left as an area for future research.

5 Limitations

This study has several limitations, primarily stemming from the simplifying assumptions made in the DSGE model and the specific mechanisms it abstracts from.

First, the scope of this study is limited to the investment vehicle aspect of CBDC, dubbed RFA, rather than as a medium of exchange. In the model, regular deposits and RFA are assumed to be equal in their provision of liquidity services within the CES aggregator. Additionally, the model abstracts from many possible real-world benefits of the policy such as reduced reliance on private banks, improved transaction services, anonymous digital payments, and increased bank competition that could lead to new services, innovation, lower or no account holding fees, and possible elimination of hidden fees (RICKS et al. 2020; THE ECONOMIST 2018). It also abstracts from potential macroeconomic advantages like true price stability, helicopter money, and the elimination of the zero lower bound as discussed by BORDO & LEVIN (2017). Some potential real-world disadvantages are not included, notably regarding the effect of RFA on the risk of bank runs (BIDDER et al. 2025).

The modelling of the financial sector presents several limitations. The banking sector's design is inadequate for studying disintermediation in detail, as banks in the model do not adjust quantities. Banks' monopoly power is exogenous, being solely

determined by fixed elasticities such as ϵ^d , unlike the endogenous market power in ASSENMACHER et al. (2024) or CHIU et al. (2023). Because of this, the bank competition channel of introducing RFA is entirely absent in the model, meaning that the markups and markdowns are wholly unaffected by the policy. Banks only adjust their rates according to changes in the marginal cost of funding, which is itself a constant in the steady state. The marginal cost of funding is determined by the central bank lending rate i^f which, in the steady state, is pinned down by i^b , itself being pinned down by β_S . It is for this reason that rates do not change from one steady state to another, and disintermediation only directly affects bank profits. The reason for the reduced amount of lending in the RFA steady state, is thus not due to a reduction in credit supply but of credit demand (M-agents). Nonetheless, this limitation only affects the steady-state analysis since, during the transition, i^f is not pinned down by β_S and is instead determined by monetary policy. Furthermore, the reduction in profits reduces S-agent income, making them decrease aggregate investment in the economy, which in a way indirectly simulates the effects of disintermediation on the supply of funds.

To properly study disintermediation, the model should be extended to feature endogenous market power and direct lending to firms. In such a setup, one might initially expect that increased competition in deposits would push their rate upwards, lower bank margins, and tighten credit supply. As firms would be reliant on loans, this would affect their production capacity directly, leading to a more severe economic contraction. However, both ASSENMACHER et al. (2024) and CHIU et al. (2023) instead find an increase in output in the steady state. This occurs as, despite the adverse mechanisms delineated above, the increased deposit rates actually attract more deposit funding, mitigating disintermediation and ultimately enabling higher lending.

Multiple shortcuts are taken when it comes to modelling Switzerland. I assume a closed economy, which is somewhat unfit for a small open economy like Switzerland. The Swiss National Bank’s monetary policy relies heavily on the exchange rate mechanism, where the interest rate affects the Swiss Franc’s value. By omitting this, the model fails to capture exchange-rate induced asset revaluations that play a role in the monetary policy transmission to Swiss household consumption (BROWN et al. 2025). Furthermore, I do not implement a complete housing market. The supply side of housing is absent, with the assumption that it is in fixed supply. This mechanically exaggerates the price responses and subsequent wealth effects for M and S agents. Moreover, I assume that wealthy households are the sole owners of banks, which concentrates the effect of collapsed bank profits on the top end of the wealth distribution. This is relatively unrealistic for Switzerland as, for instance, cantonal banks (owned by the cantons) represent a considerable proportion of the financial sector. Another assumption lies in the calibration of the policy rate, which is set to be higher than the historical average of the past years. This likely leads to overestimating the wealth-accumulation benefits of RFA for the current zero interest rate environment.

Finally, the absence of a fiscal authority removes government redistribution mechanisms from the economy. The model features no government debt, no taxes, and no transfer systems. Transfers, particularly social security pensions, constitute a significant income share for the bottom of the Swiss income distribution (MARTÍNEZ

2021). By omitting these, I likely underestimate the baseline level of income stability of H-agents, which in this model derive their income solely through wages or deposit interest. Furthermore, any central bank profits (seigniorage) generated in this model are not rebated to households or used in any way. They are assumed to constitute waste. Considering that BACCHETTA & PERAZZI (2025) describes seigniorage as one of three welfare-improving channels of CBDC, this assumption likely results in underestimating the welfare impact of RFA.

6 Conclusion

This paper examines the distributive and welfare implications of introducing “Reserves for All” (RFA), a theoretical policy opening up central bank reserves to everyone and comparable to a remunerated retail CBDC. While there exists a growing amount of CBDC literature modelling its impacts on welfare and macroeconomic aggregates, its effects on wealth and income inequality have remained largely unexplored. To address this gap, this study develops a Three-Agent New Keynesian (THRANK) model calibrated to the Swiss economy. Featuring heterogeneous households segmented mainly by portfolio composition, the model is designed to broadly reflect the Swiss wealth and income distributions. The model additionally includes a financial sector, characterised by imperfectly competitive banks, so as to enable rate spreads and simulate the core mechanisms and consequences of disintermediation.

The quantitative analysis demonstrates that RFA greatly improves the wealth accumulation capacity of poorer agents, leading to a reduction in both income and wealth inequality. The policy additionally causes bank disintermediation, lowering dividend payouts to the wealthy. These two mechanisms explain the decrease in the income Gini by 0.68 pp and the wealth Gini by 3.19 pp. Furthermore, the introduction of RFA provides a small macroeconomic stabilisation effect, lowering the standard deviations of aggregate output, consumption, and inflation.

This redistribution of wealth translates to an unequal impact on welfare, where H and M agents benefit while S-agents suffer a welfare loss. In aggregate, the policy presents clear benefits with welfare gains of 1.16% when comparing steady states. These steady-state gains do not take into account the transition path, found to be costly with all agents facing welfare losses in the short-run caused by a negative demand shock. When taking the full transition path into account, aggregate welfare gains reach 0.46% indicating the policy is still beneficial. Extending the model to include tiered remuneration, a trade-off appears in reducing disintermediation without overly constraining the wealth accumulation potential of RFA. Under a certain threshold calibration, this extension increases the welfare gains of RFA. While successful in mitigating steady-state disintermediation, this extension does not ease the transition costs. Alternative policies such as gradual introduction of RFA could be explored as a solution.

While this framework isolates the role of RFA as an investment vehicle, it abstracts from retail CBDC’s potential to improve payment efficiency and reduce transaction costs. A key limitation of the current model is its simplified banking sector, which abstracts from bank equity, loans to firms, and endogenous rate spreads. Extending this model to include these features would allow to assess whether increased deposit

competition and potential resulting increased lending could offset the distributional costs of disintermediation observed here. Integrating small open-economy dynamics, an endogenous housing supply, and government would additionally provide a more precise representation of the Swiss macroeconomic environment and further clarify the distributive mechanisms of RFA.

These findings imply that retail CBDCs hold more potential than simply serving as digital cash. When remunerated at the policy rate, they fundamentally alter the transmission mechanisms of monetary policy and the wealth accumulation capacity of financially constrained households, and thus lead to welfare improvements.

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A Model Definitions

A.1 Full Variable & Parameter List

Table 7: Endogenous Variables Definitions

Dynare	L ^A T _E X	Description
C_H	C_H	H Consumption
L_H	L_H	H Labour
m_H	m_H	H Bank Deposits
m_cH	m_H^c	H RFA Deposits
m_tilde_H	$\tilde{m}_H$	H Money Bundle
C_M	C_M	M Consumption
L_M	L_M	M Labour
H_M	H_M	M Housing
n_M	n_M	M Mortgage Debt
m_M	m_M	M Bank Deposits
m_cM	m_M^c	M RFA Deposits
m_tilde_M	$\tilde{m}_M$	M Money Bundle
C_S	C_S	S Consumption
H_S	H_S	S Housing
b_S	b_S	S Bonds
K_S	K_S	S Capital
I_S	I_S	S Investment
d_S	d_S	S Dividends
m_S	m_S	S Bank Deposits
m_cS	m_S^c	S RFA Deposits
m_tilde_S	$\tilde{m}_S$	S Money Bundle
C	C	Aggregate Consumption
H	H	Aggregate Housing
d	d	Aggregate Profits
m	m	Aggregate Deposits
m_c	m^c	Aggregate CB Reserves
n	n	Aggregate Mortgages
b	b	Aggregate Bonds
Y	Y	Aggregate Output
K	K	Aggregate Capital
I	I	Aggregate Investment
L	L	Aggregate Labour
mc	mc	Marginal Cost
d_i	d^i	Intermediate Goods Firm Profits
d_k	d^k	Capital Producer Firm Profits
d_b	d^b	Commercial Bank Profits
DELTA	Δ	Price Dispersion

Table 7 – Continued

Dynare	L ^A T _E X	Description
X_1	X_1	Auxiliary Variable 1
X_2	X_2	Auxiliary Variable 2
T_H	T_H	H Union Fees
T_M	T_M	M Union Fees
f	f	Central Bank Lending
m_cB	m_B^c	Central Bank Reserves
G	G	Seigniorage
i_d	i^d	Interest on Bank Deposits
i_b	i^b	Interest on Bonds
i_n	i^n	Interest on Mortgages
i	i	Policy Interest Rate
i_f	i^f	Central Bank Lending Rate
p_tilde	$\tilde{p}$	Reset Price
w	w	Wage
w_H	w_H	H Wage
w_M	w_M	M Wage
s	s	Price of Housing
q	q	Tobins Q (shadow value of installed capital)
r	r	Rental Rate of Capital
pi_var	π	Price Inflation
pi_w_H	π_H^w	H Wage Inflation
pi_w_M	π_M^w	M Wage Inflation
mu	μ	Borrowing Constraint Multiplier
Z	Z	Technology
xi	ξ	Discretionary Monetary Policy
INC_H	INC_H	H Income
NW_H	NW_H	H Net Wealth
INC_M	INC_M	M Income
NW_M	NW_M	M Net Wealth
INC_S	INC_S	S Income
NW_S	NW_S	S Net Wealth
GINI_W	$GINI_W$	Gini Coefficient for Wealth
GINI_I	$GINI_I$	Gini Coefficient for Income
i_r	i^r	Real Policy Rate
SR_H	SR_H	H Savings Rate
SR_M	SR_M	M Savings Rate
SR_S	SR_S	S Savings Rate

Table 8: Exogenous Variables Definitions

Dynare	L ^A T _E X	Description
varepsilon_z	ε_z	Technological Shock
varepsilon_xi	ε_ξ	Monetary Policy Shock

Table 9: Parameter Definitions

Dynare	L ^A T _E X	Description
lambda_H	λ_H	H Population Share
lambda_M	λ_M	M Population Share
lambda_S	λ_S	S Population Share
betta_H	β_H	H Discount Factor
betta_M	β_M	M Discount Factor
betta_S	β_S	S Discount Factor
sigma	σ	Relative Risk Aversion
psi	ψ	Inverse Elasticity of Money Demand
eta	η	Inverse Elasticity of substitution for Housing
varphi	φ	Inverse Frisch Elasticity of Labour Supply
epsilon_c	ϵ^c	Elasticity of Substitution between Deposits and RFA
chi_H	χ_H	H Labour Disutility Parameter
chi_M	χ_M	M Labour Disutility Parameter
PSI	Ψ	Money Utility Parameter
THETA	Θ	Housing Utility Parameter
Gamma	Γ	LTV Ratio
H_bar	$\bar{H}$	Fixed Housing Supply
alppha	α	Capital Share of Production
theta	θ	Calvo Sticky Price Parameter
delta	δ	Capital Depreciation Rate
kappa_y	κ^y	Investment Adjustment Costs Parameter
epsilon_y	ϵ^y	Elasticity of Substitution between Good varieties
kappa_w	κ^w	Wage Adjustment Cost Parameter
epsilon_w	ϵ_w	Elasticity of Substitution between Labour varieties
alpha_L	α_l	H Share in Labour Aggregator
kappa_n	κ^n	Mortgage Rate Adjustment Costs
kappa_d	κ^d	Deposit Rate Adjustment Costs
kappa_b	κ^b	Bond Rate Adjustment Costs
epsilon_n	ϵ^n	Elasticity of Substitution between Mortgage varieties
epsilon_d	ϵ^d	Elasticity of Substitution between Bank Deposit varieties
epsilon_b	ϵ^b	Elasticity of Substitution between Bond varieties
phi_pi	ϕ_π	Taylor Rule Inflation Reaction Parameter
phi_f	ϕ_f	Central Bank Lending Rate Penalty

Table 9 – Continued

Dynare	L ^A T _E X	Description
varpi	ϖ	Money Bundle Weight
omega	ω	Reserve Requirement
rho_i	ρ_i	Taylor Rule Interest Rate Smoothing Parameter
rho_z	ρ_z	Technology Shock Persistence
rho_xi	ρ_ξ	Monetary Policy Shock Persistence
sigma_z	σ_z	Standard Deviation of Technological Shock
sigma_xi	σ_ξ	Standard Deviation of Monetary Policy Shock

A.2 61 Real Equilibrium Equations

Since the model features a nominal trend, the system is rendered stationary by expressing all nominal variables in real terms. Real variables are defined as $x_t \equiv \frac{X_t}{P_t}$ except for the collateral constraint Lagrange multiplier, where $\mu_t \equiv \Omega_{M,t} P_t$

0) Goods Market Clearing (omitted per Walras):

$$\begin{aligned}
P_t Y_t = & P_t C_t + P_t I_t + P_t G_t + \frac{\kappa^y}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 P_t I_t \\
& + \lambda_H P_t T_{H,t} + \lambda_M P_t T_{M,t} + \\
& + \frac{\kappa^n}{2} \left(\frac{i_t^n}{i_{t-1}^n} - 1 \right)^2 i_t^n N_t \\
& + \frac{\kappa^d}{2} \left(\frac{i_t^d}{i_{t-1}^d} - 1 \right)^2 i_t^d M_t \\
& + \frac{\kappa^b}{2} \left(\frac{i_t^b}{i_{t-1}^b} - 1 \right)^2 i_t^b B_t
\end{aligned}$$

1) H Budget Constraint:

$$C_{H,t} + m_{H,t} + m_{H,t}^c + T_{H,t} = w_{H,t} L_{H,t} + \frac{1 + i_{t-1}^d}{1 + \pi_t} m_{H,t-1} + \frac{1 + i_{t-1}^d}{1 + \pi_t} m_{H,t-1}^c$$

2) H Money CES bundle:

$$\widetilde{m}_{H,t} = \left[\varpi^{\frac{1}{\epsilon_c}} (m_{H,t})^{\frac{\epsilon_c-1}{\epsilon_c}} + (1 - \varpi)^{\frac{1}{\epsilon_c}} (m_{H,t}^c)^{\frac{\epsilon_c-1}{\epsilon_c}} \right]^{\frac{\epsilon_c}{\epsilon_c-1}}$$

3) H Deposit Euler:

$$C_{H,t}^{-\sigma} = \Psi \widetilde{m}_{H,t}^{-\psi} \varpi^{\frac{1}{\epsilon_c}} \left(\frac{\widetilde{m}_{H,t}}{m_{H,t}} \right)^{\frac{1}{\epsilon_c}} + \beta_H \mathbb{E}_t \left[\frac{1 + i_t^d}{1 + \pi_{t+1}} C_{H,t+1}^{-\sigma} \right]$$

4) H RFA Deposit Euler:

$$C_{H,t}^{-\sigma} = \Psi \tilde{m}_{H,t}^{-\psi} (1 - \varpi)^{\frac{1}{\epsilon_c}} \left(\frac{\tilde{m}_{H,t}}{m_{H,t}^c} \right)^{\frac{1}{\epsilon_c}} + \beta_H \mathbb{E}_t \left[\frac{1 + i_t}{1 + \pi_{t+1}} C_{H,t+1}^{-\sigma} \right]$$

5) M Budget Constraint:

$$C_{M,t} + s_t H_{M,t} + m_{M,t} + m_{M,t}^c + \frac{1 + i_{t-1}^n}{1 + \pi_t} n_{M,t-1} + T_{M,t} \leq$$

$$w_{M,t} L_{M,t} + s_t H_{M,t-1} + n_{M,t} + \frac{1 + i_{t-1}^d}{1 + \pi_t} m_{M,t-1} + \frac{1 + i_{t-1}}{1 + \pi_t} m_{M,t-1}^c$$

6) M Collateral Constraint:

$$n_{M,t} = \Gamma \mathbb{E}_t \left[s_{t+1} \frac{1 + \pi_{t+1}}{1 + i_t^n} H_{M,t} \right]$$

7) M Housing Euler:

$$s_t C_{M,t}^{-\sigma} = \Theta H_{M,t}^{-\eta} + \beta_M \mathbb{E}_t \left[s_{t+1} C_{M,t+1}^{-\sigma} \right] + \Gamma \mu_t \mathbb{E}_t \left[s_{t+1} \frac{1 + \pi_{t+1}}{1 + i_t^n} \right]$$

8) M Debt Euler:

$$C_{M,t}^{-\sigma} = \mu_t + \beta_M \mathbb{E}_t \left[\frac{1 + i_t^n}{1 + \pi_{t+1}} C_{M,t+1}^{-\sigma} \right]$$

9) M Money CES bundle:

$$\tilde{m}_{M,t} = \left[\varpi^{\frac{1}{\epsilon_c}} (m_{M,t})^{\frac{\epsilon_c - 1}{\epsilon_c}} + (1 - \varpi)^{\frac{1}{\epsilon_c}} (m_{M,t}^c)^{\frac{\epsilon_c - 1}{\epsilon_c}} \right]^{\frac{\epsilon_c}{\epsilon_c - 1}}$$

10) M Deposit Euler:

$$C_{M,t}^{-\sigma} = \Psi \tilde{m}_{M,t}^{-\psi} \varpi^{\frac{1}{\epsilon_c}} \left(\frac{\tilde{m}_{M,t}}{m_{M,t}} \right)^{\frac{1}{\epsilon_c}} + \beta_M \mathbb{E}_t \left[\frac{1 + i_t^d}{1 + \pi_{t+1}} C_{M,t+1}^{-\sigma} \right]$$

11) M RFA Deposit Euler:

$$C_{M,t}^{-\sigma} = \Psi \tilde{m}_{M,t}^{-\psi} (1 - \varpi)^{\frac{1}{\epsilon_c}} \left(\frac{\tilde{m}_{M,t}}{m_{M,t}^c} \right)^{\frac{1}{\epsilon_c}} + \beta_M \mathbb{E}_t \left[\frac{1 + i_t}{1 + \pi_{t+1}} C_{M,t+1}^{-\sigma} \right]$$

12) S Budget Constraint:

$$C_{S,t} + s_t H_{S,t} + q_t I_{S,t} + b_{S,t} + m_{S,t} + m_{S,t}^c \leq$$

$$r_t K_{S,t-1} + d_{S,t} + \frac{1 + i_{t-1}^b}{1 + \pi_t} b_{S,t-1} + s_t H_{S,t-1} + \frac{1 + i_{t-1}^d}{1 + \pi_t} m_{S,t-1} + \frac{1 + i_{t-1}}{1 + \pi_t} m_{S,t-1}^c$$

13) S Capital Accumulation:

$$K_{S,t} = (1 - \delta) K_{S,t-1} + I_{S,t}$$

14) S Housing Euler:

$$s_t C_{S,t}^{-\sigma} = \Theta H_{S,t}^{-\eta} + \beta_S \mathbb{E}_t [s_{t+1} C_{S,t+1}^{-\sigma}]$$

15) S Capital Euler:

$$q_t C_{S,t}^{-\sigma} = \beta_S \mathbb{E}_t [C_{S,t+1}^{-\sigma} (r_{t+1} + (1 - \delta)q_{t+1})]$$

16) S Bond Euler:

$$C_{S,t}^{-\sigma} = \beta_S \mathbb{E}_t \left[\frac{1 + i_t^b}{1 + \pi_{t+1}} C_{S,t+1}^{-\sigma} \right]$$

17) S Money CES bundle:

$$\tilde{m}_{S,t} = \left[\varpi^{\frac{1}{\epsilon_c}} (m_{S,t})^{\frac{\epsilon_c-1}{\epsilon_c}} + (1 - \varpi)^{\frac{1}{\epsilon_c}} (m_{S,t}^c)^{\frac{\epsilon_c-1}{\epsilon_c}} \right]^{\frac{\epsilon_c}{\epsilon_c-1}}$$

18) S Deposit Euler:

$$C_{S,t}^{-\sigma} = \Psi \tilde{m}_{S,t}^{-\psi} \varpi^{\frac{1}{\epsilon_c}} \left(\frac{\tilde{m}_{S,t}}{m_{S,t}} \right)^{\frac{1}{\epsilon_c}} + \beta_S \mathbb{E}_t \left[\frac{1 + i_t^d}{1 + \pi_{t+1}} C_{S,t+1}^{-\sigma} \right]$$

19) S RFA Deposit Euler:

$$C_{S,t}^{-\sigma} = \Psi \tilde{m}_{S,t}^{-\psi} (1 - \varpi)^{\frac{1}{\epsilon_c}} \left(\frac{\tilde{m}_{S,t}}{m_{S,t}^c} \right)^{\frac{1}{\epsilon_c}} + \beta_S \mathbb{E}_t \left[\frac{1 + i_t}{1 + \pi_{t+1}} C_{S,t+1}^{-\sigma} \right]$$

20) H Union Fees:

$$T_H = \frac{\kappa^w}{2} (\pi_{H,t}^w)^2 w_{H,t}$$

21) M Union Fees:

$$T_M = \frac{\kappa^w}{2} (\pi_{M,t}^w)^2 w_{M,t}$$

22) H Wage Inflation:

$$\frac{1 + \pi_{H,t}^w}{1 + \pi_t} = \frac{w_{H,t}}{w_{H,t-1}}$$

23) M Wage Inflation:

$$\frac{1 + \pi_{M,t}^w}{1 + \pi_t} = \frac{w_{M,t}}{w_{M,t-1}}$$

24) H Wage Phillips Curve:

$$\kappa^w \pi_{H,t}^w (1 + \pi_{H,t}^w) = (1 - \epsilon_w) L_{H,t} + \epsilon_w \frac{\chi_H L_{H,t}^{1+\varphi}}{w_{H,t} C_{H,t}^{-\sigma}} + \beta_H \mathbb{E}_t \left[\frac{C_{H,t+1}^{-\sigma}}{C_{H,t}^{-\sigma}} \kappa^w \pi_{H,t+1}^w \frac{(1 + \pi_{H,t+1}^w)^2}{1 + \pi_{t+1}} \right]$$

25) M Wage Phillips Curve:

$$\kappa^w \pi_{M,t}^w (1 + \pi_{M,t}^w) = (1 - \epsilon_w) L_{M,t} + \epsilon_w \frac{\chi_M L_{M,t}^{1+\varphi}}{w_{M,t} C_{M,t}^{-\sigma}} + \beta_M \mathbb{E}_t \left[\frac{C_{M,t+1}^{-\sigma}}{C_{M,t}^{-\sigma}} \kappa^w \pi_{M,t+1}^w \frac{(1 + \pi_{M,t+1}^w)^2}{1 + \pi_{t+1}} \right]$$

26) Goods Producer Profits:

$$d_t^i = Y_t - w_t L_t - r_t K_{t-1}$$

27) Production:

$$Y_t \Delta_t = Z_t K_{t-1}^\alpha L_t^{1-\alpha}$$

28) Aggregate Labour:

$$L_t = (\lambda_H L_{H,t})^{\alpha_l} (\lambda_M L_{M,t})^{1-\alpha_l}$$

29) H Labour Demand:

$$\lambda_H w_{H,t} L_{H,t} = \alpha_l w_t L_t$$

30) M Labour Demand:

$$\lambda_M w_{M,t} L_{M,t} = (1 - \alpha_l) w_t L_t$$

31) Marginal Cost:

$$mc_t = \frac{1}{Z_t} \left(\frac{w_t}{1 - \alpha} \right)^{1-\alpha} \left(\frac{r_t}{\alpha} \right)^\alpha$$

32) Optimal Input Ratio:

$$\frac{r_t}{w_t} = \frac{\alpha}{1 - \alpha} \frac{L_t}{K_{t-1}}$$

33) Auxiliary equation 1:

$$X_{1,t} = C_{S,t}^{-\sigma} Y_t mc_t + \theta \beta_S \mathbb{E}_t \left[(1 + \pi_{t+1})^{\epsilon_y} X_{1,t+1} \right]$$

34) Auxiliary equation 2:

$$X_{2,t} = C_{S,t}^{-\sigma} Y_t + \theta \beta_S \mathbb{E}_t \left[(1 + \pi_{t+1})^{\epsilon_y - 1} X_{2,t+1} \right]$$

35) Optimal Pricing Rule:

$$\tilde{p}_t = \frac{\epsilon_y}{\epsilon_y - 1} \frac{X_{1,t}}{X_{2,t}}$$

36) Aggregate Price Evolution:

$$\tilde{p}_t = \left(\frac{1 - \theta(1 + \pi_t)^{\epsilon_y - 1}}{1 - \theta} \right)^{\frac{1}{1 - \epsilon_y}}$$

37) Price Dispersion Evolution:

$$\Delta_t = (1 - \theta) (\tilde{p}_t)^{-\epsilon_y} + \theta(1 + \pi_t)^{\epsilon_y} \Delta_{t-1}$$

38) Capital Producer Profits:

$$d_t^k = q_t I_t - I_t \left[1 + \frac{\kappa^y}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 \right]$$

39) Tobin's Q:

$$q_t = 1 + \kappa^y \left(\frac{I_t}{I_{t-1}} - 1 \right) \frac{I_t}{I_{t-1}} + \frac{\kappa^y}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 \\ - \mathbb{E}_t \left[\beta_S \frac{C_{S,t+1}^{-\sigma}}{C_{S,t}^{-\sigma}} \kappa^y \left(\frac{I_{t+1}}{I_t} - 1 \right) \left(\frac{I_{t+1}}{I_t} \right)^2 \right]$$

40) Banks Balance Sheet:

$$m_{B,t}^c + n_t = f_t + m_t + b_t$$

41) Banks Reserve Requirement:

$$m_{B,t}^c = \omega m_t$$

42) Banks Profit:

$$d_t^b = \frac{i_{t-1}^n - i_{t-1}^f}{1 + \pi_t} n_{t-1} - \frac{\kappa^n}{2} \left(\frac{i_t^n}{i_{t-1}^n} - 1 \right)^2 i_t^n n_t \\ - \frac{i_{t-1}^d - \omega i_{t-1} - (1 - \omega) i_{t-1}^f}{1 + \pi_t} m_{t-1} - \frac{\kappa^d}{2} \left(\frac{i_t^d}{i_{t-1}^d} - 1 \right)^2 i_t^d m_t \\ - \frac{i_{t-1}^b - i_{t-1}^f}{1 + \pi_t} b_{t-1} - \frac{\kappa^b}{2} \left(\frac{i_t^b}{i_{t-1}^b} - 1 \right)^2 i_t^b b_t$$

43) Mortgage Rate Setting:

$$i_t^n = \frac{\epsilon^n}{\epsilon^n - 1} i_t^f - (1 + i_t^b) \frac{\kappa^n}{\epsilon^n - 1} \left(\frac{i_t^n}{i_{t-1}^n} - 1 \right) \frac{(i_t^n)^2}{i_{t-1}^n} \\ + (1 + i_t^b) \mathbb{E}_t \left[\beta_S \frac{C_{S,t+1}^{-\sigma}}{C_{S,t}^{-\sigma}} \frac{\kappa^n}{\epsilon^n - 1} \left(\frac{i_{t+1}^n}{i_t^n} - 1 \right) \frac{(i_{t+1}^n)^2}{i_t^n} \frac{n_{t+1}}{n_t} \right]$$

44) Deposit Rate Setting:

$$i_t^d = \frac{\epsilon^d}{\epsilon^d - 1} (\omega i_t + (1 - \omega) i_t^f) + (1 + i_t^b) \frac{\kappa^d}{\epsilon^d - 1} \left(\frac{i_t^d}{i_{t-1}^d} - 1 \right) \frac{(i_t^d)^2}{i_{t-1}^d} \\ - (1 + i_t^b) \mathbb{E}_t \left[\beta_S \frac{C_{S,t+1}^{-\sigma}}{C_{S,t}^{-\sigma}} \frac{\kappa^d}{\epsilon^d - 1} \left(\frac{i_{t+1}^d}{i_t^d} - 1 \right) \frac{(i_{t+1}^d)^2}{i_t^d} \frac{m_{t+1}}{m_t} \right]$$

45) Bond Rate Setting:

$$i_t^b = \frac{\epsilon^b}{\epsilon^b - 1} i_t^f + (1 + i_t^b) \frac{\kappa^b}{\epsilon^b - 1} \left(\frac{i_t^b}{i_{t-1}^b} - 1 \right) \frac{(i_t^b)^2}{i_{t-1}^b} \\ - (1 + i_t^b) \mathbb{E}_t \left[\beta_S \frac{C_{S,t+1}^{-\sigma}}{C_{S,t}^{-\sigma}} \frac{\kappa^b}{\epsilon^b - 1} \left(\frac{i_{t+1}^b}{i_t^b} - 1 \right) \frac{(i_{t+1}^b)^2}{i_t^b} \frac{b_{t+1}}{b_t} \right]$$

46) Central Bank Balance Sheet:

$$f_t = \lambda_H m_{H,t}^c + \lambda_M m_{M,t}^c + \lambda_S m_{S,t}^c + m_{B,t}^c$$

47) Central Bank Profits:

$$G_t = \frac{i_{t-1}^f}{1 + \pi_t} f_{t-1} - \frac{i_{t-1}}{1 + \pi_t} (m_{B,t-1}^c + \lambda_H m_{H,t-1}^c + \lambda_M m_{M,t-1}^c + \lambda_S m_{S,t-1}^c)$$

48) Taylor Rule:

$$\ln \left(\frac{1 + i_t}{1 + \bar{i}} \right) = \rho_i \ln \left(\frac{1 + i_{t-1}}{1 + \bar{i}} \right) + (1 - \rho_i) \phi_\pi \ln(1 + \pi_t) + \xi_t$$

49) Lending Rule:

$$i_t^f = \phi_f + i_t$$

50) Total Profits:

$$d_t = d_t^i + d_t^k + d_t^b$$

51) Aggregate Consumption:

$$C_t = \lambda_H C_{H,t} + \lambda_M C_{M,t} + \lambda_S C_{S,t}$$

52) Aggregate Deposits:

$$m_t = \lambda_H m_{H,t} + \lambda_M m_{M,t} + \lambda_S m_{S,t}$$

53) Aggregate Housing:

$$H_t = \lambda_M H_{M,t} + \lambda_S H_{S,t}$$

54) Aggregate Mortgages:

$$n_t = \lambda_M n_{M,t}$$

55) Aggregate Investment:

$$I_t = \lambda_S I_{S,t}$$

56) Aggregate Capital:

$$K_t = \lambda_S K_{S,t}$$

57) Aggregate Bonds:

$$b_t = \lambda_S b_{S,t}$$

58) Aggregate Dividends:

$$d_t = \lambda_S d_{S,t}$$

59) Housing Market Clearing:

$$H_t = \bar{H}$$

60) Technological Process:

$$\ln Z_t = \rho_Z \ln Z_{t-1} + \varepsilon_{z,t}$$

61) Monetary Policy Shock:

$$\xi_t = \rho_\xi \xi_{t-1} + \varepsilon_{\xi,t}$$

A.3 Measurement Variables

Additional to the 61 endogenous variables delineated above, I include a number of “measurement variables” which do not affect the model, but serve only to quantify. These include variables measuring income, net wealth, income and wealth inequality (through Gini coefficients), and welfare (through consumption equivalent variation).

Income is defined as the flow of all revenues including capital gains, minus interest on debt and union fees:

H Income:

$$INC_{H,t} \equiv w_{H,t}L_{H,t} + \frac{i_{t-1}^d}{1 + \pi_t}m_{H,t-1} + \frac{i_{t-1}}{1 + \pi_t}m_{H,t-1}^c - T_{H,t}$$

M Income:

$$INC_{M,t} \equiv w_{M,t}L_{M,t} + \frac{i_{t-1}^d}{1 + \pi_t}m_{M,t-1} + \frac{i_{t-1}}{1 + \pi_t}m_{M,t-1}^c + (s_t - s_{t-1})H_{M,t-1} - \frac{i_{t-1}^n}{1 + \pi_t}n_{M,t-1} - T_{M,t}$$

S Income:

$$INC_{S,t} \equiv \frac{i_{t-1}^d}{1 + \pi_t}m_{S,t-1} + \frac{i_{t-1}}{1 + \pi_t}m_{S,t-1}^c + (s_t - s_{t-1})H_{S,t-1} + (q_t - q_{t-1} + r_t)K_{S,t-1} + \frac{i_{t-1}^b}{1 + \pi_t}b_{S,t-1} + d_{S,t}$$

These income definitions are then used to measure the savings rates as well:

$$SR_{i,t} \equiv \frac{INC_{i,t} - C_{i,t}}{INC_{i,t}}, \quad i \in \{H, M, S\}$$

Net wealth is defined as all financial and physical assets, minus all debt:

H Net Wealth:

$$NW_{H,t} \equiv m_{H,t} + m_{H,t}^c$$

M Net Wealth:

$$NW_{M,t} \equiv m_{M,t} + m_{M,t}^c + s_t H_{M,t} - n_{M,t}$$

S Net Wealth:

$$NW_{S,t} \equiv m_{S,t} + m_{S,t}^c + q_t K_{S,t} + b_{S,t}$$

Inequality is measured using discrete Gini coefficients for income ($GINI_t^I$) and wealth ($GINI_t^W$). Because the three-agent structure renders the distributions discontinuous, the standard Gini formula simplifies to the population-weighted sum of pairwise income or wealth differences, normalised by the aggregate income or wealth. In order to keep this variable differentiable for Dynare, I omit the standard absolute value operators and replace them with a strict rank-ordering assumption where $INC_{H,t} \leq INC_{M,t} \leq INC_{S,t}$ (and identically for net worth). If a simulated shock induces a rank reversal between these classes, the equations become mathematically invalid and will understate inequality. Furthermore, because this discrete approach

isolates between-group inequality and assumes zero within-group variance, the resulting coefficients will likely underestimate true real-world inequality.

$$GINI_t^I \equiv \frac{\lambda_H \lambda_M (INC_{M,t} - INC_{H,t}) + \lambda_S \lambda_H (INC_{S,t} - INC_{H,t}) + \lambda_S \lambda_M (INC_{S,t} - INC_{M,t})}{\lambda_H INC_{H,t} + \lambda_M INC_{M,t} + \lambda_S INC_{S,t}}$$

$$GINI_t^W \equiv \frac{\lambda_H \lambda_M (NW_{M,t} - NW_{H,t}) + \lambda_S \lambda_H (NW_{S,t} - NW_{H,t}) + \lambda_S \lambda_M (NW_{S,t} - NW_{M,t})}{\lambda_H NW_{H,t} + \lambda_M NW_{M,t} + \lambda_S NW_{S,t}}$$

The definition of the backward-looking CEV is taken from AURAY, DEVEREUX et al. (2026). It serves to evaluate welfare along the transition path of one steady state to another. BL-CEV_{i,t} denotes the percent amount of individual's $i \in \{H, M, S\}$ pre-shock consumption that they are willing to give up each period from 0 (the initial steady state) to t to be as well off during the RFA transition. I call it backward-looking, as it evaluates utility from t backwards to 0. As the utility function is additively separable, the individual BL-CEV_{i,t} can be defined as following:

$$\sum_{s=0}^{s=t} \beta_i^s U(C_{i,s}, X_{i,s}) \stackrel{!}{=} \sum_{s=0}^{s=t} \beta_i^s U([1 + \text{BL-CEV}_{i,t}] C_{i,0}, X_{i,0})$$

$$\text{BL-CEV}_{i,t} = \left[1 + \left(\frac{\sum_{s=0}^{s=t} \beta_i^s U(C_{i,s}, X_{i,s})}{\sum_{s=0}^{s=t} \beta_i^s} - U(C_{i,0}, X_{i,0}) \right) \frac{1 - \sigma}{C_{i,0}^{1-\sigma}} \right]^{\frac{1}{1-\sigma}} - 1$$

where X_i represents the other variables in the utility function for the relevant agent, i.e. labour, the money bundle, and housing.

In order to compute the aggregate welfare gain, we define the weighted sum of utilities in proportion to their population shares. The aggregate BL-CEV denotes the uniform percent amount of pre-shock consumption that every agent is willing to give up each period from 0 to t for aggregate utility to remain unchanged. It is defined as:

$$\sum_i \left\{ \lambda_i \sum_{s=0}^{s=t} \beta_i^s U(C_{i,s}, X_{i,s}) \right\} \stackrel{!}{=} \sum_i \left\{ \lambda_i \sum_{s=0}^{s=t} \beta_i^s U([1 + \text{BL-CEV}_t] C_{i,0}, X_{i,0}) \right\}$$

$$\text{BL-CEV}_t = \left[1 + \frac{\sum_i \lambda_i \sum_{s=0}^{s=t} \beta_i^s (U(C_{i,s}, X_{i,s}) - U(C_{i,0}, X_{i,0}))}{\sum_i \lambda_i C_{i,0}^{1-\sigma} / (1 - \sigma) \sum_{s=0}^{s=t} \beta_i^s} \right]^{\frac{1}{1-\sigma}} - 1$$

The definition of the forward-looking CEV is taken from PAUL et al. (2026). It serves to evaluate the welfare gains of one steady state to another. FL-CEV_i denotes the percent amount of consumption required to keep an individual indifferent between an initial scenario (no-RFA steady state, with subscript 0) and a new scenario (post-RFA steady state, with subscript T). I call it forward-looking as it evaluates utility from

$s = 0$ to infinity. It is defined as following:

$$\sum_{s=0}^{s=\infty} \beta_i^s U(C_{i,T}, X_{i,T}) \stackrel{!}{=} \sum_{s=0}^{s=\infty} \beta_i^s U([1 + \text{FL-CEV}_i] C_{i,0}, X_{i,0})$$

$$\text{FL-CEV}_i = \left[1 + \frac{(U(C_{i,T}, X_{i,T}) - U(C_{i,0}, X_{i,0})) (1 - \sigma)}{C_{i,0}^{1-\sigma}} \right]^{\frac{1}{1-\sigma}} - 1$$

The aggregate version follows the same logic described earlier, and is defined as:

$$\sum_i \left\{ \lambda_i \sum_{s=0}^{s=\infty} \beta_i^s U(C_{i,T}, X_{i,T}) \right\} \stackrel{!}{=} \sum_i \left\{ \lambda_i \sum_{s=0}^{s=\infty} \beta_i^s U([1 + \text{FL-CEV}] C_{i,0}, X_{i,0}) \right\}$$

$$\text{FL-CEV} = \left[1 + \frac{\sum_i \frac{\lambda_i}{1-\beta_i} (U(C_{i,T}, X_{i,T}) - U(C_{i,0}, X_{i,0}))}{\sum_i \frac{\lambda_i}{1-\beta_i} C_{i,0}^{1-\sigma} / (1 - \sigma)} \right]^{\frac{1}{1-\sigma}} - 1$$

B Model with Derivations

B.1 Households

B.1.1 H-agents

Nominal Budget Constraint:

$$P_t C_{H,t} + M_{H,t} + M_{H,t}^c + P_t T_{H,t} \leq W_{H,t} L_{H,t} + (1 + i_{t-1}^d) M_{H,t-1} + (1 + i_{t-1}) M_{H,t-1}^c$$

Preferences:

$$U\left(C_{H,t}, \frac{\widetilde{M}_{H,t}}{P_t}, L_{H,t}\right) = \frac{C_{H,t}^{1-\sigma}}{1-\sigma} + \Psi \frac{(\widetilde{M}_{H,t}/P_t)^{1-\psi}}{1-\psi} - \chi_H \frac{L_{H,t}^{1+\varphi}}{1+\varphi}$$

Welfare:

$$V\left(C_{H,t}, \frac{\widetilde{M}_{H,t}}{P_t}, L_{H,t}\right) = \mathbb{E}_0 \sum_{t=0}^{+\infty} \beta_H^t \left(\frac{C_{H,t}^{1-\sigma}}{1-\sigma} + \Psi \frac{(\widetilde{M}_{H,t}/P_t)^{1-\psi}}{1-\psi} - \chi_H \frac{L_{H,t}^{1+\varphi}}{1+\varphi} \right)$$

Bellman:

$$\mathcal{V}(M_{H,t-1}, M_{H,t-1}^c) = \max_{\{C_{H,t}, M_{H,t}, M_{H,t}^c\}} \frac{C_{H,t}^{1-\sigma}}{1-\sigma} + \Psi \frac{(\widetilde{M}_{H,t}/P_t)^{1-\psi}}{1-\psi} - \chi_H \frac{L_{H,t}^{1+\varphi}}{1+\varphi}$$

$$+ \beta_H \mathbb{E}_t \mathcal{V}(M_{H,t}, M_{H,t}^c) + \Lambda_{H,t} [W_{H,t} L_{H,t} + (1 + i_{t-1}^d) M_{H,t-1} + (1 + i_{t-1}) M_{H,t-1}^c - P_t T_{H,t} - P_t C_{H,t} - M_{H,t} - M_{H,t}^c]$$

$$\text{s.t.} \quad \widetilde{M}_{H,t} = \left[\varpi^{\frac{1}{\epsilon_c}} (M_{H,t})^{\frac{\epsilon_c-1}{\epsilon_c}} + (1 - \varpi)^{\frac{1}{\epsilon_c}} (M_{H,t}^c)^{\frac{\epsilon_c-1}{\epsilon_c}} \right]^{\frac{\epsilon_c}{\epsilon_c-1}}$$

FOCs:

$$\begin{aligned}
0 &\stackrel{!}{=} \frac{\partial \mathcal{V}(\widetilde{M}_{H,t-1})}{\partial C_{H,t}} = C_{H,t}^{-\sigma} - P_t \Lambda_{H,t} \\
0 &\stackrel{!}{=} \frac{\partial \mathcal{V}(\widetilde{M}_{H,t-1})}{\partial M_{H,t}} = \Psi \frac{\widetilde{M}_{H,t}^{-\psi}}{P_t^{1-\psi}} \varpi^{\frac{1}{\epsilon_c}} \left(\frac{\widetilde{M}_{H,t}}{M_{H,t}} \right)^{\frac{1}{\epsilon_c}} - \Lambda_{H,t} + \beta_H \mathbb{E}_t [\mathcal{V}_{M_{H,t}}] \\
0 &\stackrel{!}{=} \frac{\partial \mathcal{V}(\widetilde{M}_{H,t-1})}{\partial M_{H,t}^c} = \Psi \frac{\widetilde{M}_{H,t}^{-\psi}}{P_t^{1-\psi}} (1 - \varpi)^{\frac{1}{\epsilon_c}} \left(\frac{\widetilde{M}_{H,t}}{M_{H,t}^c} \right)^{\frac{1}{\epsilon_c}} - \Lambda_{H,t} + \beta_H \mathbb{E}_t [\mathcal{V}_{M_{H,t}^c}]
\end{aligned}$$

where $\mathcal{V}_{X_t} \equiv \frac{\partial \mathcal{V}(X_t)}{X_t}$.

Envelope theorem:

$$\begin{aligned}
\mathcal{V}_{M_{H,t-1}} &= (1 + i_{t-1}^d) \Lambda_{H,t} \implies \mathcal{V}_{M_{H,t}} = (1 + i_t^d) \Lambda_{H,t+1} \\
\mathcal{V}_{M_{H,t-1}^c} &= (1 + i_{t-1}) \Lambda_{H,t} \implies \mathcal{V}_{M_{H,t}^c} = (1 + i_t) \Lambda_{H,t+1}
\end{aligned}$$

Deposit Euler:

$$\begin{aligned}
\frac{C_{H,t}^{-\sigma}}{P_t} &= \Psi \frac{\widetilde{M}_{H,t}^{-\psi}}{P_t^{1-\psi}} \varpi^{\frac{1}{\epsilon_c}} \left(\frac{\widetilde{M}_{H,t}}{M_{H,t}} \right)^{\frac{1}{\epsilon_c}} + \beta_H \mathbb{E}_t \left[(1 + i_t^d) \frac{C_{H,t+1}^{-\sigma}}{P_{t+1}} \right] \\
\frac{C_{H,t}^{-\sigma}}{P_t} &= \Psi \frac{\widetilde{M}_{H,t}^{-\psi}}{P_t^{1-\psi}} (1 - \varpi)^{\frac{1}{\epsilon_c}} \left(\frac{\widetilde{M}_{H,t}}{M_{H,t}^c} \right)^{\frac{1}{\epsilon_c}} + \beta_H \mathbb{E}_t \left[(1 + i_t) \frac{C_{H,t+1}^{-\sigma}}{P_{t+1}} \right]
\end{aligned}$$

B.1.2 M-agents

Nominal Budget Constraint:

$$\begin{aligned}
P_t C_{M,t} + S_t H_{M,t} + M_{M,t} + M_{M,t}^c + (1 + i_{t-1}^n) N_{M,t-1} + P_t T_{M,t} &\leq \\
W_{M,t} L_{M,t} + S_t H_{M,t-1} + N_{M,t} + (1 + i_{t-1}^d) M_{M,t-1} + (1 + i_{t-1}) M_{M,t-1}^c &
\end{aligned}$$

Borrowing Constraint:

$$N_{M,t} \leq \Gamma \mathbb{E}_t \left[\frac{S_{t+1} H_{M,t}}{1 + i_t^n} \right]$$

Preferences:

$$U \left(C_{M,t}, \frac{\widetilde{M}_{M,t}}{P_t}, H_{M,t}, L_{M,t} \right) = \frac{C_{M,t}^{1-\sigma}}{1-\sigma} + \Psi \frac{(\widetilde{M}_{M,t}/P_t)^{1-\psi}}{1-\psi} + \Theta \frac{H_{M,t}^{1-\eta}}{1-\eta} - \chi_M \frac{L_{M,t}^{1+\varphi}}{1+\varphi}$$

Welfare:

$$V \left(C_{M,t}, \frac{\widetilde{M}_{M,t}}{P_t}, H_{M,t}, L_{M,t} \right) = \mathbb{E}_0 \sum_{t=0}^{+\infty} \beta_M^t \left(\frac{C_{M,t}^{1-\sigma}}{1-\sigma} + \Psi \frac{(\widetilde{M}_{M,t}/P_t)^{1-\psi}}{1-\psi} + \Theta \frac{H_{M,t}^{1-\eta}}{1-\eta} - \chi_M \frac{L_{M,t}^{1+\varphi}}{1+\varphi} \right)$$

Bellman (ignoring deposits for readability):

$$\begin{aligned}
\mathcal{V}(H_{M,t-1}, N_{M,t-1}) &= \max_{\{C_{M,t}; H_{M,t}; N_{M,t}\}} \frac{C_{M,t}^{1-\sigma}}{1-\sigma} + \Theta \frac{H_{M,t}^{1-\eta}}{1-\eta} - \chi_M \frac{L_{M,t}^{1+\varphi}}{1+\varphi} \\
&+ \beta_M \mathbb{E}_t \mathcal{V}(H_{M,t}, N_{M,t}) + \Lambda_{M,t} [W_{M,t} L_{M,t} \\
&+ S_t (H_{M,t-1} - H_{M,t}) + N_{M,t} - (1 + i_{t-1}^n) N_{M,t-1} \\
&- P_t T_{M,t} - P_t C_{M,t}] + \Omega_{M,t} \left[\Gamma \mathbb{E}_t \left(\frac{S_{t+1} H_{M,t}}{1 + i_t^n} \right) - N_{M,t} \right]
\end{aligned}$$

FOCs:

$$\begin{aligned}
0 &\stackrel{!}{=} \frac{\partial \mathcal{V}(H_{M,t-1}, N_{M,t-1})}{\partial C_{M,t}} = C_{M,t}^{-\sigma} - P_t \Lambda_{M,t} \\
0 &\stackrel{!}{=} \frac{\partial \mathcal{V}(H_{M,t-1}, N_{M,t-1})}{\partial H_{M,t}} = \Theta H_{M,t}^{-\eta} + \beta_M \mathbb{E}_t [\mathcal{V}_{H_{M,t}}] - S_t \Lambda_{M,t} + \Omega_{M,t} \Gamma \mathbb{E}_t \left[\frac{S_{t+1}}{1 + i_t^n} \right] \\
0 &\stackrel{!}{=} \frac{\partial \mathcal{V}(H_{M,t-1}, N_{M,t-1})}{\partial N_{M,t}} = \Lambda_{M,t} - \Omega_{M,t} + \beta_M \mathbb{E}_t [\mathcal{V}_{N_{M,t}}]
\end{aligned}$$

Housing Envelope Theorem:

$$\mathcal{V}_{H_{M,t-1}} = S_t \Lambda_{M,t} \implies \mathcal{V}_{H_{M,t}} = S_{t+1} \Lambda_{M,t+1}$$

M Housing Euler:

$$S_t \frac{C_{M,t}^{-\sigma}}{P_t} = \Theta H_{M,t}^{-\eta} + \beta_M \mathbb{E}_t \left[S_{t+1} \frac{C_{M,t+1}^{-\sigma}}{P_{t+1}} \right] + \Gamma \Omega_{M,t} \mathbb{E}_t \left[\frac{S_{t+1}}{1 + i_t^n} \right]$$

Debt Envelope Theorem:

$$\mathcal{V}_{N_{M,t-1}} = -(1 + i_{t-1}^n) \Lambda_{M,t} \implies \mathcal{V}_{N_{M,t}} = -(1 + i_t^n) \Lambda_{M,t+1}$$

Debt Euler:

$$\frac{C_{M,t}^{-\sigma}}{P_t} = \Omega_{M,t} + \beta_M \mathbb{E}_t \left[(1 + i_t^n) \frac{C_{M,t+1}^{-\sigma}}{P_{t+1}} \right]$$

M Deposit Eulers:

$$\begin{aligned}
\frac{C_{M,t}^{-\sigma}}{P_t} &= \Psi \frac{\widetilde{M}_{M,t}^{-\psi}}{P_t^{1-\psi}} \varpi^{\frac{1}{\epsilon_c}} \left(\frac{\widetilde{M}_{M,t}}{M_{M,t}} \right)^{\frac{1}{\epsilon_c}} + \beta_M \mathbb{E}_t \left[(1 + i_t^d) \frac{C_{M,t+1}^{-\sigma}}{P_{t+1}} \right] \\
\frac{C_{M,t}^{-\sigma}}{P_t} &= \Psi \frac{\widetilde{M}_{M,t}^{-\psi}}{P_t^{1-\psi}} (1 - \varpi)^{\frac{1}{\epsilon_c}} \left(\frac{\widetilde{M}_{M,t}}{M_{M,t}^c} \right)^{\frac{1}{\epsilon_c}} + \beta_M \mathbb{E}_t \left[(1 + i_t) \frac{C_{M,t+1}^{-\sigma}}{P_{t+1}} \right]
\end{aligned}$$

B.1.3 S-agents

Nominal Budget Constraint:

$$P_t C_{S,t} + S_t H_{S,t} + Q_t I_{S,t} + B_{S,t} + M_{S,t} + M_{S,t}^c \leq$$

$$R_t K_{S,t-1} + D_{S,t} + (1 + i_{t-1}^b) B_{S,t-1} + S_t H_{S,t-1} + (1 + i_{t-1}^d) M_{S,t-1} + (1 + i_{t-1}) M_{S,t-1}^c$$

Capital Accumulation:

$$K_{S,t} = (1 - \delta) K_{S,t-1} + I_{S,t}$$

Preferences:

$$U\left(C_{S,t}, \frac{\widetilde{M}_{S,t}}{P_t}, H_{S,t}\right) = \frac{C_{S,t}^{1-\sigma}}{1-\sigma} + \Psi \frac{(\widetilde{M}_{S,t}/P_t)^{1-\psi}}{1-\psi} + \Theta \frac{H_{S,t}^{1-\eta}}{1-\eta}$$

Welfare

$$V\left(C_{S,t}, \frac{\widetilde{M}_{S,t}}{P_t}, H_{S,t}\right) = \mathbb{E}_0 \sum_{t=0}^{+\infty} \beta_S^t \left(\frac{C_{S,t}^{1-\sigma}}{1-\sigma} + \Psi \frac{(\widetilde{M}_{S,t}/P_t)^{1-\psi}}{1-\psi} + \Theta \frac{H_{S,t}^{1-\eta}}{1-\eta} \right)$$

Bellman (ignoring deposits):

$$\begin{aligned} \mathcal{V}(H_{S,t-1}, K_{S,t-1}, B_{S,t-1}) = & \max_{\{C_{S,t}; H_{S,t}; K_{S,t}; B_{S,t}\}} \frac{C_{S,t}^{1-\sigma}}{1-\sigma} + \Theta \frac{H_{S,t}^{1-\eta}}{1-\eta} + \beta_S \mathbb{E}_t \mathcal{V}(H_{S,t}, K_{S,t}, B_{S,t}) \\ & + \Lambda_{S,t} [R_t K_{S,t-1} + D_{S,t} + (1 + i_{t-1}^b) B_{S,t-1} \\ & + S_t (H_{S,t-1} - H_{S,t}) - P_t C_{S,t} - Q_t (K_{S,t} - (1 - \delta) K_{S,t-1}) - B_{S,t}] \end{aligned}$$

FOCs:

$$\begin{aligned} 0 & \stackrel{!}{=} \frac{\partial \mathcal{V}(H_{S,t-1}, K_{S,t-1}, B_{S,t-1})}{\partial C_{S,t}} = C_{S,t}^{-\sigma} - P_t \Lambda_{S,t} \\ 0 & \stackrel{!}{=} \frac{\partial \mathcal{V}(H_{S,t-1}, K_{S,t-1}, B_{S,t-1})}{\partial H_{S,t}} = \Theta H_{S,t}^{-\eta} + \beta_S \mathbb{E}_t [\mathcal{V}_{H_{S,t}}] - S_t \Lambda_{S,t} \\ 0 & \stackrel{!}{=} \frac{\partial \mathcal{V}(H_{S,t-1}, K_{S,t-1}, B_{S,t-1})}{\partial K_{S,t}} = \beta_S \mathbb{E}_t [\mathcal{V}_{K_{S,t}}] - Q_t \Lambda_{S,t} \\ 0 & \stackrel{!}{=} \frac{\partial \mathcal{V}(H_{S,t-1}, K_{S,t-1}, B_{S,t-1})}{\partial B_{S,t}} = \beta_S \mathbb{E}_t [\mathcal{V}_{B_{S,t}}] - \Lambda_{S,t} \end{aligned}$$

Housing Envelope Theorem:

$$\mathcal{V}_{H_{S,t-1}} = S_t \Lambda_{S,t} \implies \mathcal{V}_{H_{S,t}} = S_{t+1} \Lambda_{S,t+1}$$

S Housing Euler:

$$S_t \frac{C_{S,t}^{-\sigma}}{P_t} = \Theta H_{S,t}^{-\eta} + \beta_S \mathbb{E}_t \left[S_{t+1} \frac{C_{S,t+1}^{-\sigma}}{P_{t+1}} \right]$$

Capital Envelope Theorem:

$$\mathcal{V}_{K_{S,t-1}} = [R_t + (1 - \delta)Q_t] \Lambda_{S,t} \implies \mathcal{V}_{K_{S,t}} = [R_{t+1} + (1 - \delta)Q_{t+1}] \Lambda_{S,t+1}$$

Capital Euler:

$$Q_t \frac{C_{S,t}^{-\sigma}}{P_t} = \beta_S \mathbb{E}_t \left[\frac{C_{S,t+1}^{-\sigma}}{P_{t+1}} (R_{t+1} + (1 - \delta)Q_{t+1}) \right]$$

Bond Envelope Theorem::

$$\mathcal{V}_{B_{S,t-1}} = (1 + i_{t-1}^b) \Lambda_{S,t} \implies \mathcal{V}_{B_{S,t}} = (1 + i_t^b) \Lambda_{S,t+1}$$

Bond Euler:

$$\frac{C_{S,t}^{-\sigma}}{P_t} = \beta_S \mathbb{E}_t \left[(1 + i_t^b) \frac{C_{S,t+1}^{-\sigma}}{P_{t+1}} \right]$$

S Deposit Eulers:

$$\begin{aligned} \frac{C_{S,t}^{-\sigma}}{P_t} &= \Psi \frac{\widetilde{M}_{S,t}^{-\psi}}{P_t^{1-\psi}} \varpi^{\frac{1}{\epsilon_c}} \left(\frac{\widetilde{M}_{S,t}}{M_{S,t}} \right)^{\frac{1}{\epsilon_c}} + \beta_S \mathbb{E}_t \left[(1 + i_t^d) \frac{C_{S,t+1}^{-\sigma}}{P_{t+1}} \right] \\ \frac{C_{S,t}^{-\sigma}}{P_t} &= \Psi \frac{\widetilde{M}_{S,t}^{-\psi}}{P_t^{1-\psi}} (1 - \varpi)^{\frac{1}{\epsilon_c}} \left(\frac{\widetilde{M}_{S,t}}{M_{S,t}^c} \right)^{\frac{1}{\epsilon_c}} + \beta_S \mathbb{E}_t \left[(1 + i_t) \frac{C_{S,t+1}^{-\sigma}}{P_{t+1}} \right] \end{aligned}$$

B.2 Labour Market

B.2.1 Labour Packers

CES Labour for each household group:

$$L_{i,t} = \left[\int_0^1 L_{i,t}(j)^{\frac{\epsilon_w - 1}{\epsilon_w}} dj \right]^{\frac{\epsilon_w}{\epsilon_w - 1}}$$

Labour Packers profit maximisation:

$$\max_{\{L_{i,t}(j)\}} D_t^u = W_{i,t} L_{i,t} - \int_0^1 W_{i,t}(j) L_{i,t}(j) dj$$

FOC:

$$\begin{aligned} 0 &\stackrel{!}{=} \frac{\partial D_t^u}{\partial L_{i,t}(j)} = \frac{\epsilon_w}{\epsilon_w - 1} W_{i,t} \left[\int_0^1 L_{i,t}(j)^{\frac{\epsilon_w - 1}{\epsilon_w}} dj \right]^{\frac{1}{\epsilon_w - 1}} \frac{\epsilon_w - 1}{\epsilon_w} L_{i,t}(j)^{-\frac{1}{\epsilon_w}} - W_{i,t}(j) \\ W_{i,t}^{\epsilon_w} \frac{L_{i,t}}{L_{i,t}(j)} &= W_{i,t}(j)^{\epsilon_w} \\ L_{i,t}(j) &= \left(\frac{W_{i,t}(j)}{W_{i,t}} \right)^{-\epsilon_w} L_{i,t} \end{aligned}$$

B.2.2 Unions

Wage Adjustment Cost:

$$\frac{\kappa^w}{2} \left(\frac{W_{i,t}(j)}{W_{i,t-1}(j)} - 1 \right)^2 W_{i,t} = \frac{\kappa^w}{2} (\pi_{i,t}^w(j))^2 W_{i,t}$$

where $\pi_{i,t}^w \equiv \frac{W_{i,t}}{W_{i,t-1}} - 1$ is the nominal wage inflation for agents i .

Union Profits:

$$\max_{\{W_{i,t}(j)\}} \mathbb{E}_0 \sum_{t=0}^{+\infty} \beta_i^t \left[\Lambda_{i,t} \left(W_{i,t}(j) L_{i,t}(j) - \frac{\kappa^w}{2} (\pi_{i,t}^w(j))^2 W_{i,t} \right) - \chi_i \frac{L_{i,t}^{1+\varphi}(j)}{1+\varphi} \right]$$

Becomes:

$$\begin{aligned} \max_{\{W_{i,t}(j)\}} \mathbb{E}_0 \sum_{t=0}^{+\infty} \beta_i^t \left[\Lambda_{i,t} \left(W_{i,t}(j) \left(\frac{W_{i,t}(j)}{W_{i,t}} \right)^{-\epsilon_w} L_{i,t} - \frac{\kappa^w}{2} (\pi_{i,t}^w(j))^2 W_{i,t} \right) \right. \\ \left. - \frac{\chi_i}{1+\varphi} \left(\frac{W_{i,t}(j)}{W_{i,t}} \right)^{-\epsilon_w(1+\varphi)} L_{i,t}^{1+\varphi} \right] \end{aligned}$$

FOC:

$$\begin{aligned} 0 = (1 - \epsilon_w) L_{i,t}(j) \Lambda_{i,t} + \epsilon_w \frac{\chi_i L_{i,t}^{1+\varphi}(j)}{W_{i,t}(j)} - \kappa^w \left(\frac{W_{i,t}(j)}{W_{i,t-1}(j)} - 1 \right) \frac{W_{i,t}}{W_{i,t-1}(j)} \Lambda_{i,t} \\ + \beta_i \mathbb{E}_t \left[\kappa^w \left(\frac{W_{i,t+1}(j)}{W_{i,t}(j)} - 1 \right) \frac{W_{i,t+1}(j)}{(W_{i,t}(j))^2} W_{i,t+1} \Lambda_{i,t+1} \right] \end{aligned}$$

Wage Phillips Curve:

$$\begin{aligned} \kappa^w \pi_{i,t}^w (1 + \pi_{i,t}^w) = (1 - \epsilon_w) L_{i,t} + \epsilon_w \frac{\chi_i L_{i,t}^{1+\varphi}}{W_{i,t} \Lambda_{i,t}} \\ + \beta_i \mathbb{E}_t \left[\frac{\Lambda_{i,t+1}}{\Lambda_{i,t}} \kappa^w \pi_{i,t+1}^w (1 + \pi_{i,t+1}^w)^2 \right] \end{aligned}$$

B.3 Firms

B.3.1 Final Goods Producers

CES:

$$Y_t = \left[\int_0^1 Y_t(j)^{\frac{\epsilon_y-1}{\epsilon_y}} dj \right]^{\frac{\epsilon_y}{\epsilon_y-1}}$$

Price index:

$$P_t = \left[\int_0^1 P_t(j)^{1-\epsilon_y} dj \right]^{\frac{1}{\epsilon_y-1}}$$

Profits:

$$\max_{\{Y_t(j)\}} D_t^f = P_t Y_t - \int_0^1 P_t(j) Y_t(j) di$$

FOC:

$$\begin{aligned} 0 &\stackrel{!}{=} \frac{\partial D_t^f}{\partial Y_t(j)} = \frac{\epsilon_y}{\epsilon_y - 1} P_t \left[\int_0^1 Y_t(j)^{\frac{\epsilon_y - 1}{\epsilon_y}} di \right]^{\frac{1}{\epsilon_y - 1}} \frac{\epsilon_y - 1}{\epsilon_y} Y_t(j)^{-\frac{1}{\epsilon_y}} - P_t(j) \\ P_t^{\epsilon_y} \frac{Y_t}{Y_t(j)} &= P_t(j)^{\epsilon_y} \\ Y_t(j) &= \left(\frac{P_t(j)}{P_t} \right)^{-\epsilon_y} Y_t \end{aligned}$$

B.3.2 Intermediate Goods Producers

Production:

$$Y_t(j) = Z_t K_{t-1}(j)^\alpha L_t(j)^{1-\alpha}$$

Labour:

$$L_t = (\lambda_H L_{H,t})^{\alpha_l} (\lambda_M L_{M,t})^{1-\alpha_l}$$

Labour Optimisation:

$$\max_{\{L_{H,t}, L_{M,t}\}} W_t L_t - \lambda_H W_{H,t} L_{H,t} - \lambda_M W_{M,t} L_{M,t}$$

which yields labour demand curves for each labour union group:

$$\lambda_H W_{H,t} L_{H,t} = \alpha_l W_t L_t$$

$$\lambda_M W_{M,t} L_{M,t} = (1 - \alpha_l) W_t L_t$$

Cost Minimisation:

$$\min_{\{L_t(j), K_{t-1}(j)\}} W_t L_t(j) + R_t K_{t-1}(j) + MC_t(j) [Y_t(j) - Z_t K_{t-1}(j)^\alpha L_t(j)^{1-\alpha}]$$

Taking the derivatives with respect to labour and capital yields the following first-order conditions:

$$W_t = MC_t(j) (1 - \alpha) \frac{Y_t(j)}{L_t(j)}$$

$$R_t = MC_t(j) \alpha \frac{Y_t(j)}{K_{t-1}(j)}$$

Marginal Cost:

$$MC_t = \frac{1}{Z_t} \left(\frac{W_t}{1 - \alpha} \right)^{1-\alpha} \left(\frac{R_t}{\alpha} \right)^\alpha$$

Optimal Input Ratio:

$$\frac{R_t}{W_t} = \frac{\alpha}{1-\alpha} \frac{L_t}{K_{t-1}}$$

Aggregate price level:

$$P_t = \left[(1-\theta)P_t^{1-\epsilon_y} + \theta P_{t-1}^{1-\epsilon_y} \right]^{\frac{1}{1-\epsilon_y}}$$

Profits:

$$\max_{\{P'_t\}} \mathbb{E}_t \sum_{k=0}^{+\infty} \theta^k \mathcal{M}_{t,t+k} D_{t+k}^i(P'_t)$$

and where future profits can be defined as a function of the new price:

$$D_{t+k}^i(P'_t) = P'_t Y_{t+k}(j) - W_{t+k} L_{t+k}(j) - R_{t+k} K_{t-1+k}(j)$$

$$D_{t+k}^i(P'_t) = (P'_t - MC_{t+k}) Y_{t+k}(j)$$

Plugging this definition gives us the following maximisation problem:

$$\max_{\{P'_t\}} \mathbb{E}_t \sum_{k=0}^{+\infty} \theta^k \mathcal{M}_{t,t+k} \left[(P'_t - MC_{t+k}) \left(\frac{P'_t}{P_{t+k}} \right)^{-\epsilon_y} Y_{t+k} \right]$$

To which we take the first-order condition with respect to the new price:

$$0 \stackrel{!}{=} \frac{\partial}{\partial P'_t} = \mathbb{E}_t \sum_{k=0}^{+\infty} \theta^k \mathcal{M}_{t,t+k} \left[\left(\frac{P'_t}{P_{t+k}} \right)^{-\epsilon_y} Y_{t+k} + (P'_t - MC_{t+k})(-\epsilon_y) P_t'^{-\epsilon_y-1} P_{t+k}^{\epsilon_y} Y_{t+k} \right]$$

which we can rewrite as:

$$\mathbb{E}_t \sum_{k=0}^{+\infty} \theta^k \mathcal{M}_{t,t+k} Y_{t+k}(j) \left[1 - \epsilon_y \frac{P'_t - MC_{t+k}}{P'_t} \right] = 0$$

Further rearranging then gives the following optimal pricing rule:

$$P'_t = \frac{\epsilon_y}{\epsilon_y - 1} \frac{\mathbb{E}_t \sum_{k=0}^{+\infty} \theta^k \mathcal{M}_{t,t+k} Y_{t+k}(j) MC_{t+k}}{\mathbb{E}_t \sum_{k=0}^{+\infty} \theta^k \mathcal{M}_{t,t+k} Y_{t+k}(j)}$$

To make this infinite-horizon problem solvable in standard macroeconomic software like Dynare, we rewrite the optimal pricing rule recursively. We define the optimal relative price $\tilde{p}_t \equiv \frac{P'_t}{P_t}$:

We rewrite:

$$P'_t = \frac{\epsilon_y}{\epsilon_y - 1} \frac{\mathbb{E}_t \sum_{k=0}^{+\infty} (\theta \beta_S)^k C_{S,t+k}^{-\sigma} (P_{t+k})^{\epsilon_y-1} Y_{t+k} MC_{t+k}}{\mathbb{E}_t \sum_{k=0}^{+\infty} (\theta \beta_S)^k C_{S,t+k}^{-\sigma} (P_{t+k})^{\epsilon_y-1} Y_{t+k}}$$

$$\frac{P'_t}{P_t} = \frac{\epsilon_y}{\epsilon_y - 1} \frac{\mathbb{E}_t \sum_{k=0}^{+\infty} (\theta \beta_S)^k C_{S,t+k}^{-\sigma} \left(\frac{P_{t+k}}{P_t} \right)^{\epsilon_y} Y_{t+k} mc_{t+k}}{\mathbb{E}_t \sum_{k=0}^{+\infty} (\theta \beta_S)^k C_{S,t+k}^{-\sigma} \left(\frac{P_{t+k}}{P_t} \right)^{\epsilon_y-1} Y_{t+k}}$$

Note that we converted the marginal cost in real terms defined as $mc_{t+k} \equiv \frac{MC_{t+k}}{P_{t+k}}$ and define the numerator and denominator as the auxiliary variables $X_{1,t}$ and $X_{2,t}$ respectively:

$$X_{1,t} \equiv \mathbb{E}_t \sum_{k=0}^{+\infty} (\theta\beta_S)^k C_{S,t+k}^{-\sigma} \left(\frac{P_{t+k}}{P_t} \right)^{\epsilon_y} Y_{t+k} mc_{t+k}$$

$$X_{2,t} \equiv \mathbb{E}_t \sum_{k=0}^{+\infty} (\theta\beta_S)^k C_{S,t+k}^{-\sigma} \left(\frac{P_{t+k}}{P_t} \right)^{\epsilon_y-1} Y_{t+k}$$

where $X_{1,t}$ captures the expected discounted marginal costs, and $X_{2,t}$ captures expected discounted revenues.

We can then write these as recursive formulations:

$$\begin{aligned} X_{1,t} &= C_{S,t}^{-\sigma} Y_t mc_t + \mathbb{E}_t \sum_{k=1}^{+\infty} (\theta\beta_S)^k C_{S,t+k}^{-\sigma} \left(\frac{P_{t+k}}{P_t} \right)^{\epsilon_y} Y_{t+k} mc_{t+k} \\ &= C_{S,t}^{-\sigma} Y_t mc_t + \theta\beta_S \mathbb{E}_t \left[\left(\frac{P_{t+1}}{P_t} \right)^{\epsilon_y} X_{1,t+1} \right] \\ X_{2,t} &= C_{S,t}^{-\sigma} Y_t + \mathbb{E}_t \sum_{k=1}^{+\infty} (\theta\beta_S)^k C_{S,t+k}^{-\sigma} \left(\frac{P_{t+k}}{P_t} \right)^{\epsilon_y-1} Y_{t+k} \\ &= C_{S,t}^{-\sigma} Y_t + \theta\beta_S \mathbb{E}_t \left[\left(\frac{P_{t+1}}{P_t} \right)^{\epsilon_y-1} X_{2,t+1} \right] \end{aligned}$$

which defines the recursive relationship for the optimal pricing rule:

$$\tilde{p}_t \equiv \frac{P'_t}{P_t} = \frac{\epsilon_y}{\epsilon_y - 1} \frac{X_{1,t}}{X_{2,t}}$$

Aggregate Price Evolution:

$$\begin{aligned} P_t^{1-\epsilon_y} &= (1-\theta)P_t'^{1-\epsilon_y} + \theta P_{t-1}^{1-\epsilon_y} \\ 1 &= (1-\theta) \left(\frac{P'_t}{P_t} \right)^{1-\epsilon_y} + \theta \left(\frac{P_{t-1}}{P_t} \right)^{1-\epsilon_y} \\ 1 &= (1-\theta) \left(\frac{P'_t}{P_t} \right)^{1-\epsilon_y} + \theta (1+\pi_t)^{\epsilon_y-1} \\ \frac{P'_t}{P_t} &= \left(\frac{1-\theta(1+\pi_t)^{\epsilon_y-1}}{1-\theta} \right)^{\frac{1}{1-\epsilon_y}} \end{aligned}$$

Due to price rigidities, different firms charge different prices, leading to inefficient resource allocation. We track this inefficiency using the price dispersion index Δ_t ,

which evolves recursively based on aggregate inflation and the optimal reset price:

$$\begin{aligned}
\Delta_t &= \int_0^1 \left(\frac{P_t(j)}{P_t} \right)^{-\epsilon_y} di \\
\Delta_t &= \int_{\text{new}} \left(\frac{P_t(j)}{P_t} \right)^{-\epsilon_y} di + \int_{\text{old}} \left(\frac{P_{t-1}(j)}{P_t} \right)^{-\epsilon_y} di \\
\Delta_t &= (1 - \theta) \left(\frac{P'_t}{P_t} \right)^{-\epsilon_y} + \int_{\text{old}} \left(\frac{P_{t-1}(j)}{P_{t-1}} \frac{P_{t-1}}{P_t} \right)^{-\epsilon_y} di \\
\Delta_t &= (1 - \theta) \left(\frac{P'_t}{P_t} \right)^{-\epsilon_y} + \theta \left(\frac{P_{t-1}}{P_t} \right)^{-\epsilon_y} \int_0^1 \left(\frac{P_{t-1}(j)}{P_{t-1}} \right)^{-\epsilon_y} di \\
\Delta_t &= (1 - \theta) \left(\frac{P'_t}{P_t} \right)^{-\epsilon_y} + \theta(1 + \pi_t)^{\epsilon_y} \Delta_{t-1}
\end{aligned}$$

B.3.3 Capital Producers

They maximize the discounted flow of present and future profits:

$$\max_{\{I_t\}} \mathbb{E}_t \sum_{k=0}^{+\infty} \mathcal{M}_{t,t+k} D_{t+k}^k$$

where:

$$D_t^k = Q_t I_t - P_t I_t - P_t \frac{\kappa^y}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 I_t$$

Taking the FOC with respect to investment yields:

$$\begin{aligned}
0 \stackrel{!}{=} \frac{\partial}{\partial I_t} &= Q_t - P_t \left[1 + \kappa^y \left(\frac{I_t}{I_{t-1}} - 1 \right) \frac{I_t}{I_{t-1}} + \frac{\kappa^y}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 \right] \\
&+ \mathbb{E}_t \left[\mathcal{M}_{t,t+1} P_{t+1} \kappa^y \left(\frac{I_{t+1}}{I_t} - 1 \right) \left(\frac{I_{t+1}}{I_t} \right)^2 \right]
\end{aligned}$$

which can then be rearranged to obtain the investment demand equation defining the Tobin's Q:

$$\begin{aligned}
\frac{Q_t}{P_t} &= 1 + \kappa^y \left(\frac{I_t}{I_{t-1}} - 1 \right) \frac{I_t}{I_{t-1}} + \frac{\kappa^y}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 \\
&- \mathbb{E}_t \left[\beta_S \frac{C_{S,t+1}^{-\sigma}}{C_{S,t}^{-\sigma}} \kappa^y \left(\frac{I_{t+1}}{I_t} - 1 \right) \left(\frac{I_{t+1}}{I_t} \right)^2 \right]
\end{aligned}$$

B.4 Financial Sector

B.4.1 Financial Packers

Aggregate debt is a CES aggregator:

$$N_t = \left[\int_0^1 N_t(j)^{\frac{\epsilon_{n-1}}{\epsilon_n}} dj \right]^{\frac{\epsilon_n}{\epsilon_{n-1}}}$$

Financial Packer Profit Maximisation:

$$\max_{\{N_t(j)\}} i_t^n N_t - \int_0^1 i_t^n(j) N_t(j) dj$$

FOC:

$$\begin{aligned} 0 &\stackrel{!}{=} \frac{\epsilon^n}{\epsilon^n - 1} i_t^n \left[\int_0^1 N_t(j)^{\frac{\epsilon^n - 1}{\epsilon^n}} dj \right]^{\frac{1}{\epsilon^n - 1}} \frac{\epsilon^n - 1}{\epsilon^n} N_t(j)^{-\frac{1}{\epsilon^n}} - i_t^n(j) \\ (i_t^n)^{\epsilon^n} \frac{N_t}{N_t(j)} &= (i_t^n(j))^{\epsilon^n} \\ N_t(j) &= \left(\frac{i_t^n(j)}{i_t^n} \right)^{-\epsilon^n} N_t \end{aligned}$$

The aggregate money bundle is determined as:

$$M_t = \left[\int_0^1 M_t(j)^{\frac{\epsilon^d - 1}{\epsilon^d}} dj \right]^{\frac{\epsilon^d}{\epsilon^d - 1}}$$

Financial Packer Profit Maximisation:

$$\max_{\{M_t(j)\}} \int_0^1 i_t^d(j) M_t(j) dj - i_t^d M_t$$

Deposit upwards-sloping demand curve:

$$M_t(j) = \left(\frac{i_t^d(j)}{i_t^d} \right)^{-\epsilon^d} M_t$$

Identical for issued bonds:

$$B_t = \left[\int_0^1 B_t(j)^{\frac{\epsilon^b - 1}{\epsilon^b}} dj \right]^{\frac{\epsilon^b}{\epsilon^b - 1}}$$

Bond upwards-sloping demand curve:

$$B_t(j) = \left(\frac{i_t^b(j)}{i_t^b} \right)^{-\epsilon^b} B_t$$

B.4.2 Commercial Banks

Bank Balance Sheet:

$$M_{B,t}^c(j) + N_t(j) = F_t(j) + M_t(j) + B_t(j)$$

Reserve Requirement:

$$M_{B,t}^c(j) = \omega M_t(j)$$

The bank's profit function is:

$$D_t^b(j) = i_{t-1} M_{B,t-1}^c(j) + i_{t-1}^n(j) N_{t-1}(j) - i_{t-1}^d(j) M_{t-1}(j) - i_{t-1}^b(j) B_{t-1}(j) - i_{t-1}^f F_{t-1}(j) \\ - \frac{\kappa^n}{2} \left(\frac{i_t^n(j)}{i_{t-1}^n(j)} - 1 \right)^2 i_t^n N_t - \frac{\kappa^d}{2} \left(\frac{i_t^d(j)}{i_{t-1}^d(j)} - 1 \right)^2 i_t^d M_t - \frac{\kappa^b}{2} \left(\frac{i_t^b(j)}{i_{t-1}^b(j)} - 1 \right)^2 i_t^b B_t$$

We first proceed with substituting both the balance sheet and the reserve requirement in the profit function:

$$D_t^b(j) = [i_{t-1}^n(j) - i_{t-1}^f] N_{t-1}(j) - \frac{\kappa^n}{2} \left(\frac{i_t^n(j)}{i_{t-1}^n(j)} - 1 \right)^2 i_t^n N_t \\ - [i_{t-1}^d(j) - \omega i_{t-1} - (1 - \omega) i_{t-1}^f] M_{t-1}(j) - \frac{\kappa^d}{2} \left(\frac{i_t^d(j)}{i_{t-1}^d(j)} - 1 \right)^2 i_t^d M_t \\ - [i_{t-1}^b - i_{t-1}^f] B_{t-1}(j) - \frac{\kappa^b}{2} \left(\frac{i_t^b(j)}{i_{t-1}^b(j)} - 1 \right)^2 i_t^b B_t$$

They maximise their discounted future earnings with respect to the interest rate on mortgages and deposits subject to the demand functions:

$$\max_{\{i_t^n(j), i_t^d(j), i_t^b(j)\}} \mathbb{E}_t \sum_{k=0}^{+\infty} \mathcal{M}_{t,t+k} D_{t+k}^b(j)$$

Taking the first-order conditions with respect to the mortgage rate obtain:

$$0 \stackrel{!}{=} \frac{\partial}{\partial i_t^n(j)} = - \left(\frac{i_t^n(j)}{i_{t-1}^n(j)} - 1 \right) \frac{\kappa^n i_t^n N_t}{i_{t-1}^n(j)} \\ + \mathbb{E}_t \left\{ \mathcal{M}_{t,t+1} \left[N_t(j) - [i_t^n(j) - i_t^f] \frac{\epsilon^n N_t(j)}{i_t^n(j)} \right. \right. \\ \left. \left. + \left(\frac{i_{t+1}^n(j)}{i_t^n(j)} - 1 \right) \left(\frac{i_{t+1}^n(j)}{[i_t^n(j)]^2} \right) \kappa^n i_{t+1}^n N_{t+1} \right] \right\}$$

which after setting a symmetric equilibrium where $i_t^n(j) = i_t^n$ and $N_t(j) = N_t \forall j$, rearranges into:

$$i_t^n = \frac{\epsilon^n}{\epsilon^n - 1} i_t^f - \frac{1}{\mathbb{E}_t[\mathcal{M}_{t,t+1}]} \frac{\kappa^n}{\epsilon^n - 1} \left(\frac{i_t^n}{i_{t-1}^n} - 1 \right) \frac{(i_t^n)^2}{i_{t-1}^n} \\ + \frac{1}{\mathbb{E}_t[\mathcal{M}_{t,t+1}]} \mathbb{E}_t \left[\mathcal{M}_{t,t+1} \frac{\kappa^n}{\epsilon^n - 1} \left(\frac{i_{t+1}^n}{i_t^n} - 1 \right) \frac{(i_{t+1}^n)^2}{i_t^n} \frac{N_{t+1}}{N_t} \right]$$

We can further replace the stochastic discount factor using the bond Euler equation (16) which states that $\frac{1}{\mathbb{E}_t[\mathcal{M}_{t,t+1}]} = (1 + i_t^b)$:

$$i_t^n = \frac{\epsilon^n}{\epsilon^n - 1} i_t^f - (1 + i_t^b) \frac{\kappa^n}{\epsilon^n - 1} \left(\frac{i_t^n}{i_{t-1}^n} - 1 \right) \frac{(i_t^n)^2}{i_{t-1}^n} \\ + (1 + i_t^b) \mathbb{E}_t \left[\mathcal{M}_{t,t+1} \frac{\kappa^n}{\epsilon^n - 1} \left(\frac{i_{t+1}^n}{i_t^n} - 1 \right) \frac{(i_{t+1}^n)^2}{i_t^n} \frac{N_{t+1}}{N_t} \right]$$

The FOC for the bond interest is quasi-identical, though with a flipped sign:

$$i_t^b = \frac{\epsilon^b}{\epsilon^b - 1} i_t^f + (1 + i_t^b) \frac{\kappa^b}{\epsilon^b - 1} \left(\frac{i_t^b}{i_{t-1}^b} - 1 \right) \frac{(i_t^b)^2}{i_{t-1}^b} \\ - (1 + i_t^b) \mathbb{E}_t \left[\mathcal{M}_{t,t+1} \frac{\kappa^b}{\epsilon^b - 1} \left(\frac{i_{t+1}^b}{i_t^b} - 1 \right) \frac{(i_{t+1}^b)^2}{i_t^b} \frac{B_{t+1}}{B_t} \right]$$

The FOC for the deposit interest just expands the marginal funding costs to take into account the reserve requirement:

$$i_t^d = \frac{\epsilon^d}{\epsilon^d - 1} (\omega i_t + (1 - \omega) i_t^f) + (1 + i_t^d) \frac{\kappa^d}{\epsilon^d - 1} \left(\frac{i_t^d}{i_{t-1}^d} - 1 \right) \frac{(i_t^d)^2}{i_{t-1}^d} \\ - (1 + i_t^d) \mathbb{E}_t \left[\mathcal{M}_{t,t+1} \frac{\kappa^d}{\epsilon^d - 1} \left(\frac{i_{t+1}^d}{i_t^d} - 1 \right) \frac{(i_{t+1}^d)^2}{i_t^d} \frac{M_{t+1}}{M_t} \right]$$

B.5 Aggregation

Aggregate macroeconomic variables are simply the weighted sum of the respective individual household variables, scaled by their population shares λ_H , λ_M and λ_S . This goes for the following variables:

$$\text{Consumption: } C_t = \lambda_H C_{H,t} + \lambda_M C_{M,t} + \lambda_S C_{S,t}$$

$$\text{Deposits: } M_t = \lambda_H M_{H,t} + \lambda_M M_{M,t} + \lambda_S M_{S,t}$$

$$\text{Housing: } H_t = \lambda_M H_{M,t} + \lambda_S H_{S,t}$$

$$\text{Mortgages: } N_t = \lambda_M N_{M,t}$$

$$\text{Investment: } I_t = \lambda_S I_{S,t}$$

$$\text{Capital: } K_t = \lambda_S K_{S,t}$$

$$\text{Bonds: } B_t = \lambda_S B_{S,t}$$

$$\text{Dividends: } D_t = \lambda_S D_{S,t}$$

C Full Results

C.1 Baseline

This section contains detailed results for the baseline analysis. Table 10 contains all steady state values for the no-RFA equilibrium (“initial SS”) as well as for the RFA equilibrium (“final SS”). Figure 6 includes the transition paths of key variables in levels, with dotted lines representing the steady-state values.

Table 10: Baseline Steady States

Variable	Initial SS	Final SS
$C_{H,M,S}$	{0.1656, 0.3602, 1.0174}	{0.1668, 0.3603, 0.9718}
$L_{H,M}$	{0.3333, 0.3333}	{0.3307, 0.3324}
$m_{H,M,S}$	{0.4179, 0.5981, 2.5679}	{0.0104, 0.0977, 0.0607}
$m_{H,M,S}^c$	{0, 0, 0.0002}	{0.6791, 0.6275, 3.956}
$H_{M,S}$	{1.3371, 4.6517}	{1.3659, 4.5366}
$w_{H,M}$	{0.495, 1.1492}	{0.4967, 1.1471}
$INC_{H,M,S}$	{0.1656, 0.3602, 1.9525}	{0.1668, 0.3603, 1.9029}
$NW_{H,M,S}$	{0.418, 1.9895, 61.4537}	{0.6895, 2.1029, 58.432}
C	0.3286	0.3247
Y	0.4222	0.4203
K	3.7406	3.7242
I	0.0935	0.0931
L	0.1441	0.1435
H	1	1
n	1.267	1.2546
s	3.4096	3.305
b	0.562	0.218
d	0.0754	0.0719
w	1.6351	1.6351
m	0.705	0.0504
m^c	0.0177	0.9874
mc	0.8333	0.8333
d^i	0.0704	0.0701
d^b	0.005	0.0018
f	0.0177	0.9874
m_B^c	0.0176	0.0013
G	0	0.0025
q	1	1
n_M	3.1675	3.1364
b_S	5.6197	2.1803
K_S	37.4058	37.2417
I_S	0.9351	0.931
d_S	0.754	0.719
i	1.495%	1.495%
i^d	0.57%	0.57%
i^b	2.4145%	2.4145%
i^n	2.994%	2.994%
i^f	2.495%	2.495%
r	0.031	0.031
π	0	0
μ	0.012	0.012

Table 10 – Continued

Variable	Initial SS	Final SS
$GINI_W$	0.8034	0.7715
$GINI_I$	0.4548	0.448

Note: “Initial SS” refers to the no-RFA steady-state equilibrium, while “Final SS” refers to the RFA steady-state equilibrium. All values have been rounded to the fourth decimal and interest rates have been annualised.

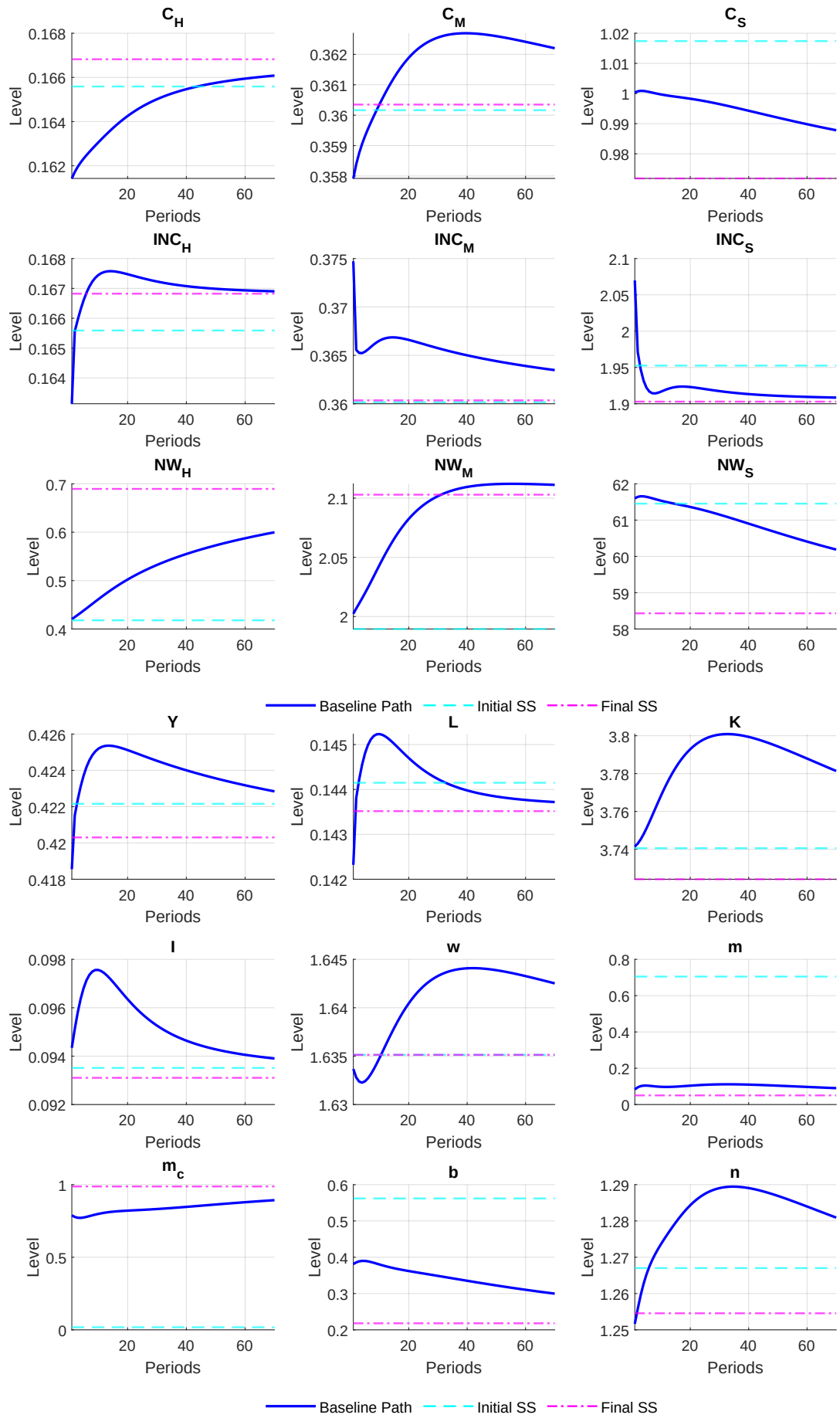
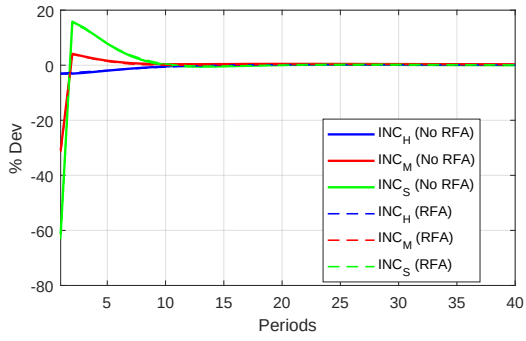


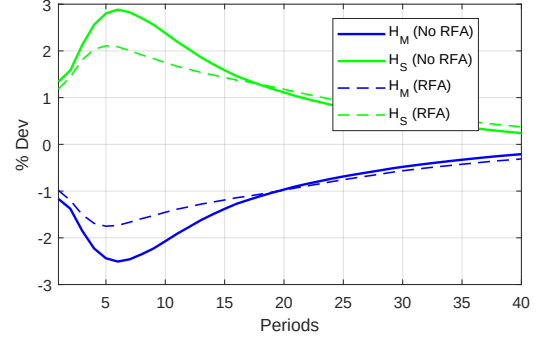
Figure 6: Transition in levels and steady states.

C.2 Stochastic IRFs

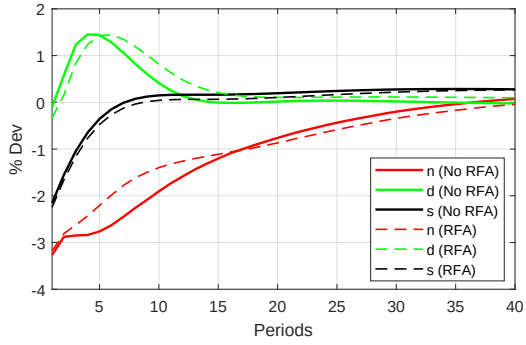
This section provides supplementary stochastic impulse response functions to both the contractionary monetary policy shock and the expansionary technology shock. Figure 7 expands upon the monetary policy analysis in the main text (Figure 4) by presenting the responses of additional variables. Figure 8 displays the complete set of IRFs following a 1% expansionary technology shock. To generate these responses, the model is solved and simulated in Dynare using a second-order Taylor approximation around the deterministic steady state. Because standard second-order perturbations can create explosive sample paths, we include pruning in the Dynare command. The stochastic moments and IRFs are computed by simulating the model for 10'000 periods across 1'000 replications plotted over a 40-period horizon.



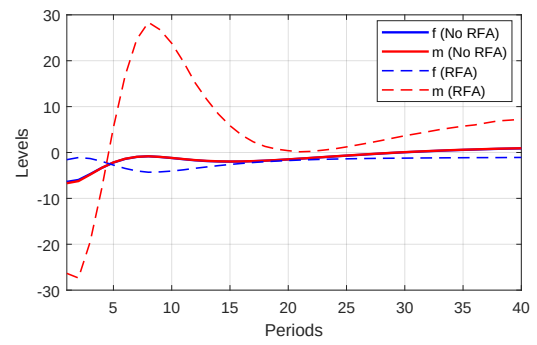
(a) Individual Incomes



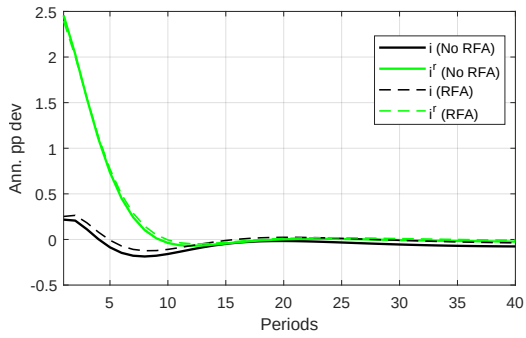
(b) Individual Housing



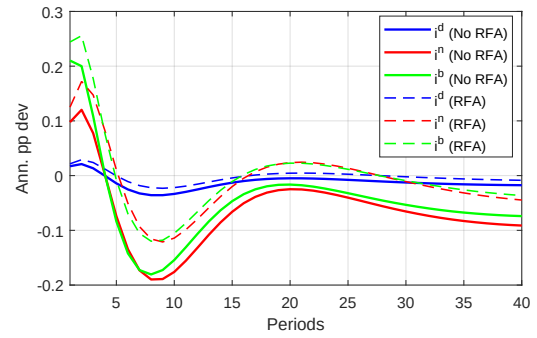
(c) Mortgages, Bonds, & House Prices



(d) CB Lending & Total Deposits

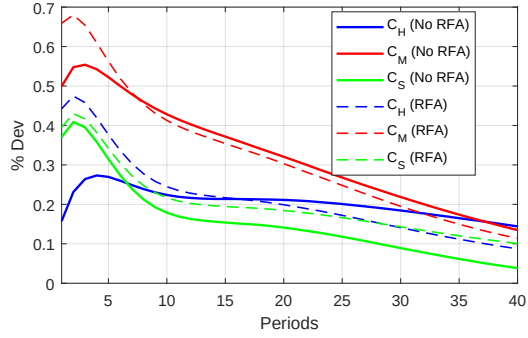


(e) Policy Rates

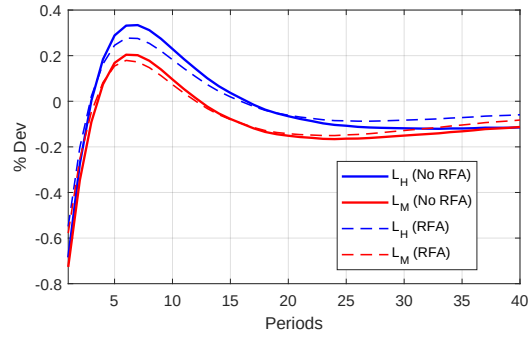


(f) Market Rates

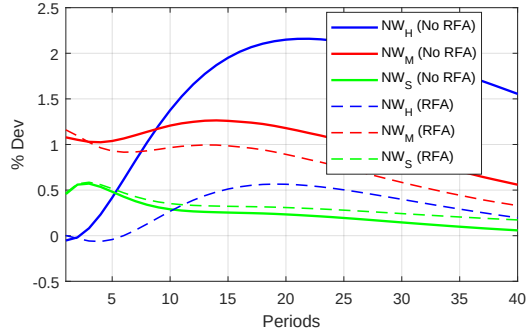
Figure 7: Additional impulse responses to a 1% annualised contractionary monetary policy shock. Time periods are measured in quarters. Variables are measured as levels or percentage deviations from the steady state, except for rates which are expressed through annualised percentage point deviations.



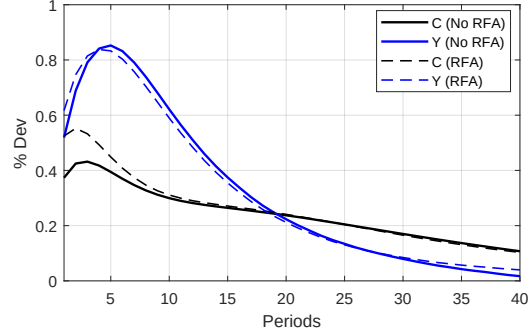
(a) Individual Consumption



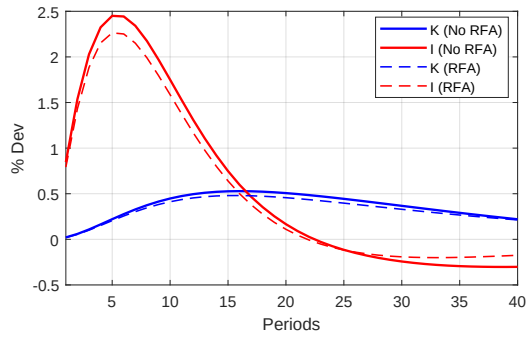
(b) Individual Labour



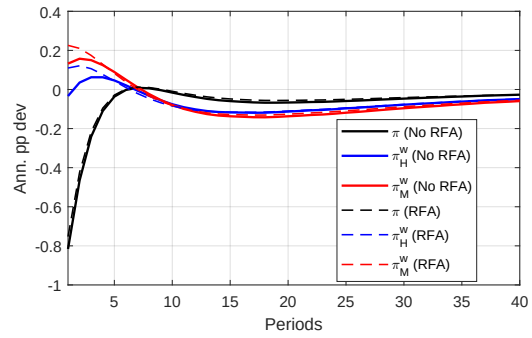
(c) Individual Wealth



(d) Consumption & Output

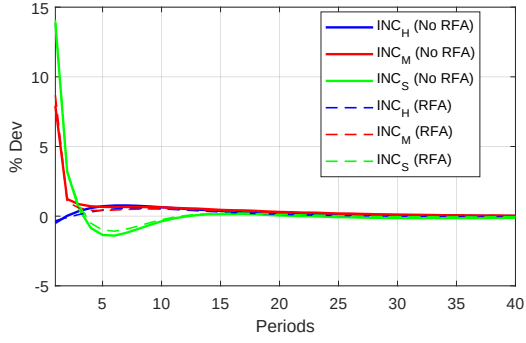


(e) Capital & Investment

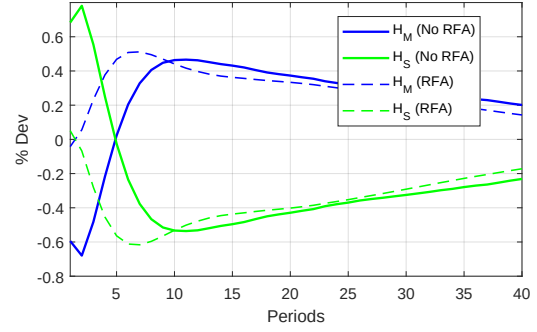


(f) Inflation

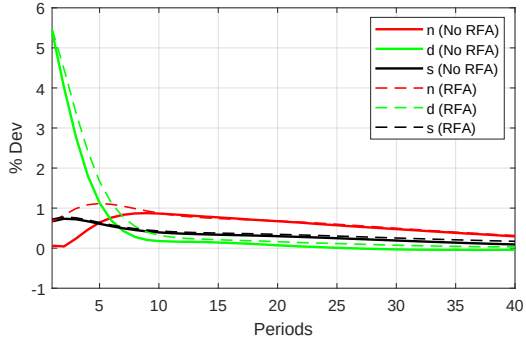
Figure 8: Impulse responses to a 1% expansionary stochastic technology shock. Time periods are measured in quarters. Variables are measured as percentage deviations from the steady state, except for inflation which is expressed through annualised percentage point deviations.



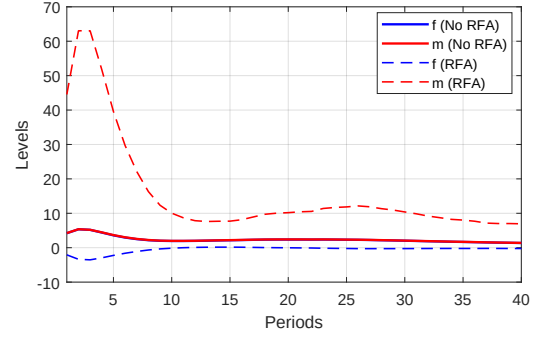
(a) Individual Incomes



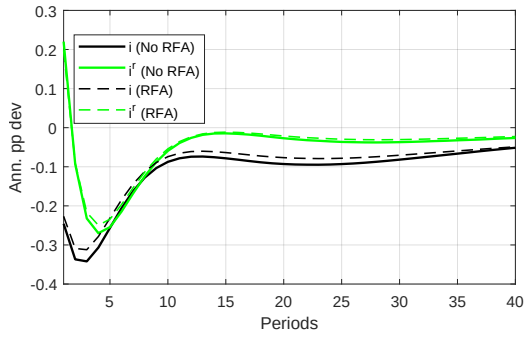
(b) Individual Housing



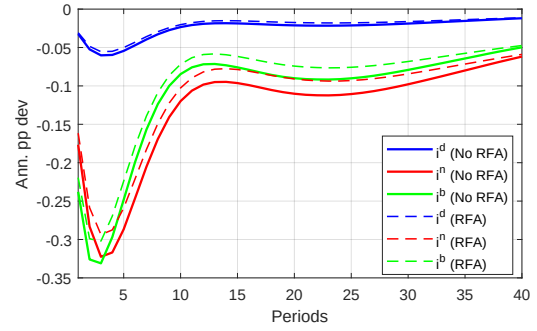
(c) Mortgages, Bonds, & House Prices



(d) CB Lending & Total Deposits



(e) Policy Rates



(f) Market Rates

Figure 9: Additional impulse responses to a 1% expansionary stochastic technology shock. Time periods are measured in quarters. Variables are measured as levels or percentage deviations from the steady state, except for rates which are expressed through annualised percentage point deviations.

C.3 Recalibration

Here we supplement section 4.4 by plotting the transition paths (in levels) of variables for each recalibration. These paths trace the economy's evolution from the initial no-RFA steady state to the final RFA steady state for the baseline calibration (blue) and the alternative calibration (red). We analyse the following parameter changes: $\beta_H = 0.994 \rightarrow \beta_H = 0.99$; $\epsilon^c = 6 \rightarrow \{3, 18\}$; $\varpi = 0.5 \rightarrow 0.01$; $\omega = 0.025 \rightarrow 0.25$; and $\{\sigma, \eta, \psi\} = 1.5 \rightarrow 1.0001$.

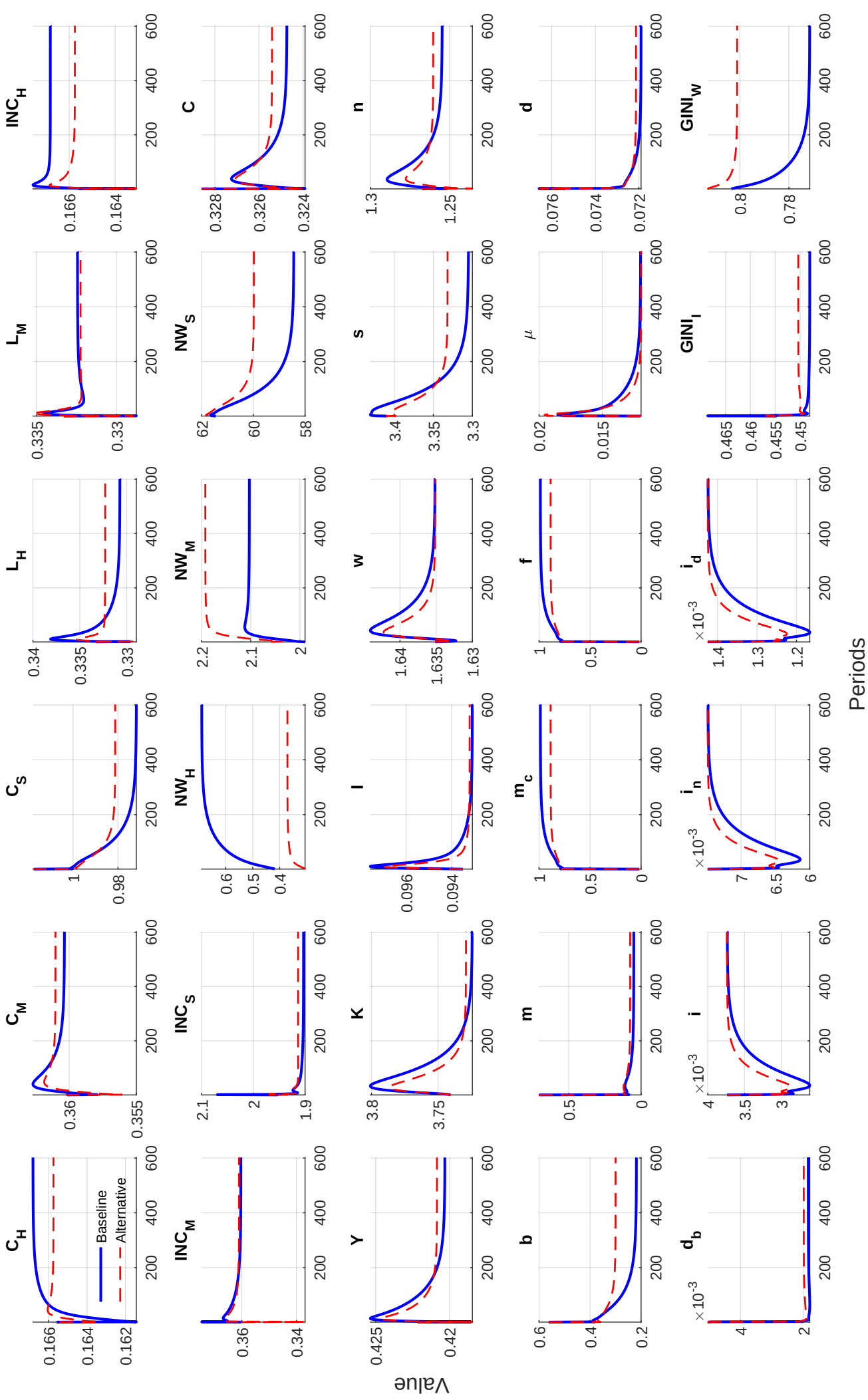


Figure 10: Full transition path (in levels) after the introduction of RFA comparing the baseline calibration of the H-agent discount factor ($\beta_H = 0.994$) against the alternative where $\beta_H = 0.99$.

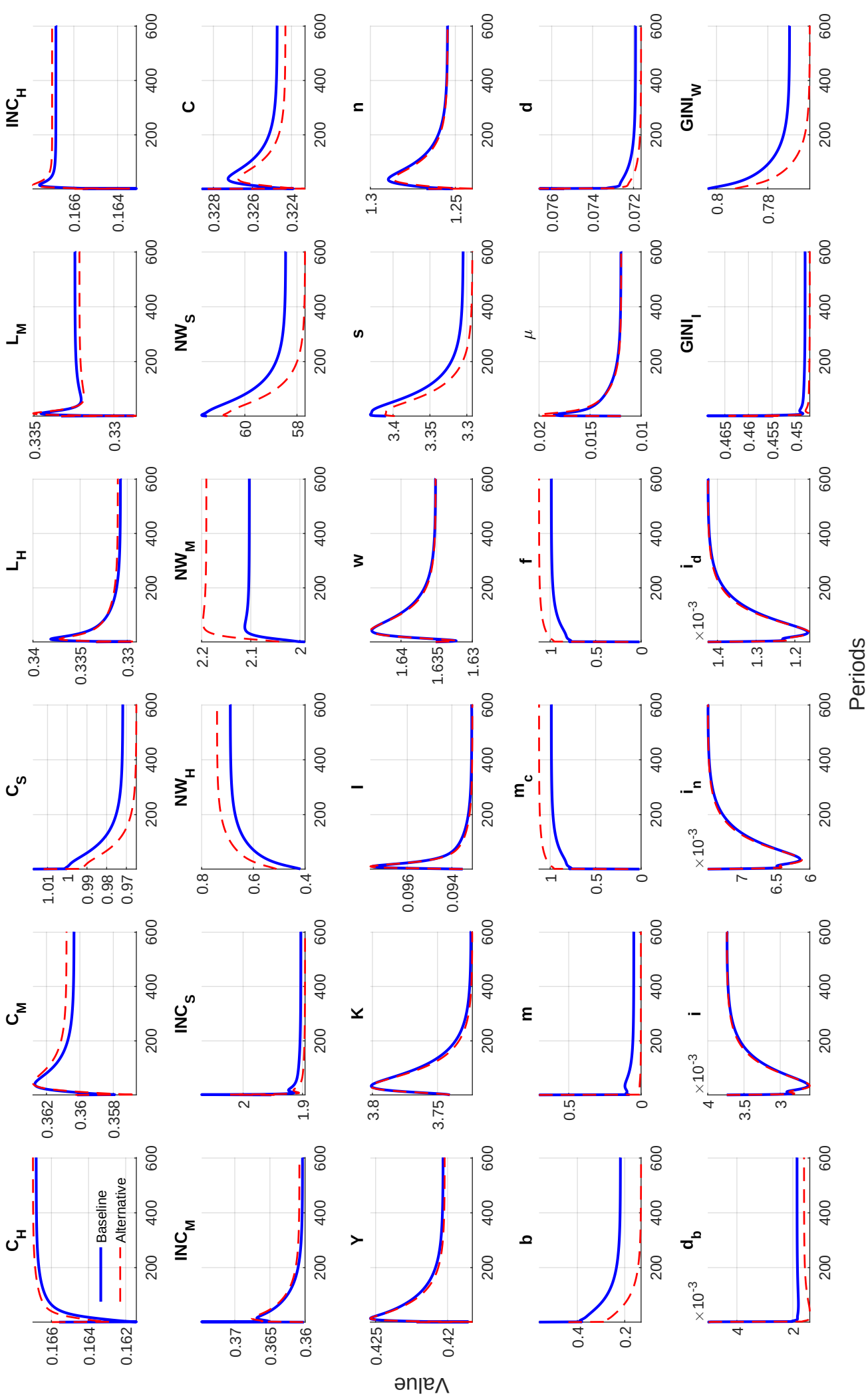


Figure 11: Full transition path (in levels) after the introduction of RFA comparing the baseline calibration of the elasticity of substitution between deposits and RFA ($\epsilon^c = 6$) against the alternative where $\epsilon^c = 18$.

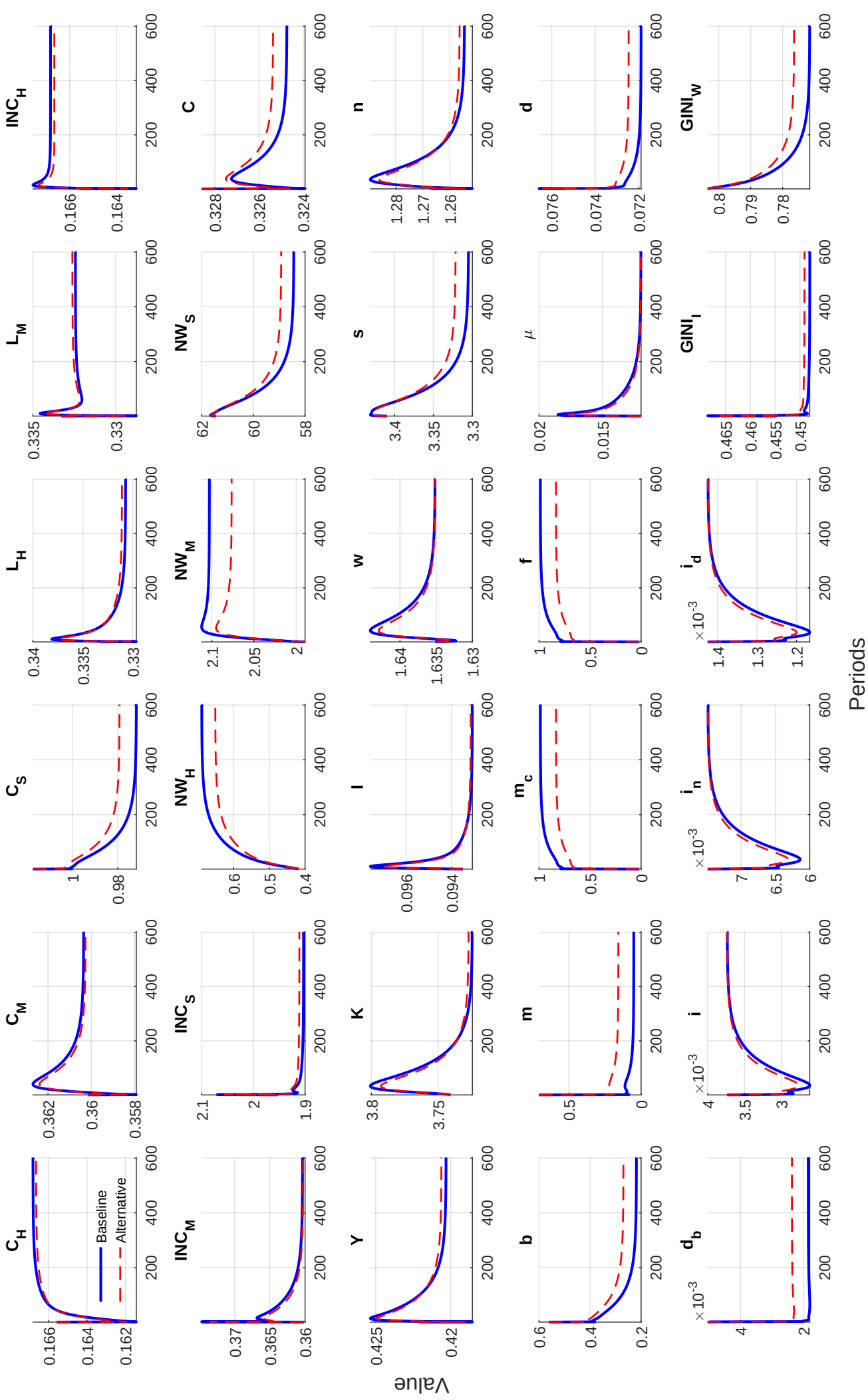


Figure 12: Full transition path (in levels) after the introduction of RFA comparing the baseline calibration of the elasticity of substitution between deposits and RFA ($\epsilon^c = 6$) against the alternative where $\epsilon^c = 3$.

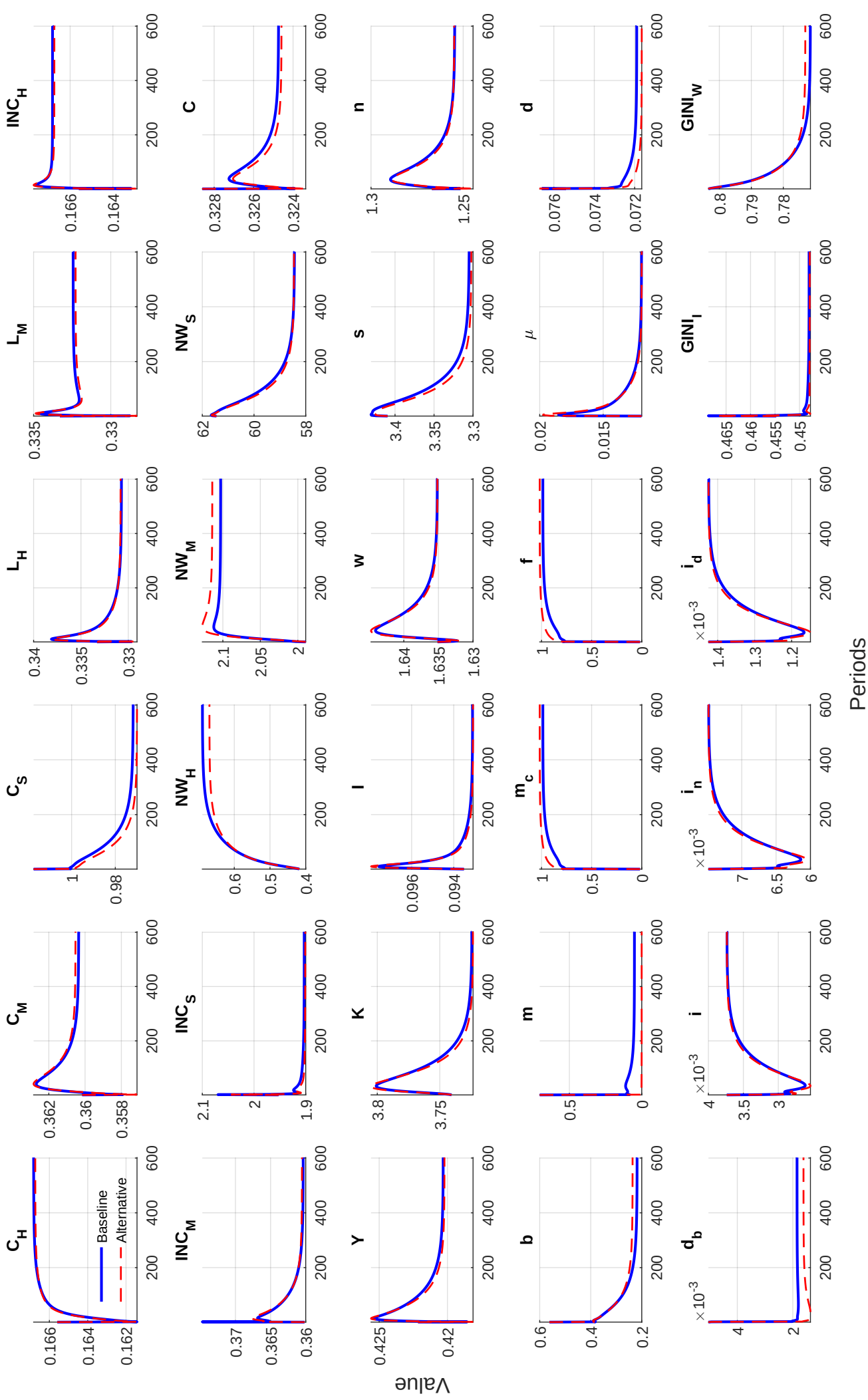


Figure 13: Full transition path (in levels) after the introduction of RFA comparing the baseline calibration of the final CES deposit share ($\varpi = 0.50$) against the alternative where $\varpi = 0.01$.

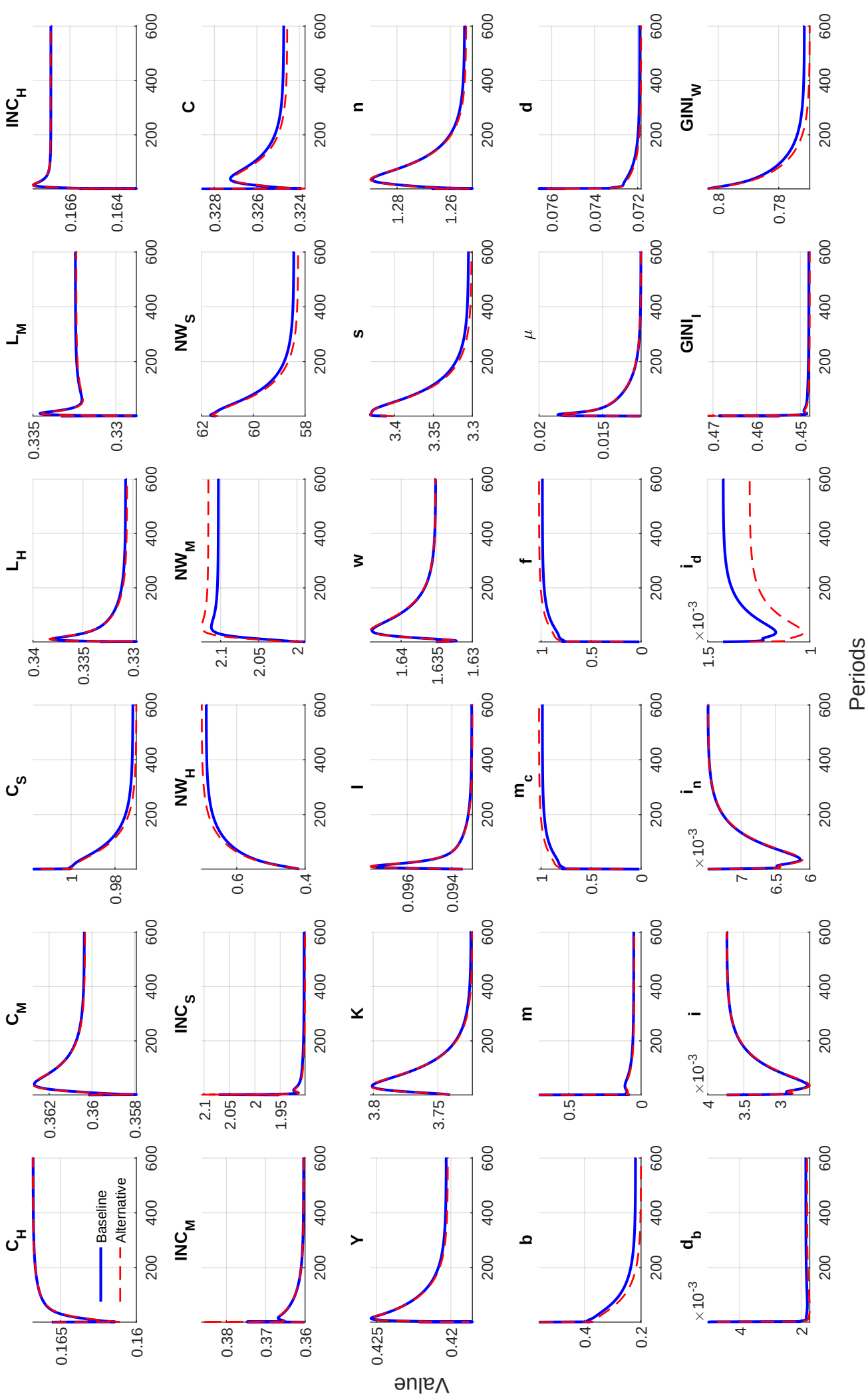


Figure 14: Full transition path (in levels) after the introduction of RFA comparing the baseline calibration of the reserve requirement ($\omega = 0.025$) against the alternative where $\omega = 0.25$.

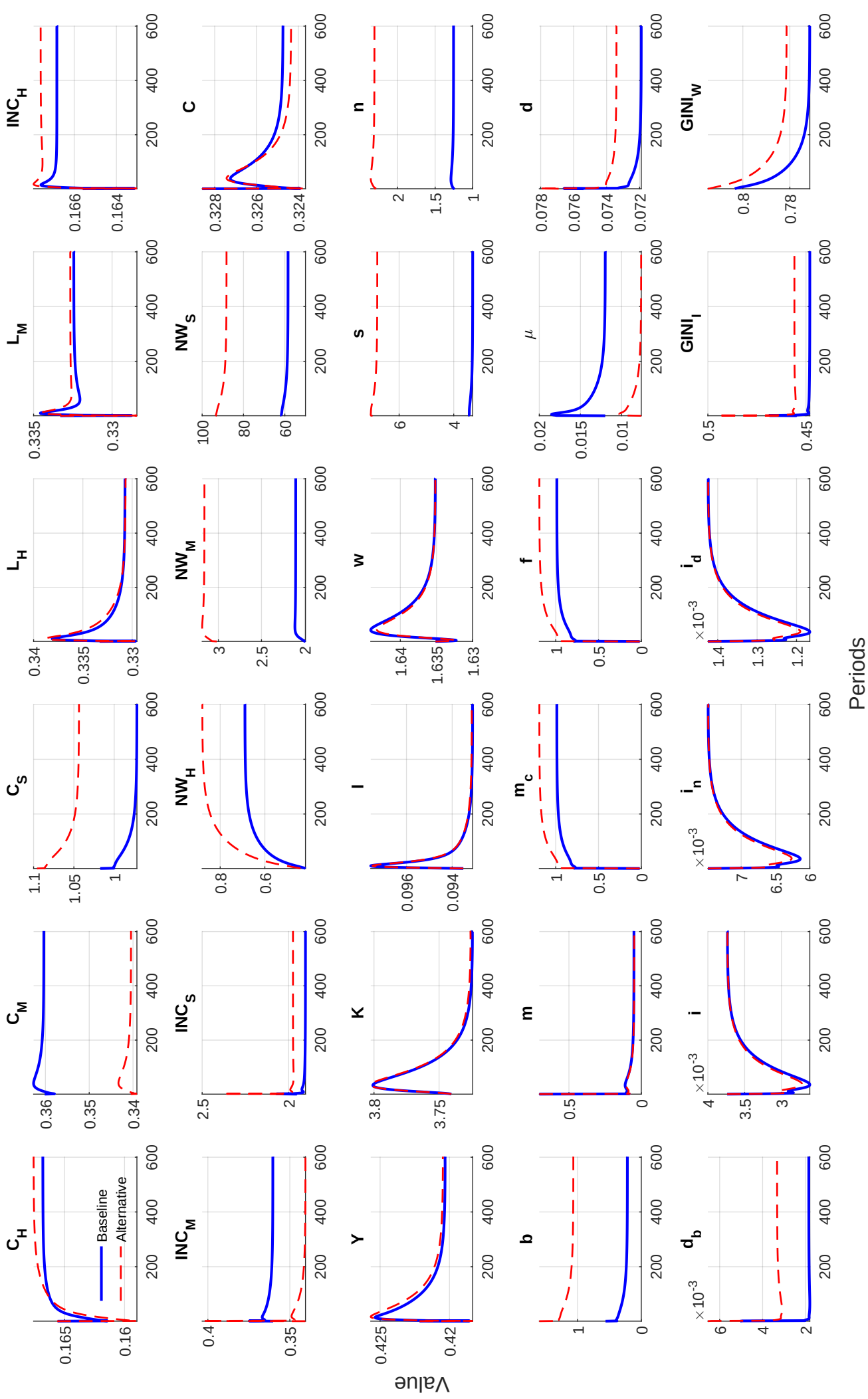


Figure 15: Full transition path (in levels) after the introduction of RFA comparing the baseline calibration of the utility elasticity parameters ($\{\sigma, \eta, \psi\} = 1.5$) against the alternative where $\{\sigma, \eta, \psi\} = 1.0001$.

C.4 Tiered Remuneration

This section includes detailed results for the tiered remuneration extension. Table 11 displays the final (RFA) steady-state values for the baseline case, and two calibrations of the threshold for the tiered remuneration cases. The initial (no-RFA) steady state has been omitted because it does not diverge from the baseline case, since tiered remuneration only affects RFA. Figure 16 and Figure 17 plot the transition paths (in levels) of the variables for $\bar{m}^c = 0.70$ and $\bar{m}^c = 0.50$, respectively. Both figures compare the baseline path (blue) with the tiered remuneration path (red).

Table 11: Tiered Remuneration Extension Steady States

Variable	Baseline Final SS	Extension Final SS	
	$\bar{m}^c = 0$	$\bar{m}^c = 0.70$	$\bar{m}^c = 0.50$
$C_{H,M,S}$	{0.1668, 0.3603, 0.9718}	{0.1668, 0.3603, 0.9776}	{0.1662, 0.3603, 0.9879}
$L_{H,M}$	{0.3307, 0.3324}	{0.3307, 0.3325}	{0.3320, 0.3328}
$m_{H,M,S}$	{0.0104, 0.0977, 0.0607}	{0.0104, 0.0977, 1.6490}	{0.0412, 0.1533, 1.8662}
$m_{H,M,S}^c$	{0.6791, 0.6275, 3.9560}	{0.6791, 0.6275, 0.7590}	{0.5146, 0.5145, 0.5665}
$H_{M,S}$	{1.3659, 4.5366}	{1.3621, 4.5515}	{1.3556, 4.5775}
$w_{H,M}$	{0.4967, 1.1471}	{0.4967, 1.1471}	{0.4958, 1.1482}
$INC_{H,M,S}$	{0.1668, 0.3603, 1.9029}	{0.1668, 0.3603, 1.9087}	{0.1662, 0.3603, 1.9208}
$NW_{H,M,S}$	{0.6895, 2.1029, 58.4320}	{0.6895, 2.1047, 58.5592}	{0.5557, 2.0506, 59.7561}
C	0.3247	0.3253	0.3260
Y	0.4203	0.4203	0.4212
K	3.7242	3.7243	3.7317
I	0.0931	0.0931	0.0933
L	0.1435	0.1435	0.1438
H	1	1	1
n	1.2546	1.2562	1.2591
s	3.3050	3.3183	3.3420
b	0.2180	0.3805	0.4708
d	0.0719	0.0727	0.0731
w	1.6351	1.6351	1.6351
m	0.0504	0.2092	0.2685
m^c	0.9874	0.6717	0.5265
mc	0.8333	0.8333	0.8333
d^i	0.0701	0.0701	0.0702
d^b	0.0018	0.0026	0.0029
f	0.9874	0.6717	0.5265
m_B^c	0.0013	0.0052	0.0067
G	0.0025	0.0019	0.0018
q	1	1	1
n_M	3.1364	3.1404	3.1478
b_S	2.1803	3.8053	4.7083

Table 11 – Continued

Variable	Baseline Final SS	Extension Final SS	
	$\bar{m}^c = 0$	$\bar{m}^c = 0.70$	$\bar{m}^c = 0.50$
K_S	37.2417	37.2429	37.3170
I_S	0.9310	0.9311	0.9329
d_S	0.7190	0.7269	0.7313
i	1.49%	1.49%	1.49%
i^d	0.57%	0.57%	0.57%
i^b	2.41%	2.41%	2.41%
i^n	2.99%	2.99%	2.99%
i^f	2.49%	2.49%	2.49%
r	0.0310	0.0310	0.0310
π	0	0	0
μ	0.0120	0.0120	0.0120
$GINI_W$	0.7715	0.7717	0.7870
$GINI_I$	0.4480	0.4487	0.4507
$i_{H,M,S}^c$	{1.49%, 1.49%, 1.49%}	{1.49%, 1.49%, 0.31%}	{1.20%, 1.20%, 0.16%}
$xs_m_{H,M,S}^c$	{0, 0, 0}	{0.0000, 0.0000, 0.0590}	{0.0146, 0.0145, 0.0665}

Note: “Final SS” refers to the RFA steady-state equilibria. The initial (no-RFA) steady-state equilibria have been omitted as they are identical across the three specifications. All values have been rounded to the fourth decimal and interest rates have been annualised. $xs_m_{H,M,S}^c$ represents the excess penalty agents incur if their RFA holdings exceed the limit, formally defined as: $xs_m_{H,M,S}^c \equiv \frac{1}{2} \left((m_i^c - \bar{m}^c) + \sqrt{(m_i^c - \bar{m}^c)^2} \right)$.

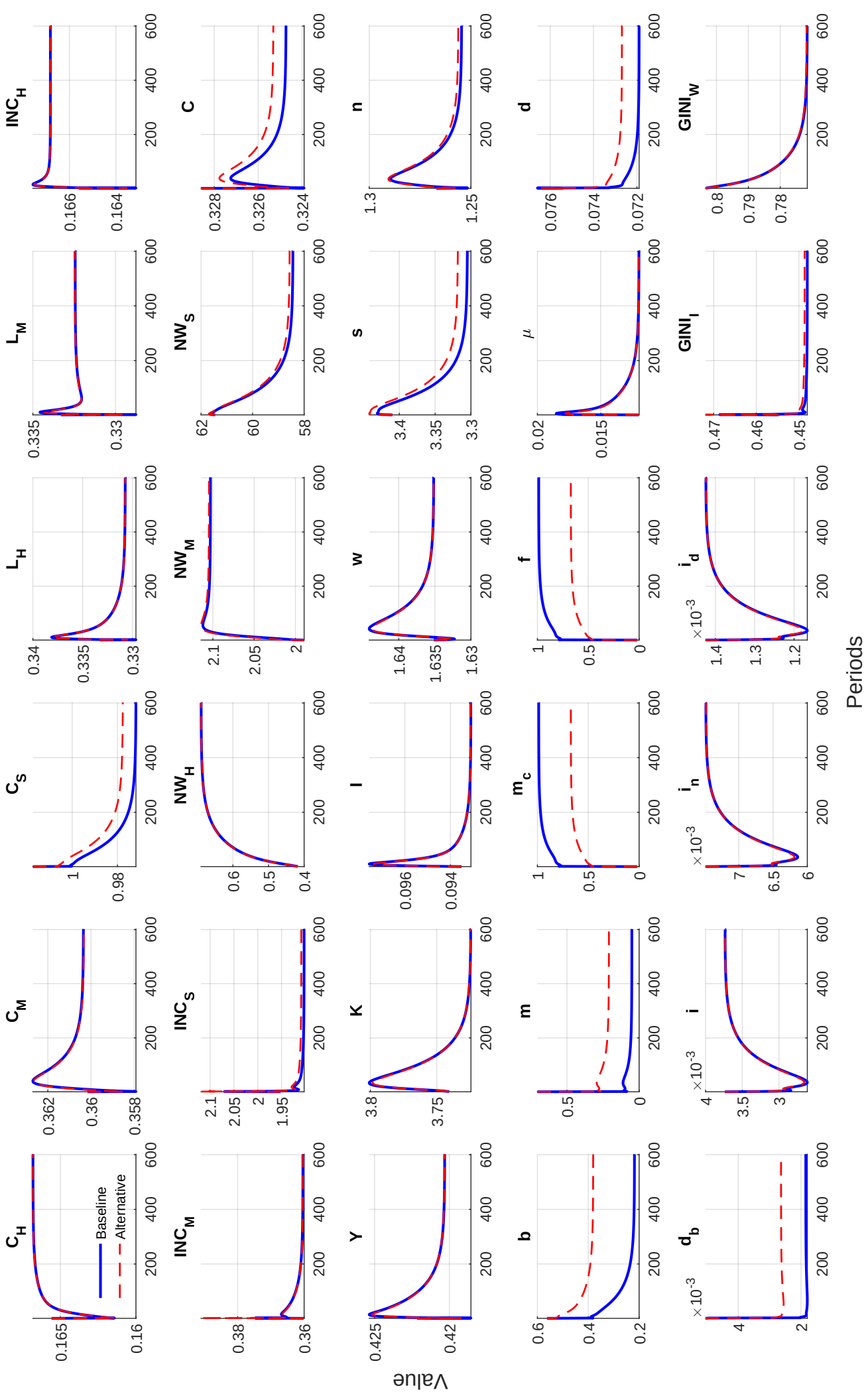


Figure 16: Full transition path (in levels) after the introduction of RFA comparing the baseline specification against the tiered remuneration extension with $\bar{m}^c = 0.70$.

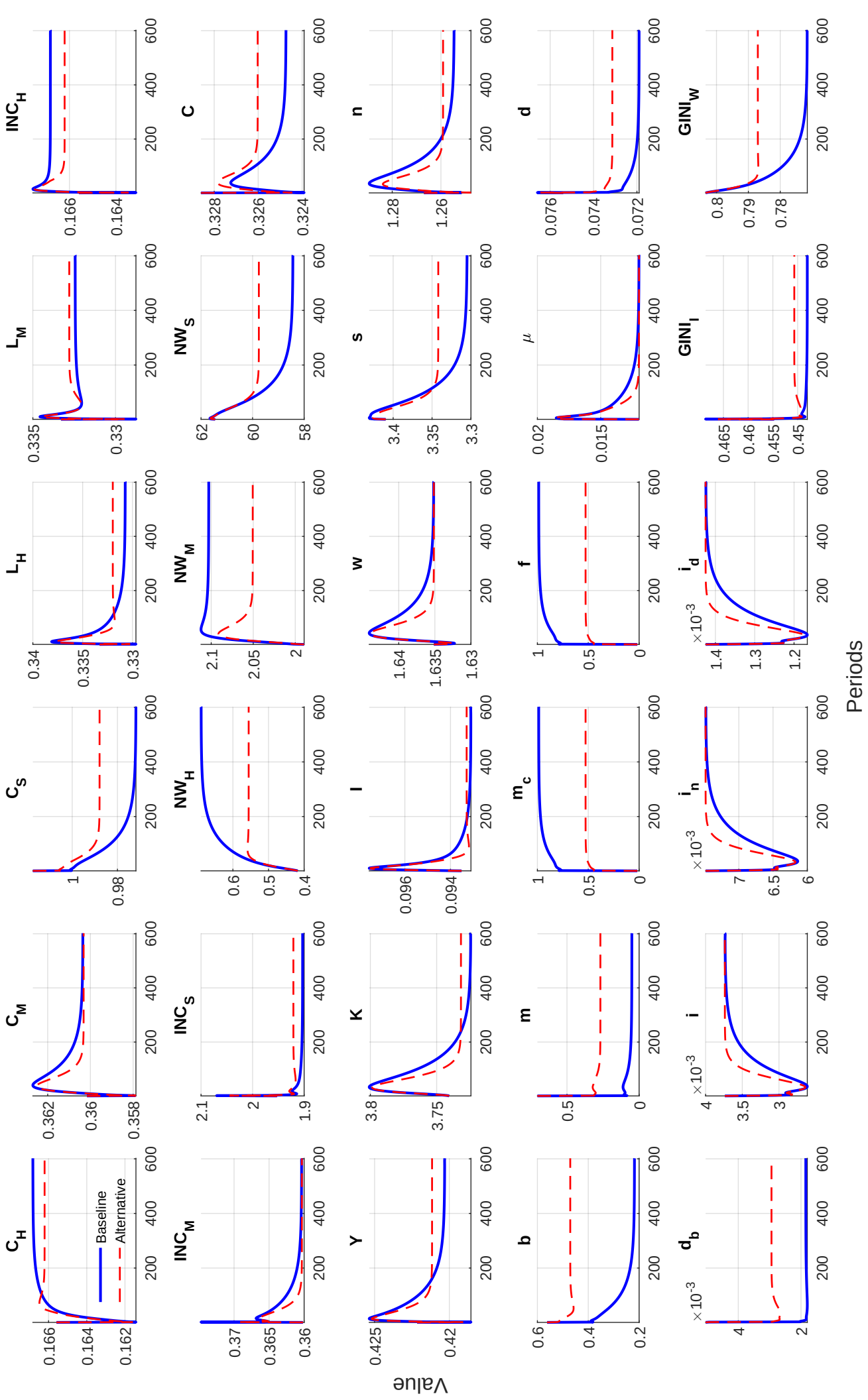


Figure 17: Full transition path (in levels) after the introduction of RFA comparing the baseline specification against the tiered remuneration extension with $\bar{m}^c = 0.50$.